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Chan et al.

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(54) **IN VIVO IMMOBILIZATION OF PROTEINS**

(56) **References Cited**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 479 days.

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(86) PCT No.: **PCT/CN2020/086652**

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PCT Pub. Date: **Oct. 29, 2020**

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C07K 14/325 (2006.01)

C12N 9/20 (2006.01)

C12N 15/70 (2006.01)

C12N 15/75 (2006.01)

(52) **U.S. Cl.**

CPC **C07K 14/325** (2013.01); **C12N 15/70** (2013.01); **C12N 15/75** (2013.01); **C07K 2319/00** (2013.01)

(58) **Field of Classification Search**

CPC **C07K 14/325**; **C07K 2319/00**; **C07K 9/20**; **C12N 15/70**; **C12N 15/75**; **C12P 21/02**

See application file for complete search history.

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(57) **ABSTRACT**

Provide is a novel system for immobilizing recombinantly produced proteins by entrapping them in crystals of co-expressed proteins that are capable of self-crystallization. Related compositions and as well as methods of making and using the immobilized proteins are also described.

23 Claims, 27 Drawing Sheets

Specification includes a Sequence Listing.

Figure 1

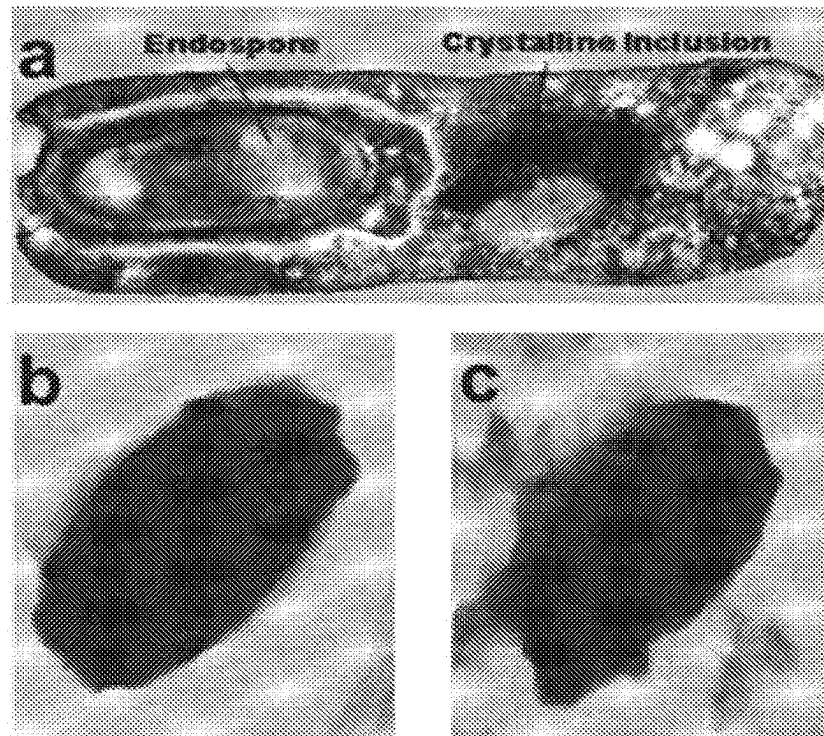


Figure 2

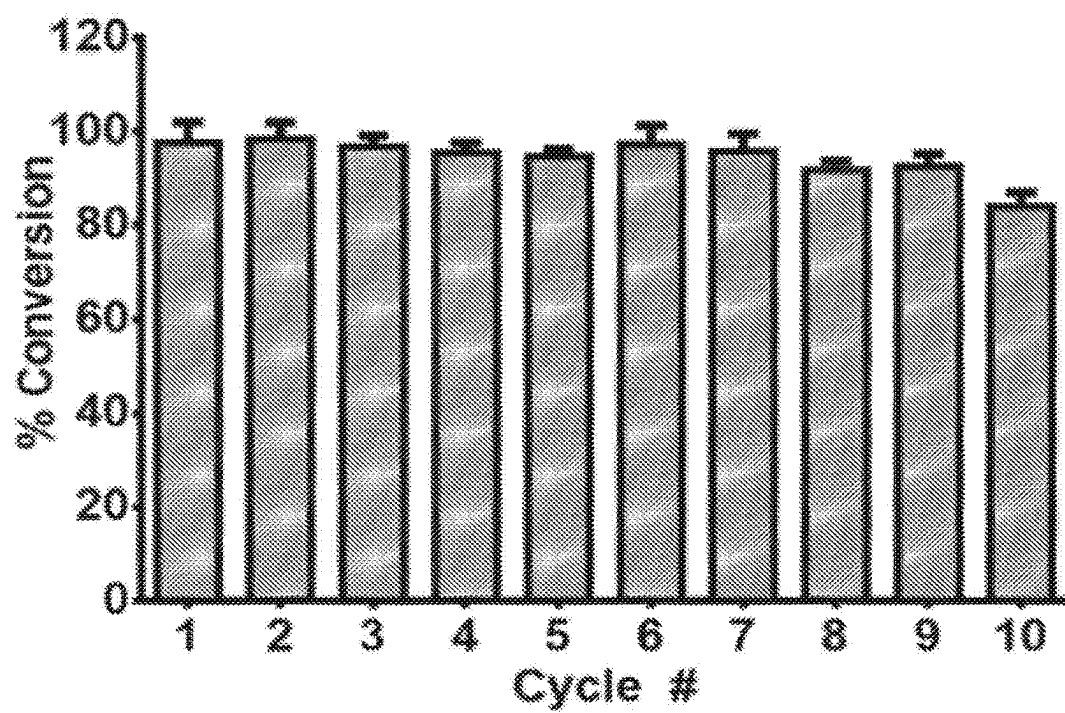


Figure 3

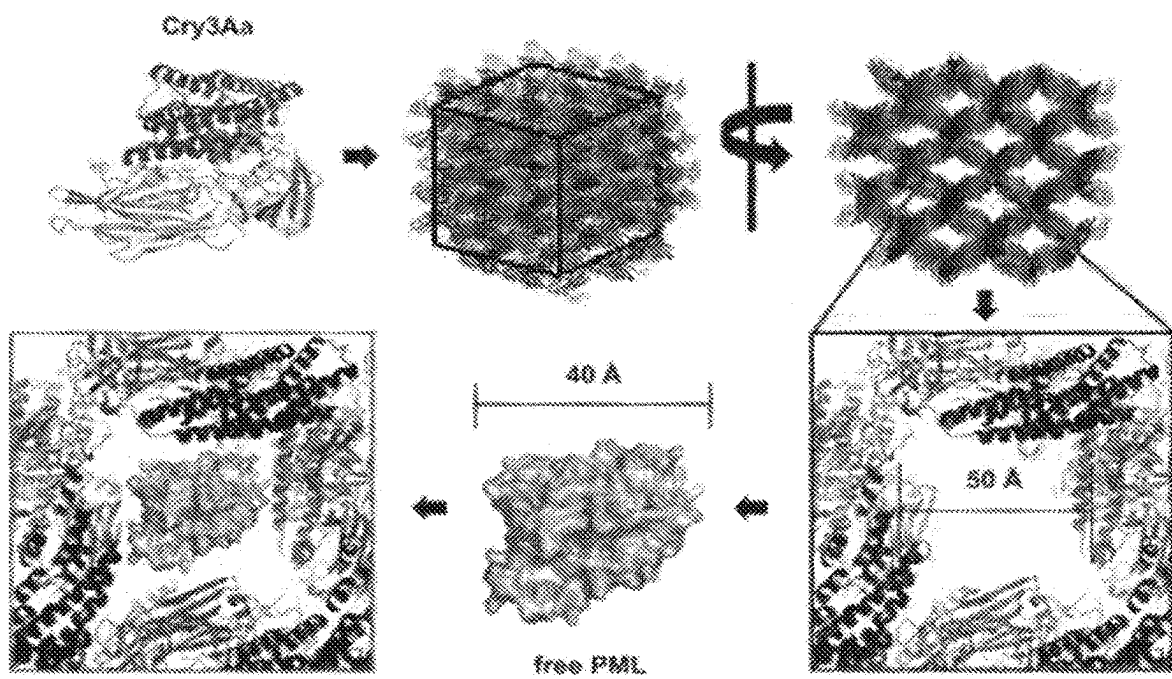


Figure 4

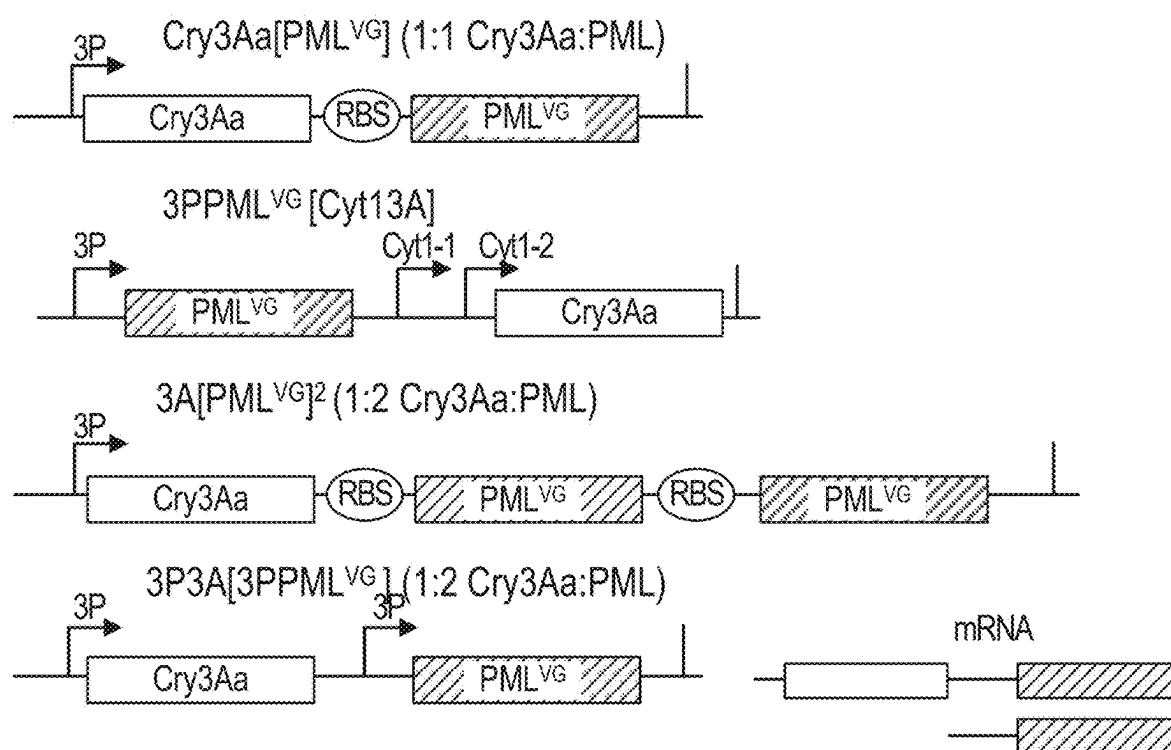


Figure 5

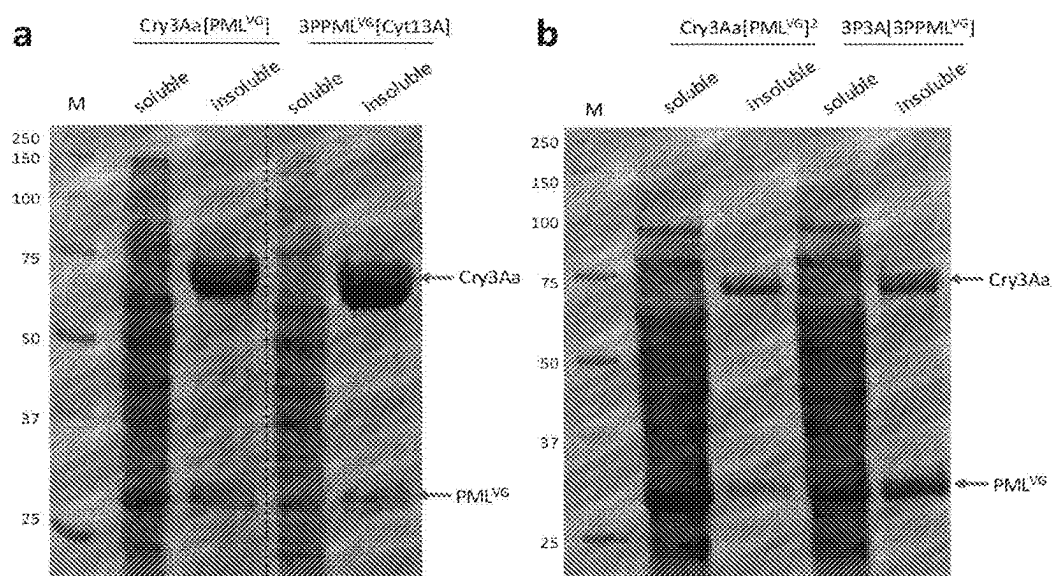


Figure 6

Cry3Aa

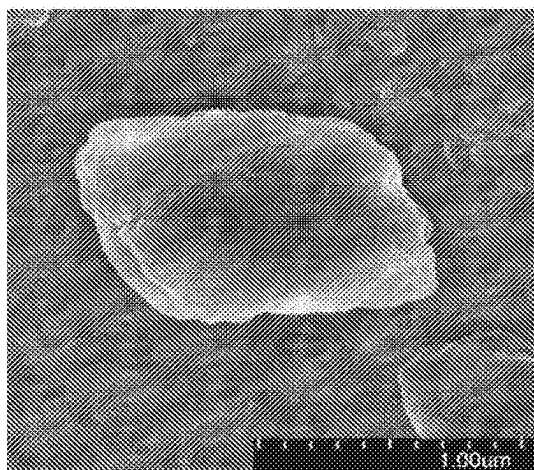
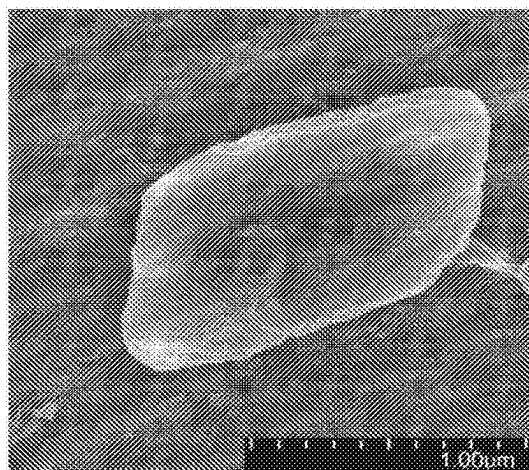
3P3A3P[PML^{VG}]

Figure 7

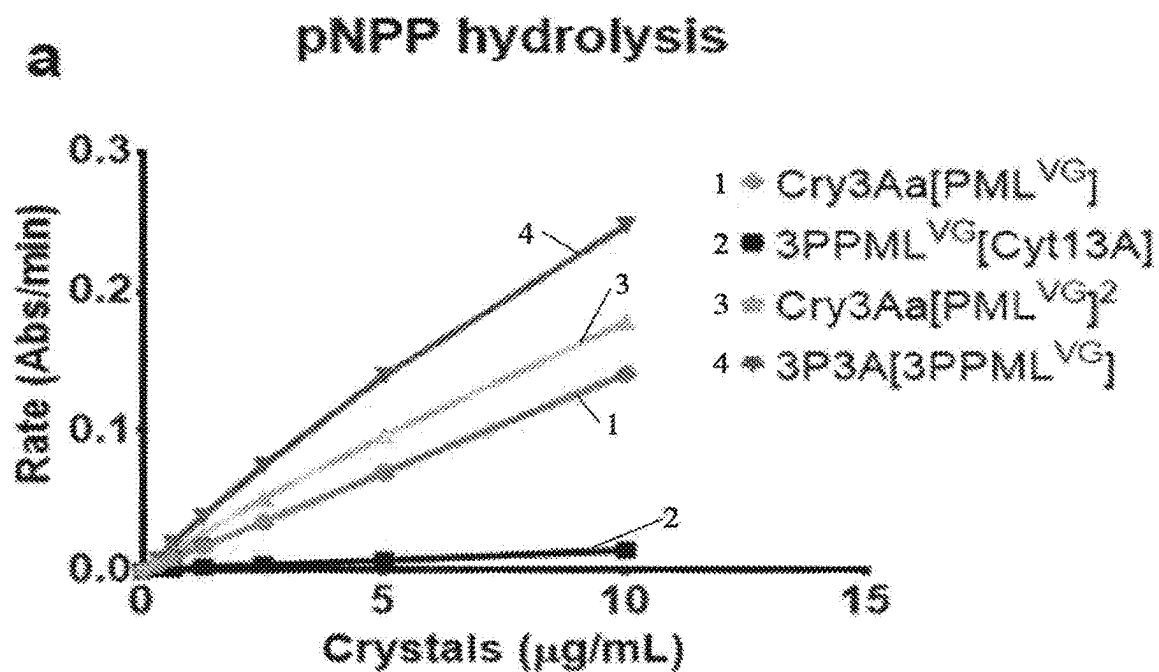
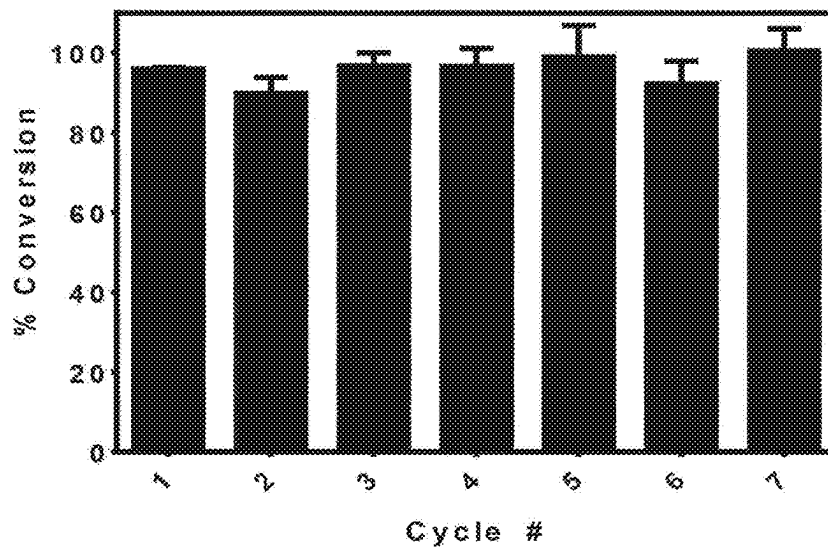


Figure 8

A



B

Thermal stability

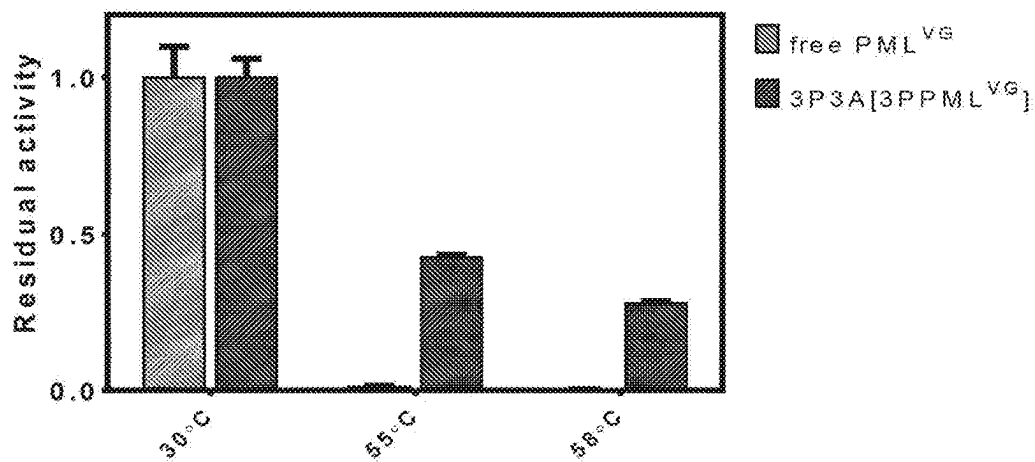


Figure 9

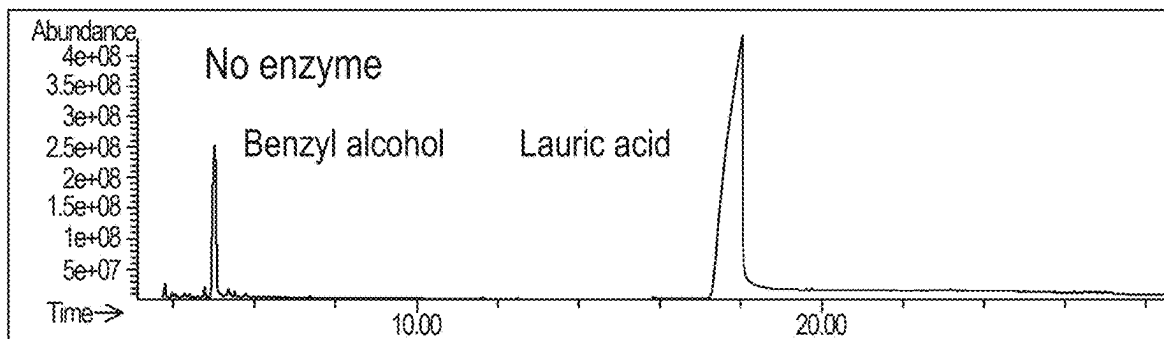
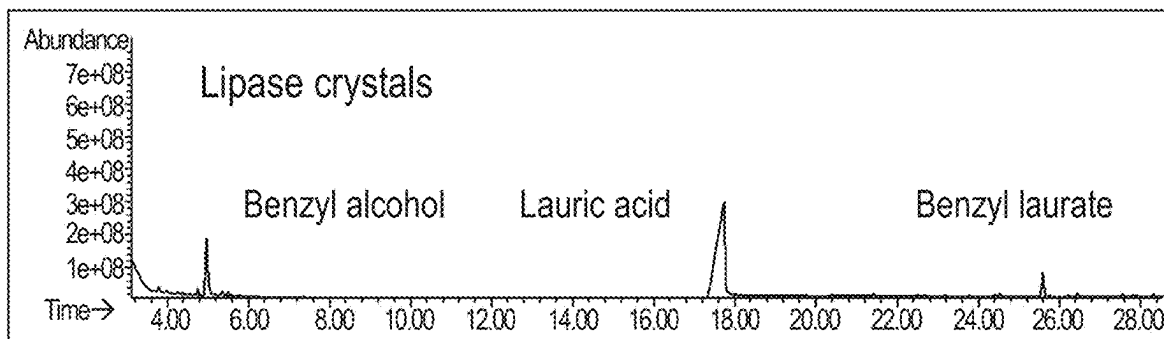
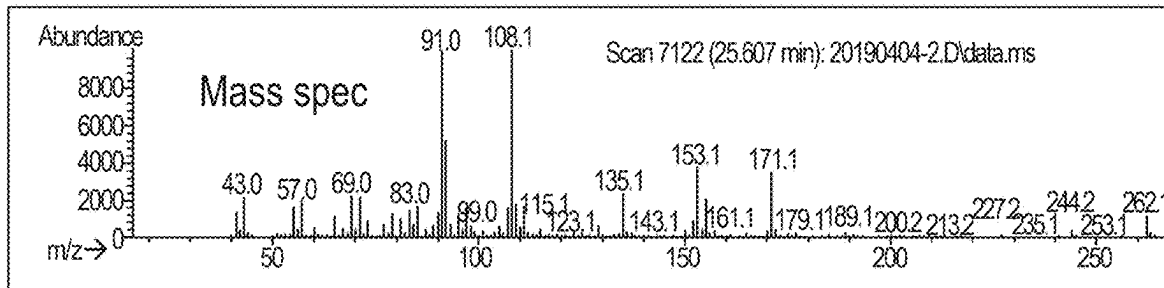
a**b****c**

Figure 10

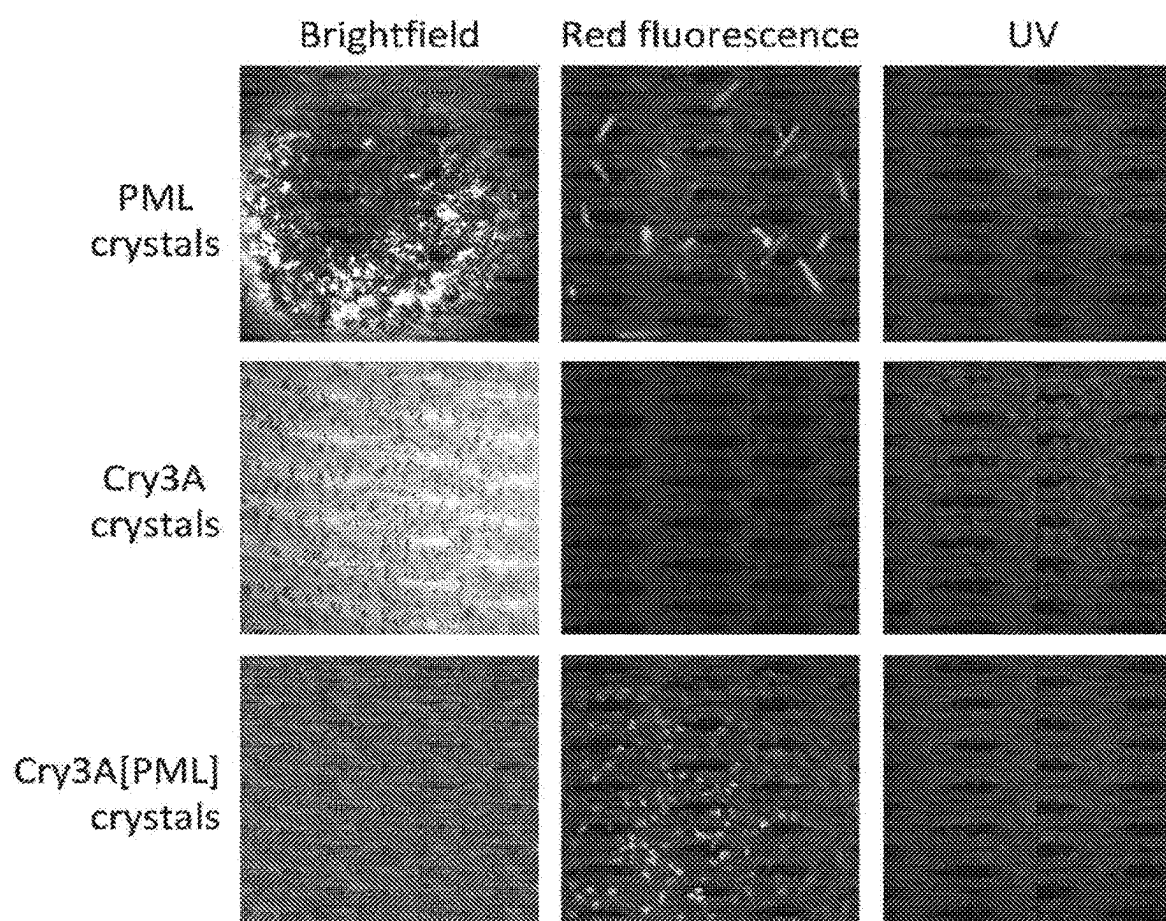


Figure 11

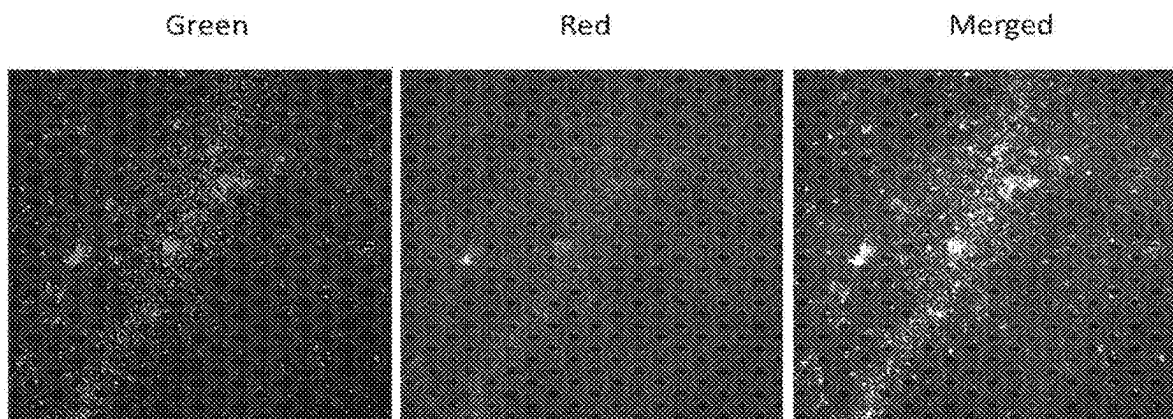


Figure 12

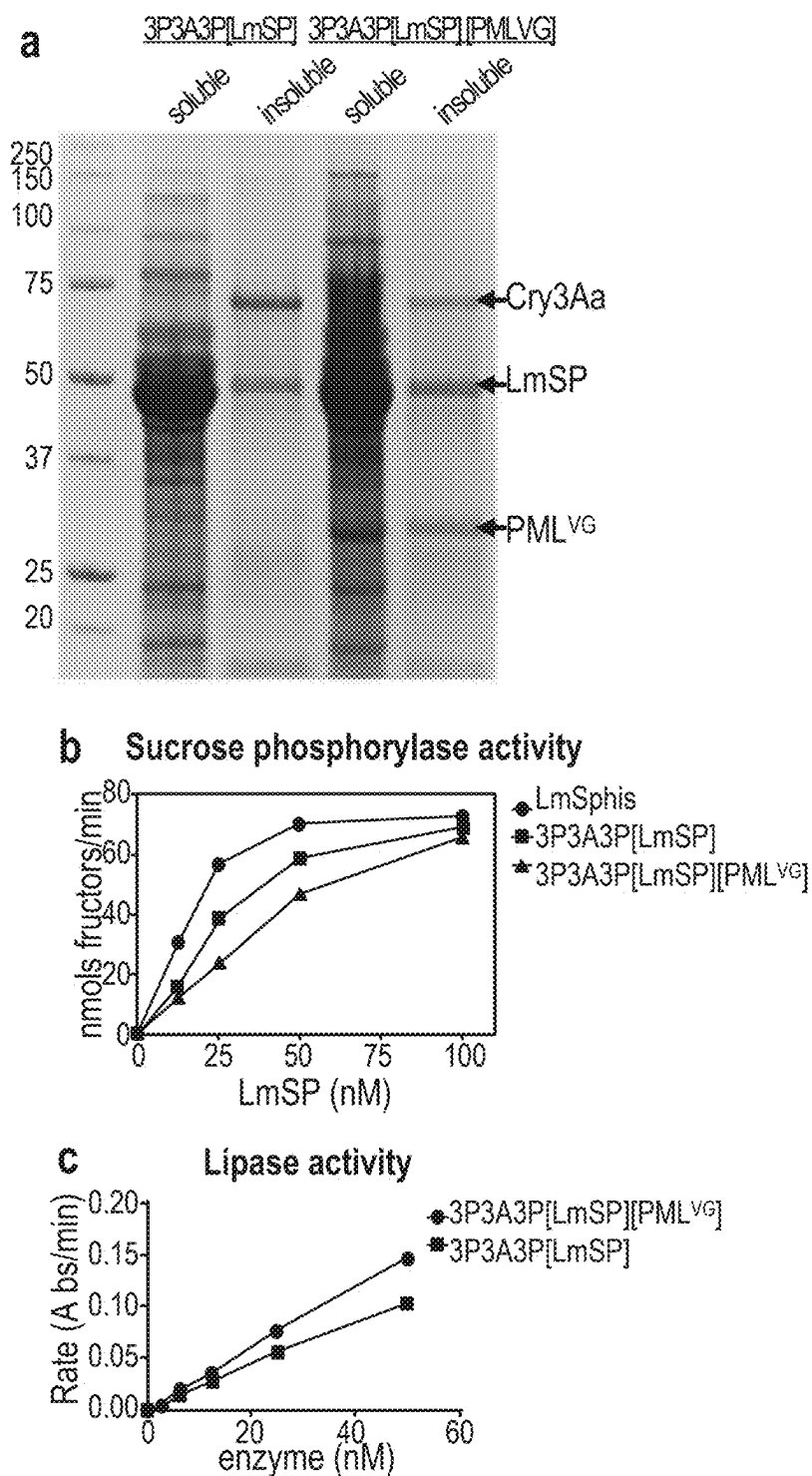


Figure 13

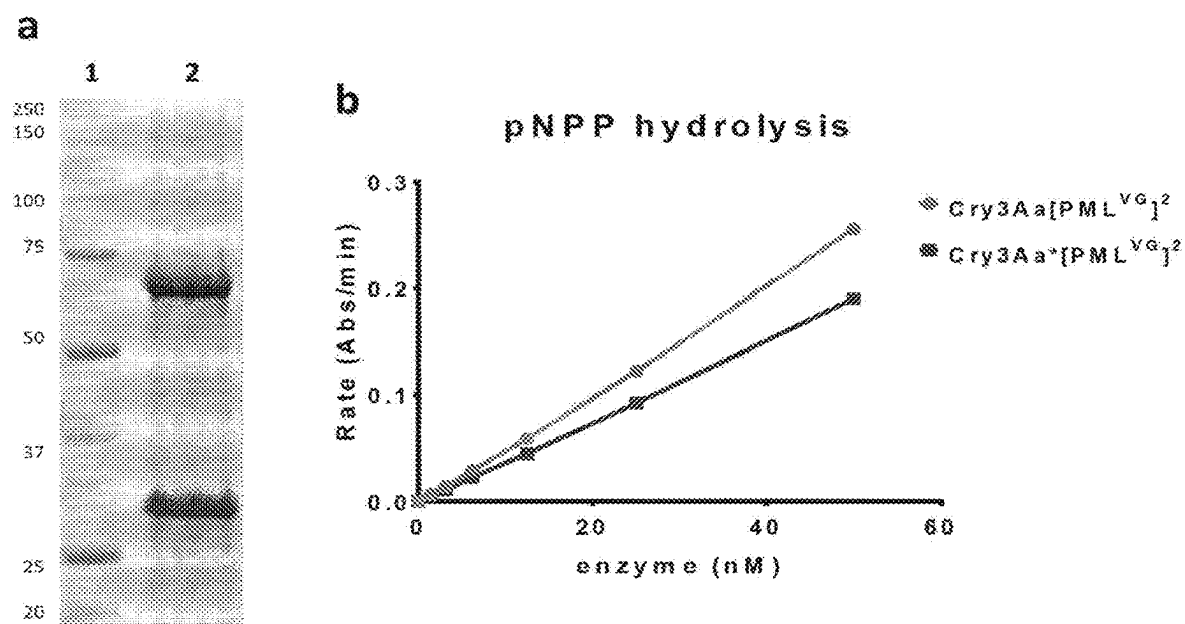


Figure 14

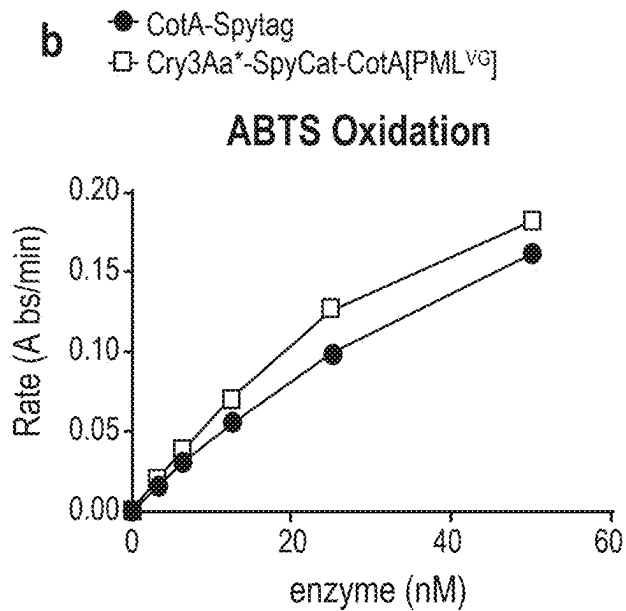
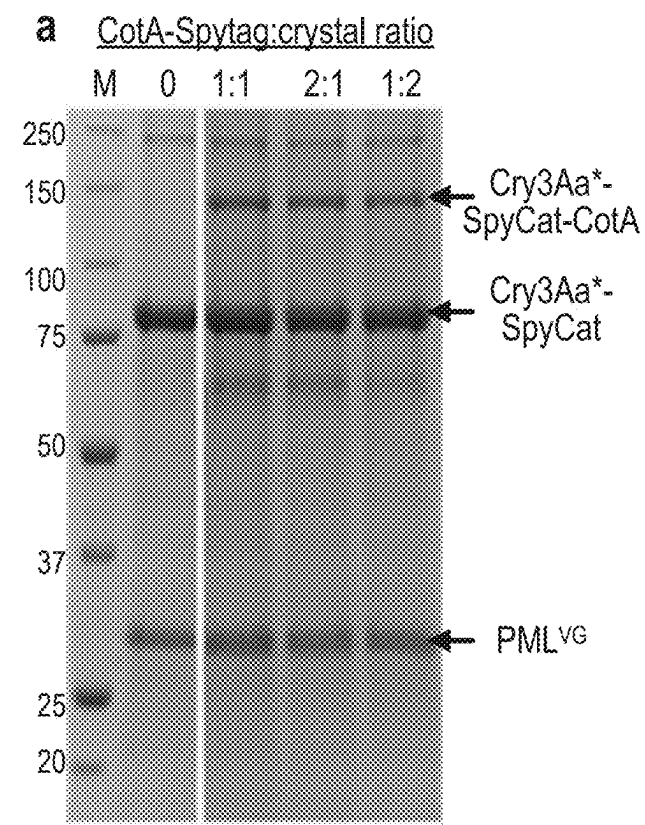


Figure 15

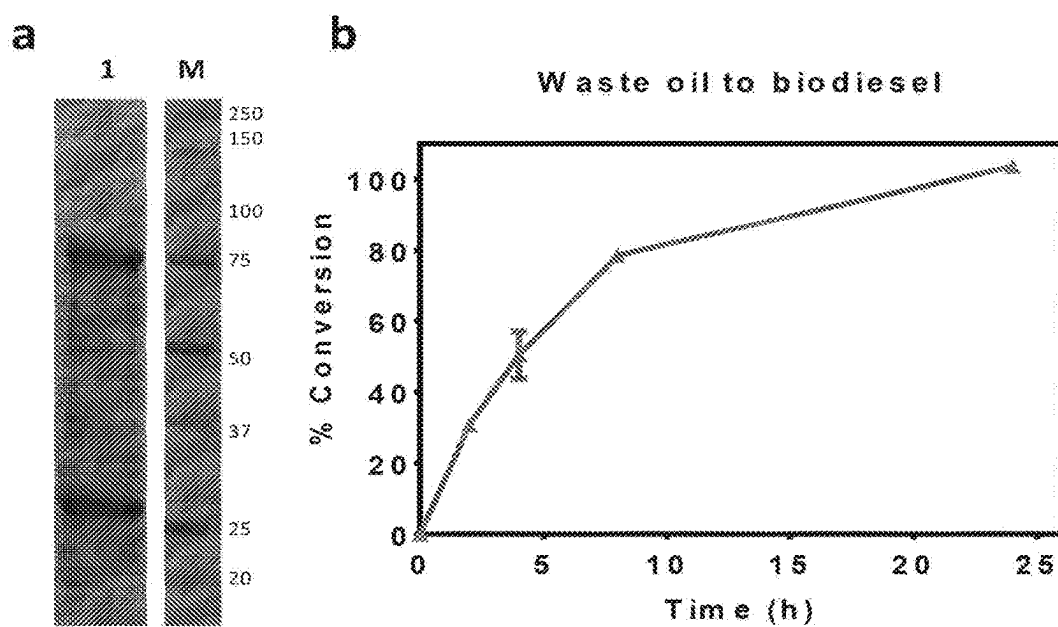
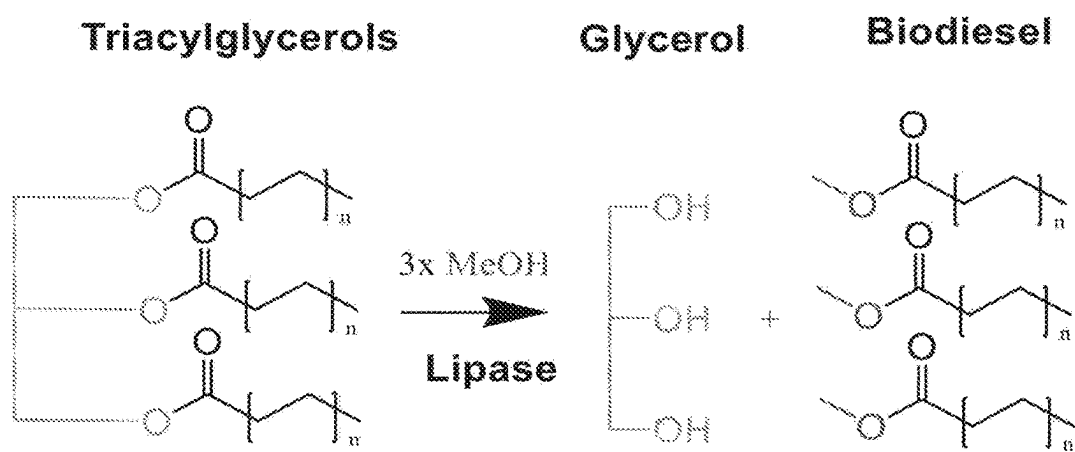


Figure 16A

Scheme 1



Scheme 2

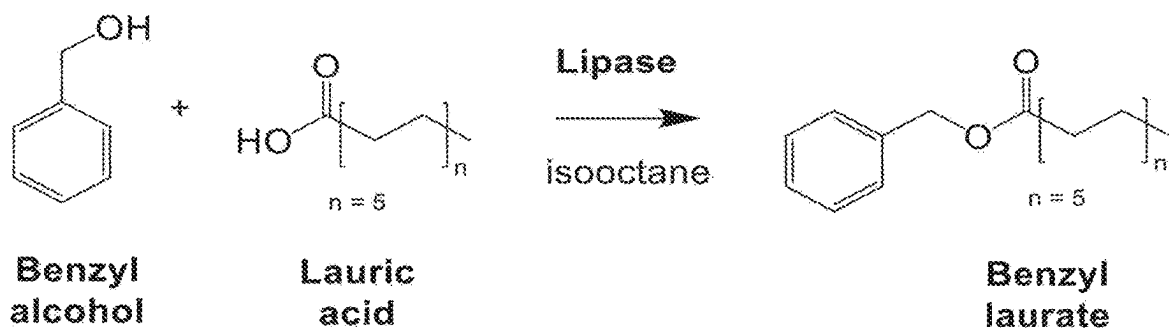
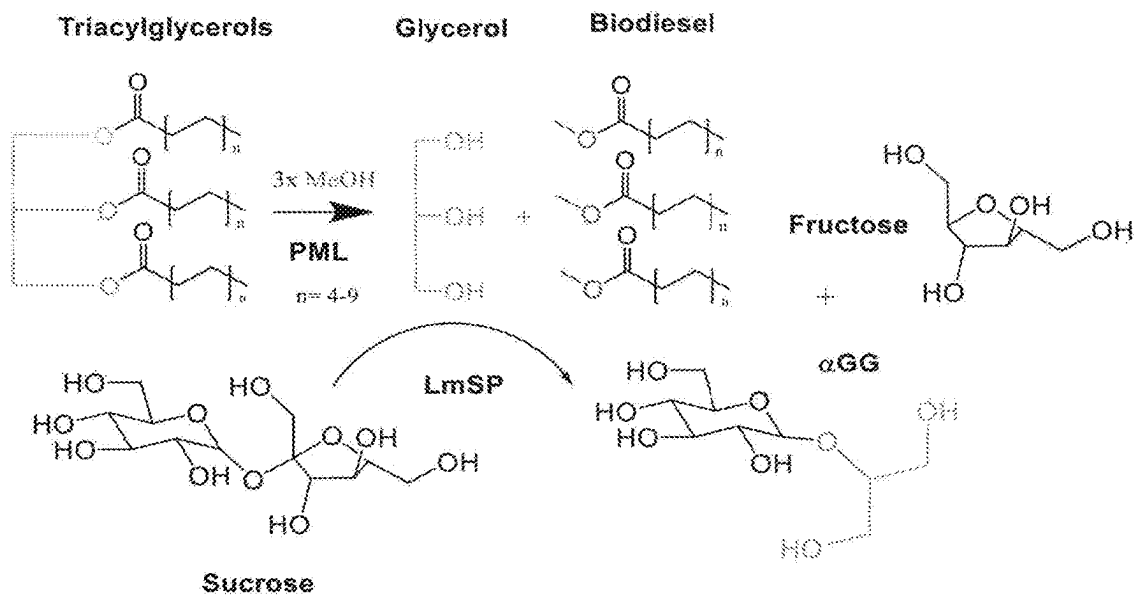


Figure 16B

Scheme 3



Scheme 4

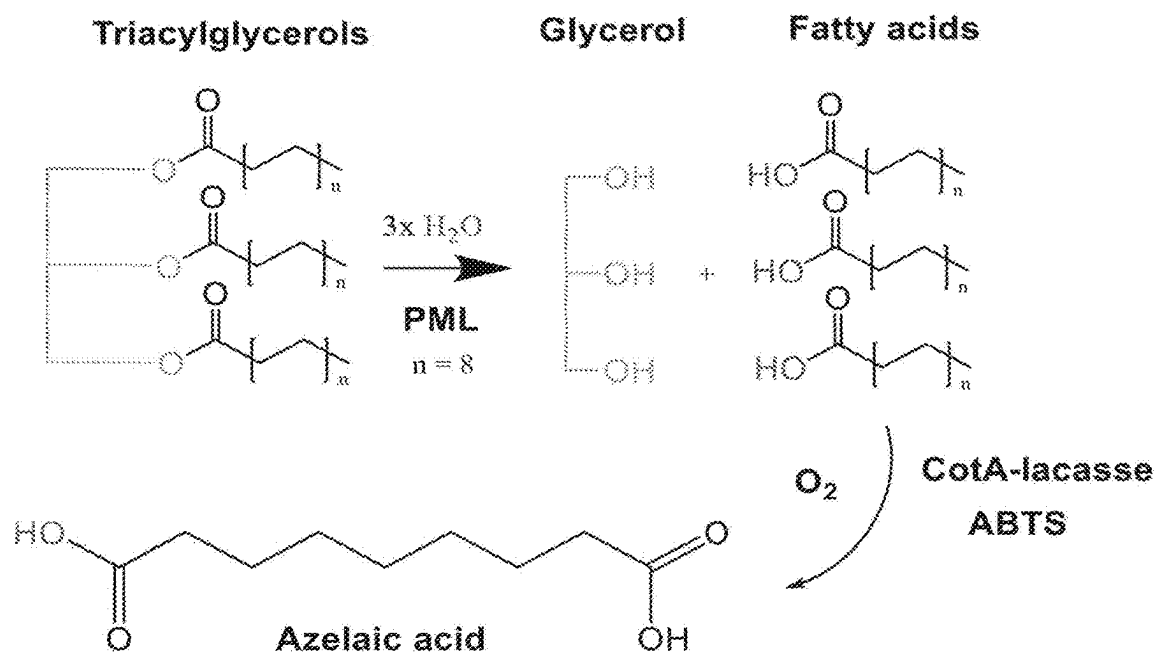


Figure 17

Cry3Aa aligned to negCry3Aa using NCBI protein blast
(Query = Cry3Aa wild-type, SEQ ID NO:3; Subject (Sbjct) = negCry3Aa, SEQ ID NO:6)

```
Query 1 MNPNRSEHDTIKTTENNEVPTNHVQYPLAETPNPTLEDLNYKEFLRMTADNNTALDSS 60
      MNPNRSEHDTIKTTENNEVPTNHVQYPLAETPNPTLEDLNYKEFLRMTADNNTALDSS
Sbjct 1 MNPNRSEHDTIKTTENNEVPTNHVQYPLAETPNPTLEDLNYKEFLRMTADNNTALDSS 60

Query 61 TTKDVIQKGISVVGDLLGVVGFPPGGALVSFYTNFLNTIWPSED PWKAFMEQVEALMDQK 120
      TTKDVIQKGISVVGDLLGVVGFPPGGALVSFYTNFLNTIWPSED PWKAFMEQVEALMDQK
Sbjct 61 TTKDVIQKGISVVGDLLGVVGFPPGGALVSFYTNFLNTIWPSED PWKAFMEQVEALMDQK 120

Query 121 IADYAKNKALAEQLQGLQNNVEDYVSALSSWQKNPVSSRNPHSQGRIPELFSQAESHFRNS 180
      IADYAKNKALAEQLQGLQNNVEDYVSALSSWQKNPVSSRNPHSQGRIPELFSQAESHFRNS
Sbjct 121 IADYAKNKALAEQLQGLQNNVEDYVSALSSWQKNPVSSRNPHSQGRIPELFSQAESHFRNS 180

Query 181 MPSFAISGIEVLFLTTYAQAANTHLFLLKDAQIYGEEWGGEYKEDIAEFYKRQLKLTQEYT 240
      MPSFAISGIEVLFLTTYAQAANTHLFLLKDAQIYGEEWGGEYKEDIAEFYKRQLKLTQEYT
Sbjct 181 MPSFAISGIEVLFLTTYAQAANTHLFLLKDAQIYGEEWGGEYKEDIAEFYKRQLKLTQEYT 240

Query 241 DHCVKWYNVGLDKLRGSSYESWVNFNRYRREMTLTVLDLIALFPLYDVRLYPEKVKTELT 300
      DHCVKWYNVGLDKLRGSSYESWVNFNRYRREMTLTVLDLIALFPLYDVRLYPEKVKTELT
Sbjct 241 DHCVKWYNVGLDKLRGSSYESWVNFNRYRREMTLTVLDLIALFPLYDVRLYPEKVKTELT 300

Query 301 RDVLTDPFIVGVNNLRGYGTTFSNIENYIRKPHLFDYLHRIQFHTRFQPGYYGNDSENYWS 360
      RDVLTDPFIVGVNNLRGYGTTFSNIENYIRKPHLFDYLHRIQFHTRFQPGYYGNDSENYWS
Sbjct 301 RDVLTDPFIVGVNNLRGYGTTFSNIENYIRKPHLFDYLHRIQFHTRFQPGYYGNDSENYWS 360

Query 361 GNYVSTRPSIGSNDIITSFFYGNSSSEPVCLEFGEKVYRAVANTNLAVWPSAVYSGVT 420
      GNYVSTRPSIGSNDIITSFFYGNSSSEPVCLEFGEKVYRAVANTNLAVWPSAVYSGVT
Sbjct 361 GNYVSTRPSIGSNDIITSFFYGNSSSEPVCLEFGEKVYRAVANTNLAVWPSAVYSGVT 420

Query 421 KVEFQYNDTDEASSTYDSNVGAVSWDSIDQLPPEDEPLEGYSHQLNYVMCFEL 480
      KVEFQYNDTDEASSTYDSNVGAVSWDSIDQLPPEDEPLEGYSHQLNYVMCFEL
Sbjct 421 KVEFQYNDTDEASSTYDSNVGAVSWDSIDQLPPEDEPLEGYSHQLNYVMCFEL 480

Query 481 MQGSRGTIPVLTWTHKSVDFNMMIDSKKITQLPLVKAYKLQSGASVVAGPRFTGGDIIQC 540
      MQGSRGTIPVLTWTHKSVDFNMMIDSKKITQLPLVKAYKLQSGASVVAGPRFTGGDIIQC
Sbjct 481 MQGSRGTIPVLTWTHKSVDFNMMIDSKKITQLPLVKAYKLQSGASVVAGPRFTGGDIIQC 540

Query 541 TENGSAATIVTPEVSYSQKYRARIHYASTSQITFTLSLDGAPFNQYYFDKTINKGDTLT 600
      TENGSAATIVTPEVSYSQKYRARIHYASTSQITFTLSLDGAPFNQYYFDKTINKGDTLT
Sbjct 541 TENGSAATIVTPEVSYSQKYRARIHYASTSQITFTLSLDGAPFNQYYFDKTINKGDTLT 600

Query 601 YNSFNLASFSTPFELSGNNLQIGVTGLSAGDKVYIDKIEFIPVN 644
      YNSFNLASFSTPFELSGNNLQIGVTGLSAGDKVYIDKIEFIPVN
Sbjct 601 YNSFNLASFSTPFELSGNNLQIGVTGLSAGDKVYIDKIEFIPVN 644
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Figure 18

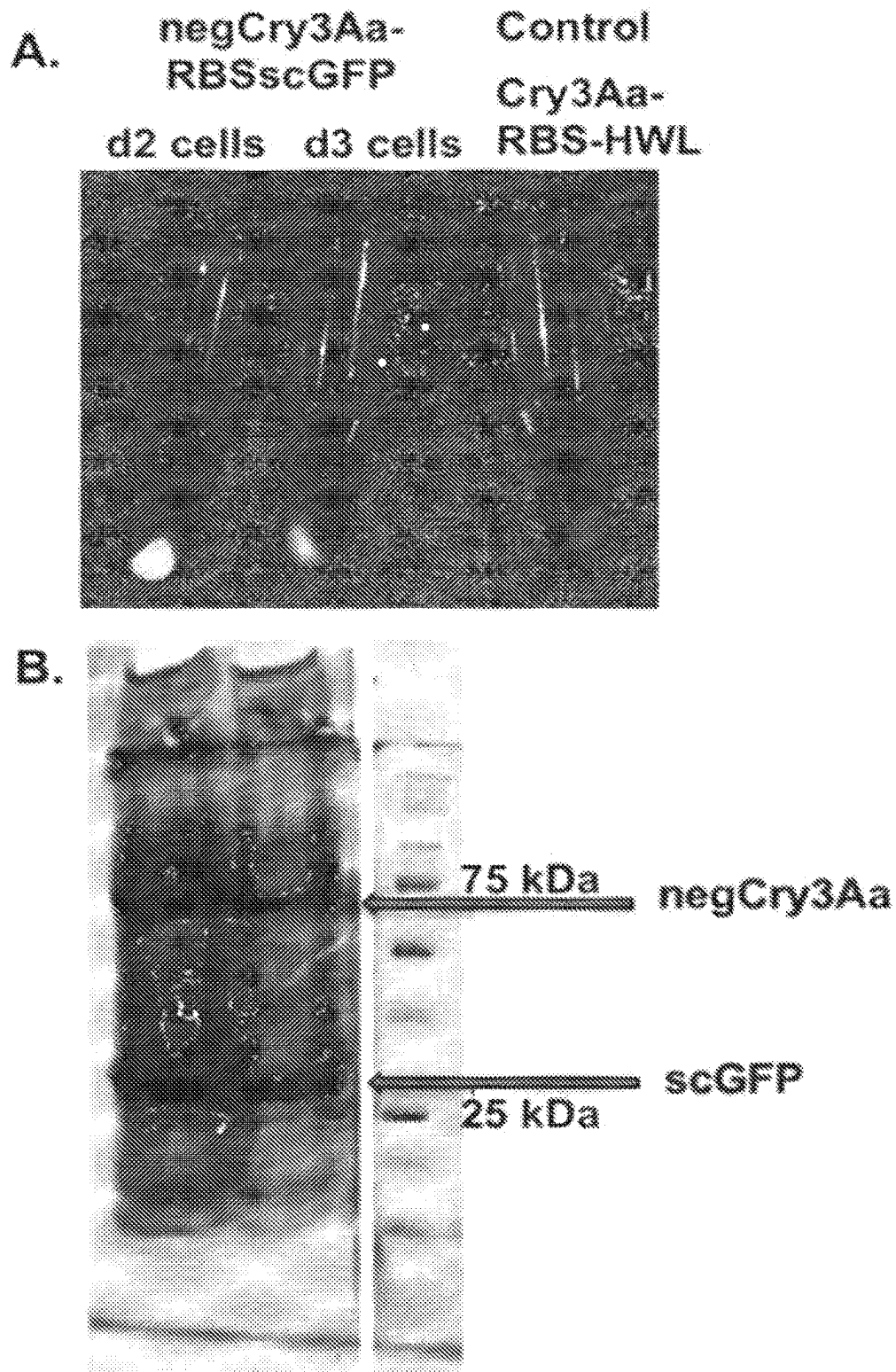


Figure 19

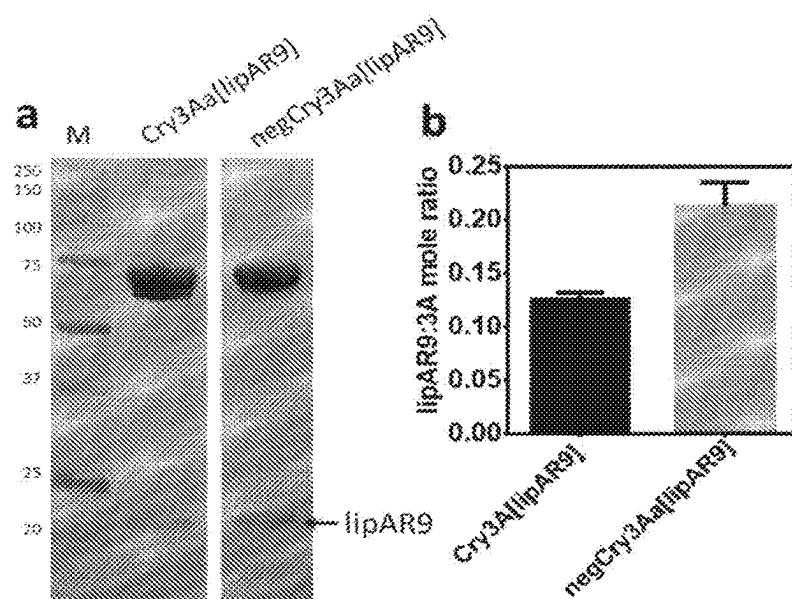


Figure 20

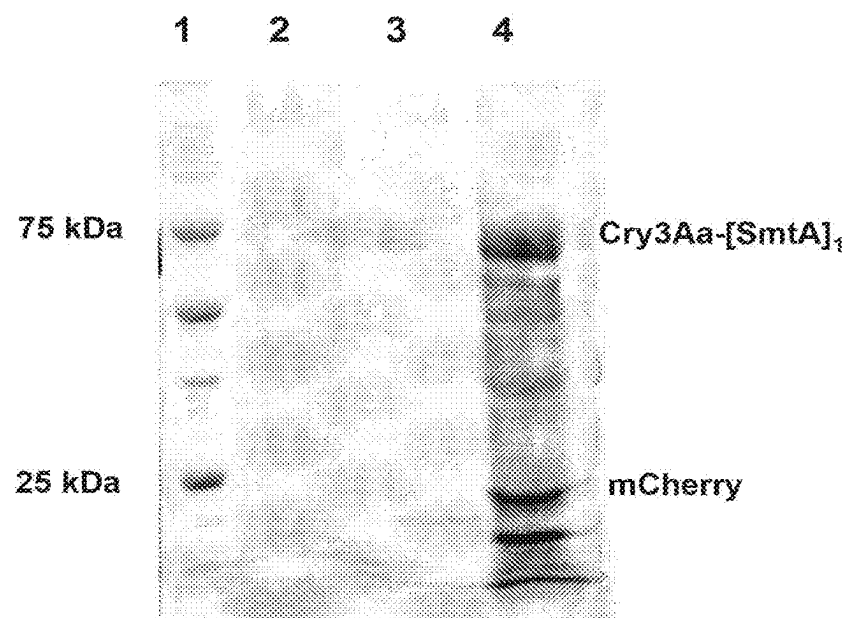


Figure 21

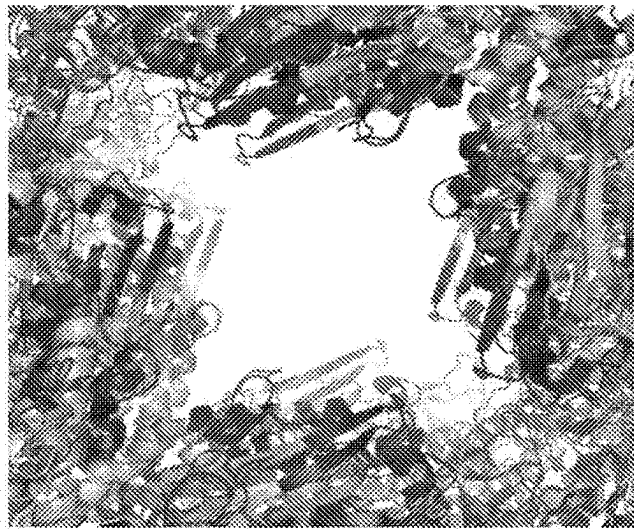


Figure 22

1 MNPNNRSEHD TIKTTENNEV PTNHVQYPLA ETPNPTLELL NYKEFLRMTA DNNTAALDSS
61 TTKDVIQKGI SVVGDLLGVV GPPFGGALVS FYTNFLNTIW PSEDPWKAFM **EQVEALMDQK**
121 **IADYAKNKAL** AELQGLQNNV EDYVSALSSW QKNFVSSRNPF HSQGRIRELF SQAESHFRNS
181 **MPSFAISGYE** VLFLTTYAQA ANTHLFLLLKD **AQIYGEEWGY** EKEDIAEFYK RQLRLTQEYT
241 DHCVKWYNVG **LDKLRGSSYE** **SWVNFNRIRR** EMTLTVLDLI ALFPLYDVRL YPKEVKTELT
301 EDVLTDFIVG VNNLRGYGTT FSNIENTYIRK PHLFDYLRRI QPHTRFQPGY YGNDSEFNYWS
361 GNYVSTRPSI GSNDIITSFF YGNKSSEFVQ NLEFNCEKVV RAVANTNLAV WPSAVYSGVT
421 **KVEFSQYNDQ** **TDEASTQTYD** SKRNVGAVSW DSIDQLPPET **TDEPLEKGYS** HQENYVMCFI
481 MQGSRGTIPV LTWTKSVDF FNMIDSKKIT QLPLVKAYKL QSGASVVAGP RFTGGDIIQC
541 TENGSAATII YTPDVSYSQK YRARIHYAST SQITFTLSLD GAPFNQYYFD KTINKGDTLT
601 YNSFNLASFS TPFELSGNNL QIGVTGLSAG **DKVYIDKIEF** IFVN

Figure 23

Region	Amino Acid Residues
110-128	M110 E111 Q112 V113 E114 A115 L116 M117 D118 Q119 K120 I121 A122 D123 Y124 A125 K126 N127 K128
181-197	M181 P182 S183 F184 A185 I186 S187 G188 Y189 E190 V191 L192 F193 L194 T195 T196 Y197
211-226	A211 Q212 I213 Y214 G215 E216 E217 W218 G219 Y220 E221 K222 E223 D224 I225 A226
250-264	G250 L251 D252 K253 L254 R255 G256 S257 S258 Y259 E260 S261 W262 V263 N264
423-436	E423 F424 S425 Q426 Y427 N428 D429 Q530 T431 D432 E433 A434 S435 T436
457-471	P457 P458 E459 T460 T461 D462 E463 P464 L465 E466 K467 G468 Y469 S470 H471
593-598	I593 N594 K595 G596 D597 T598
627-632	L627 S628 A629 G630 D631 K632

Figure 24

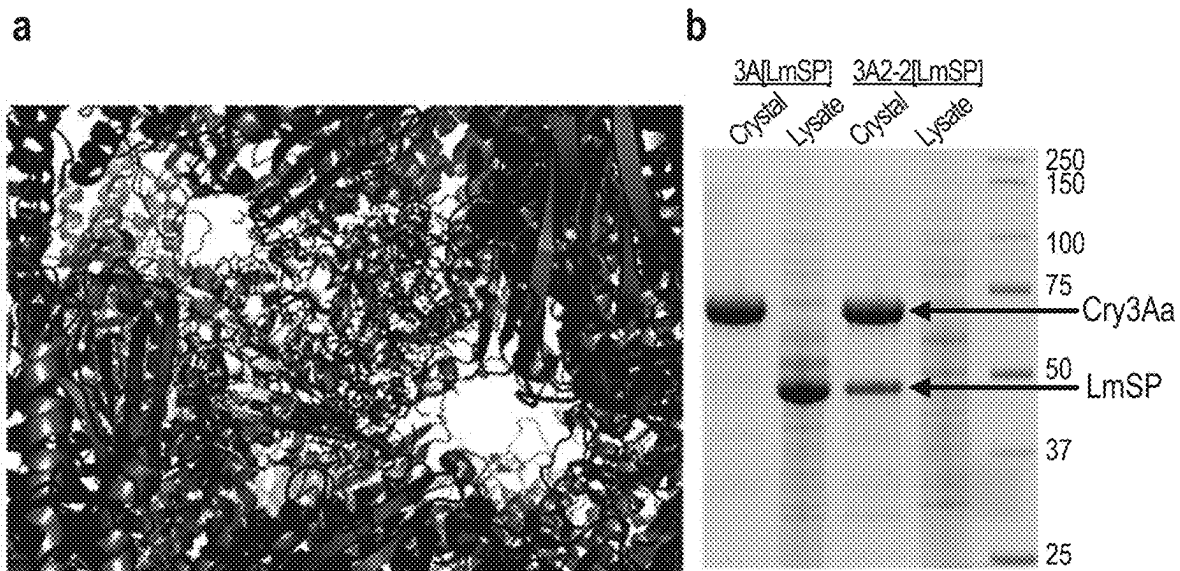


Figure 25

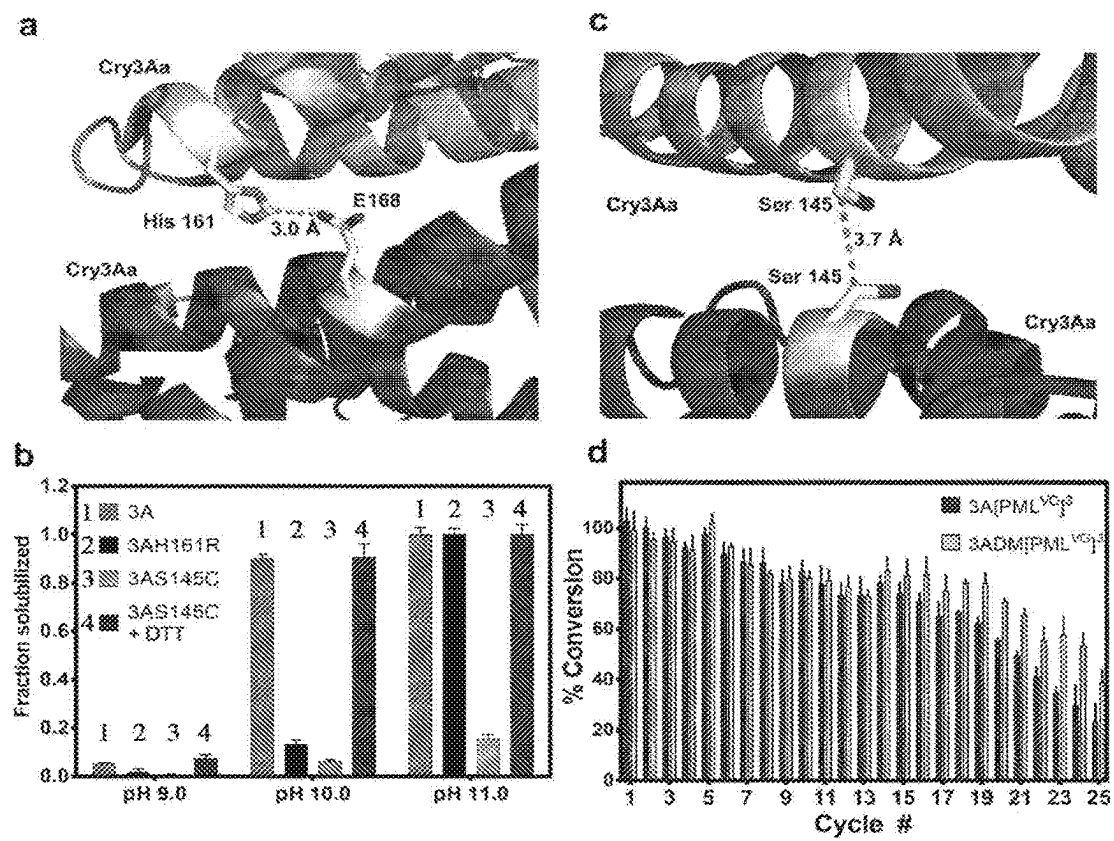
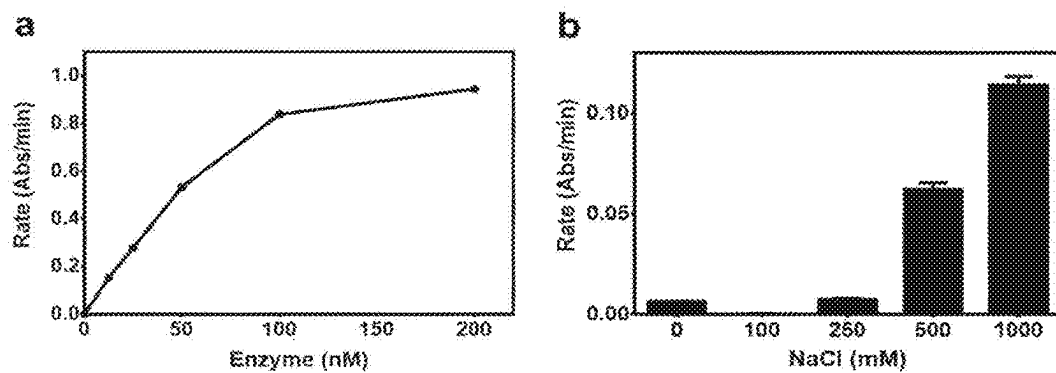


Figure 26



IN VIVO IMMOBILIZATION OF PROTEINS

RELATED APPLICATIONS

This application is a 371 National Stage Entry of PCT Patent Application No. PCT/CN2020/086652, filed Apr. 24, 2020, which claims priority to U.S. Provisional Patent Application No. 62/839,400, filed Apr. 26, 2019, the contents of which are hereby incorporated by reference in the entirety for all purposes.

REFERENCE TO SUBMISSION OF A SEQUENCE LISTING AS A TEXT FILE

The Sequence Listing written in file 080015-1258878-027710US_SL.txt created on Sep. 20, 2022, 111,479 bytes, machine format IBM-PC, MS-Windows operating system, is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND OF THE INVENTION

In many medical and industrial contexts the use of recombinant proteins is becoming increasingly more important. The production, isolation, use, and potential reuse of recombinant proteins, especially enzymes, remain in need of improvement for better quality, efficiency, and stability. For instance, enzyme immobilization can facilitate easy removal and subsequent reuse of enzymes during multiple rounds of catalysis. In many cases, immobilization also improves the stability of enzymes against many industrial conditions such as high temperatures and organic solvents. Immobilization by entrapment is particularly attractive since it does not involve any modifications to the enzyme structure, increasing the chance for high activity retention and native enzyme conformation. Thus, there exists a need for new and effective means for producing recombinant proteins, such as enzymes, in immobilized form. The present invention fulfills this and other related needs.

BRIEF SUMMARY OF THE INVENTION

This invention provides a novel approach to improve the physical properties and stability of recombinant proteins by immobilizing recombinant proteins such as enzymes useful in various industrial applications. Thus, in a first aspect, this invention provides a method for recombinantly co-expressing a protein of interest with a crystal-forming protein. The method includes these steps: (1) providing bacterial cells comprising an expression cassette encoding the protein of interest and an expression cassette encoding a Cry protein, a crystal-forming fragment of the Cry protein, or a fusion protein capable of forming crystals comprising the Cry protein or the crystal-forming fragment thereof, and (2) culturing the bacterial cells under conditions permissible for the expression of the protein of interest as well as the Cry protein, the crystal-forming fragment thereof, or the fusion protein, wherein the Cry protein, the crystal-forming fragment thereof, or the fusion protein forms crystal containing the protein of interest upon both being expressed in the bacterial cells.

In some embodiments, the protein of interest is an enzyme, such as a lipase (e.g., *Proteus mirabilis* lipase, or PML, including a PML variant with modifications of residues 118 and 130, for example, I118V+E130G, and lipA or lipAR9), ligase, hydrolase, esterase, protease, or glycosidase. In some embodiments, the Cry protein is Cry3Aa. In

some embodiments, the crystal-forming fragment is the N-terminal 290 amino acids of Cry3Aa, or the N-terminal 625 or 626 amino acids of Cry3Aa, or the 498-644 fragment of Cry3Aa. In some embodiments, the expression cassette encoding the protein of interest and the expression cassette encoding the Cry protein, crystal-forming fragment thereof, or the fusion protein is one and the same expression cassette. In some embodiments, the one single expression cassette comprises (1) one copy of polynucleotide sequence encoding the Cry protein, crystal-forming fragment thereof, or the fusion protein, and (2) one copy or two or more copies of polynucleotide sequence encoding the protein of interest. In some embodiments, (1) the polynucleotide sequence encoding the Cry protein, crystal-forming fragment thereof, or the fusion protein, and (2) the polynucleotide sequence encoding the protein of interest are operably linked to one single promoter. In some embodiments, the one single promoter is operably linked to (1) one copy of the polynucleotide sequence encoding the Cry protein, crystal-forming fragment thereof, or the fusion protein, followed by (2) one copy of the polynucleotide sequence encoding the protein of interest, with one ribosome binding site between (1) and (2). In some embodiments, the one single promoter is operably linked to (1) one copy of the polynucleotide sequence encoding the Cry protein, crystal-forming fragment thereof, or the fusion protein, followed by (2) two or more copies of the polynucleotide sequence encoding the protein of interest, with one ribosome binding site between (1) and (2) and between two copies of polynucleotide sequence encoding the protein of interest.

In some embodiments, (1) the polynucleotide sequence encoding the Cry protein, crystal-forming fragment thereof, or the fusion protein; and (2) the polynucleotide sequence encoding the protein of interest are operably linked to two separate promoters. In some embodiments, the two separate promoters are two different kinds of promoters, for example, cyt1Aa promoter and cry3Aa promoter. In some embodiments, (1) the polynucleotide sequence encoding the Cry protein, crystal-forming fragment thereof, or the fusion protein; and (2) the polynucleotide sequence encoding the protein of interest share one single termination codon, resulting in one copy of the Cry protein, crystal-forming fragment thereof, or the fusion protein and two copies of the protein of interest.

In some embodiments, the expression cassette encoding the protein of interest and the expression cassette encoding the Cry protein, crystal-forming fragment thereof, or the fusion protein are two separate expression cassettes. In some embodiments, the fusion protein comprises the Cry protein or crystal-forming fragment thereof and one or more heterologous polypeptides (such as a lipase lipA or SEQ ID NO:8 in 1-3 repeats) at the N- and/or C-terminus. In some embodiments, the fusion protein is Cry3Aa-[SmtA]₁₋₃. In some embodiments, two or more proteins of interest are co-expressed with the Cry protein, the crystal-forming fragment thereof, or the fusion protein and are contained within the crystal formed by the Cry protein, the crystal-forming fragment thereof, or the fusion protein. In some embodiments, the bacterial cells are *Bacillus subtilis* (Bs) or *Bacillus thuringiensis* (Bt) cell or *E. coli* cells. In some embodiments, the method of this invention further includes a step, prior to step (1), of introducing into the bacterial cells the expression cassette encoding the protein of interest and the expression cassette encoding the Cry protein, crystal-forming fragment thereof, or the fusion protein. In some embodiments, more than one protein of interest, e.g., two or more proteins, are recombinantly co-expressed with the Cry protein, crystal-

forming fragment thereof, or the fusion protein. These proteins of interest may be the same protein (e.g., both are PML) or different proteins (e.g., one lipase and one ligase). In some embodiments, the method of this invention further includes a step, after step (2), of isolating the crystal formed by the Cry protein, the crystal-forming fragment thereof, or the fusion protein and entrapping the protein or proteins of interest, which may be more than one protein, e.g., two or more proteins. In some embodiments, after it is isolated, the crystal containing the protein(s) of interest is washed under appropriate conditions such as choosing appropriate salt concentration etc. to permit the protein(s) entrapped within the crystal to be released from the crystal, preferably without dissolving the crystal to any substantial degree. In some embodiments, after being isolated, the crystal containing the protein(s) of interest is dissolved to release the protein(s) entrapped within the crystal. In some embodiments, the protein is an enzyme, such as PML. In some embodiments, the protein of interest is a fluorescent protein such as mCherry.

In a second aspect, the present invention provides a protein crystal produced by the method described above and herein for co-expression of one or more recombinant proteins of interest with a crystal-forming protein, such as a Cry protein, a crystal-forming fragment of the Cry protein, or a fusion protein capable of forming crystals comprising the Cry protein or the crystal-forming fragment thereof. In some embodiments, the protein of interest is an enzyme, such as a lipase (e.g., *Proteus mirabilis* lipase, or PML, including a PML variant with modifications of residues 118 and 130, for example, 1118V+E130G, and lipA or lipAR9), ligase, hydrolase, esterase, protease, or glycosidase. In some embodiments, the protein of interest is a fluorescent protein such as mCherry.

In a third aspect, the present invention provides a method for performing a reaction. The method comprises the step of incubating the protein crystal produced by the method of this invention entrapping an enzyme therein with a substrate to be catalyzed by the enzyme. In some embodiments, the enzyme is a lipase (e.g., *Proteus mirabilis* lipase, or PML, including a PML variant with modifications of residues 118 and 130, for example, 1118V+E130G, and lipA or lipAR9), ligase, hydrolase, esterase, protease, or glycosidase. In some embodiments, the method further comprises a step, after the reaction is completed, of removing the reaction product and cleaning, e.g., washing the crystal to remove any detectable amount of reaction mixture including reaction agents and/or product(s), preferably without dissolving the crystal to any substantial degree, and then reusing the protein crystal in a second reaction.

In a fourth aspect, the present invention provides an in vitro method for co-crystallizing (1) a Cry protein, a crystal-forming fragment of the Cry protein, or a fusion protein capable of forming crystals comprising the Cry protein or the crystal-forming fragment thereof with (2) one or more proteins (e.g., enzymes) by mixing a soluble protein described in (1) with the protein or proteins of (2), thus allowing the protein or protein(s) of (2) to be entrapped within a protein crystal formed by the crystal-forming protein of (1). In some embodiments, the protein is an enzyme. In some embodiments, the enzyme-entrapped protein crystal so formed is used for performing a reaction where the protein crystal is incubated with a substrate to the enzyme under conditions permissible for the substrate to be catalyzed by the enzyme. In some embodiments, the protein crystal containing entrapped protein(s) of interest formed by

the methods described above and herein is used for delivering the protein(s) to cells, such as macrophages, lymphocytes, cancer cells, red blood cells, epithelial cells, stem cells, and liver cells.

In a fifth aspect, the present invention provides various modified Cry proteins, their fragments or fusion proteins, all of which still retaining the crystalizing capability (for example, Cry3Aa*, Cry3Aa*-lipA, Cry3Aa*-SpyCat, Cry3Aa-[SmtA]₁₋₃, NegCry3Aa, 3A2-2, Cry3Aa (S145C, H161R), as well as various modified proteins with retained enzymatic activities (such as PML, PML^{VG}, LmSP, LipA, and LipAR9), a polynucleotide sequence encoding each of such proteins, a nucleic acid comprising the polynucleotide sequence encoding each of the proteins, especially an expression cassette comprising the polynucleotide coding sequence operably linked to a promoter (e.g., a heterologous promoter from an origin different from that of the wild-type base protein) or a vector comprising such an expression cassette, a host cell comprising such a vector or expression cassette, which is able to express the protein under permissible culture conditions. Also provided are methods for recombinantly producing any of these proteins by culturing the host cells under conditions permissible for the recombinant protein expression and optionally further concentrating or isolating/purifying the proteins.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1. (a) Electron micrograph of Bt. highlighting its crystal inclusion. Electron micrographs of purified (b) Cry3Aa crystals, and (c) Cry3Aa-GFP fusion crystals (reproduced from reference 15).

FIG. 2. Recyclability of Cry3Aa*-lipA for the conversion of coconut oil to biodiesel.¹⁶

FIG. 3. Schematic illustrating the potential mode of entrapment of PML within Cry3Aa crystals (modified from reference 16).

FIG. 4. Plasmid variations constructed for optimization of PML entrapment in Cry3Aa crystals. 3P-cry3Aa promoter; RBS-ribosome binding site.

FIG. 5. SDS-PAGE analysis of PML^{VG} entrapped in Cry3Aa crystals by different coexpression methods. (a) Analysis of Cry3Aa[PML^{VG}] and 3PPML^{VG}[Cyt13A] crystals and (b) Cry3Aa[PML^{VG}]² and 3P3A[3PPML^{VG}] crystals. Lane (M) molecular weight marker (kDa).

FIG. 6. Scanning electron micrographs of Cry3Aa and 3P3A3P[PML^{VG}] crystals.

FIG. 7. p-Nitrophenyl palmitate (pNPP) hydrolysis activity of PML^{VG} entrapped in Cry3Aa crystals by different coexpression methods.

FIG. 8A. Recyclability of optimized Cry3Aa[PML^{VG}] crystals during the conversion of waste cooking oil to biodiesel. Cycles were 10 h each. FIG. 8B. Thermal stability of free PML^{VG} and 3P3A[3PPML^{VG}] crystals. Samples were incubated in triplicate at various temperatures for 1 h and then tested for residual activity. Error bars show the standard deviation of the mean.

FIG. 9. GCMS analysis of benzyl laurate synthesis reaction. (a) Elution profile of no enzyme control shows no benzyl laurate peak. (b) Elution profile of lipase crystals reaction to produce benzyl laurate. (c) Mass-spec of benzyl laurate peak confirms product formation.

FIG. 10. In vitro co-crystallization of Cry3Aa and carboxy rhodamine labelled PML by vapor diffusion. Crystals with red fluorescence but no UV fluorescence are crystals comprised of PML only. Crystals with UV fluorescence but not red fluorescence are Cry3Aa crystals that do not contain

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PML. Entrapment of PML inside Cry3Aa crystals is confirmed by finding crystals with both red and UV fluorescence.

FIG. 11. Fluorescence microscopy of Cry3Aa co-entrapment of GFP and mCherry fluorescent proteins. The data demonstrate that GFP and mCherry proteins are entrapped in single Cry3Aa crystals.

FIG. 12. Production and analysis of LmSP entrapment in Cry3Aa crystals. (a) SDS-PAGE analysis of 3P3A3P [LmSP] and 3P3A3P[LmSP][PML^{VG}] crystals. (b) Sucrose phosphorylase activity and (c) lipase activity of coentrapped crystals.

FIG. 13. Production and characterization of Cry3Aa*[PML^{VG}]² crystals. (a) SDS-PAGE gel of purified Cry3Aa*[PML^{VG}]² crystals. Lane (1) molecular weight marker (kDa), Lane (2) crystals. (b) pNPP hydrolysis activity of Cry3Aa*[PML^{VG}]² crystals.

FIG. 14. Production and activity of the lipase:lacasse catalyst. (a) SDS-PAGE gel showing conjugation of CotA-Spytag to Cry3Aa*-SpyCat[PML^{VG}] crystals at different ratios. Lane (M) molecular weight marker (kDa). (b) 2,2'-azino-bis(3-ethylbenzothiazoline-6-sulphonic acid) (ABTS) oxidation by free CotA-Spytag and Cry3Aa*-SpyCat [PML^{VG}] crystals.

FIG. 15. Production and characterization of Cry3Aa*-lipA[PML^{VG}] crystals. (a) SDS-PAGE analysis of purified Cry3Aa*-lipA[PML^{VG}] crystals. Lane (1) Pure crystals; lane (M) molecular weight marker (kDa). (b) Time-point study of waste cooking oil conversion to biodiesel.

FIG. 16A. Scheme 1: Lipase catalyzed tranesterification of triacylglycerols with methanol (MeOH) to produce biodiesel. Scheme 2: Lipase catalyzed esterification of benzyl alcohol and lauric acid in neat isooctane to produce the cosmetic compound benzyl laurate.

FIG. 16B. Scheme 3: Reaction scheme for the one-pot synthesis of biodiesel and αGG using PML lipase and LmSP sucrose phosphorylase. Scheme 4: Reactions scheme for one-pot production of azelaic acid using PML lipase and CotA-lacasse.

FIG. 17. Amino acid sequence alignment between Cry3Aa (SEQ ID NO:4) and NegCry3Aa (SEQ ID NO:6).

FIG. 18. Purified negCry3Aa-RBS-scGFP protein crystals. (A) NegCry3Aa-RBS-scGFP crystal pellets purified from Bt cells grown for 48 h (left) and 72 h (right) were imaged using BioRad ChemiDoc MP System at the GFP excitation wavelength (ex=488 nm). (B) SDS-PAGE gel showing the corresponding purified NegCry3Aa-RBS-scGFP crystals in (A).

FIG. 19. Entrapment of lipAR9 in negCry3Aa crystals. (a) SDS-PAGE analysis of Cry3Aa[lipAR9] and negCry3Aa [lipAR9] crystals. (b) Quantitation of loading of lipAR9 in Cry3Aa and negCry3Aa crystals.

FIG. 20. SDS-PAGE of Cry3Aa-[SmtA]₁ produced in Bt and purified by sucrose density gradient centrifugation. Lane 1: BioRad Precision plus molecular weight marker; lane 2-3: Extracts from <40% sucrose layers; lane 4: crystal extracts from the interface of 40%/60% sucrose layer.

FIG. 21. Structure of Cry3Aa crystal channel. Residues highlighted in red are regions exposed to the solvent channel and can be deleted to expand the channel size.

FIG. 22 Cry3Aa protein sequence. Residues in bold are regions exposed to the solvent channel and can be deleted to expand the channel size. (SEQ ID NO: 4).

FIG. 23. Amino acids in specific regions exposed to the Cry3Aa (SEQ ID NO:4) crystal channel.

FIG. 24. Entrapment of LmSP in 3A2-2 crystal. (a) Model of LmSP homolog *Thermoanaerobacter thermosaccharo-*

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lyticum 6F-phosphate phosphorylase (PDB ID: 6S9V) inside the Cry3Aa crystal (PDB ID: 1DLC) channel. Loop regions that are oriented in the channel with potential steric clash with LmSP are colored red. LmSP is colored yellow with its amino acids colored in black and portrayed as lines. Cry3Aa is colored blue. Model was produced in COOT and images were prepared in PyMol. (b) SDS-PAGE analysis after entrapment of LmSP in the wild-type Cry3Aa crystal (3A [LmSP]) and in the mutant 3A2-2 crystal (3A2-2[LmSP]).

FIG. 25. Stabilizing mutations in Cry3Aa protein. (a) Structure of two Cry3Aa monomers showing His161-E168 electrostatic interaction. (b) Solubilization resistance of Cry3Aa mutants. (c) Structure of two Cry3Aa monomers showing potential site for disulfide formation. (d) Recyclability of PML^{VG} triple copy constructs 3A[PML^{VG}]³ and double mutant 3ADM[PML^{VG}]³ during the synthesis of biodiesel from waste cooking oil.

FIG. 26. Catch and release of lipAR9 by Cry3Aa crystals. (a) p-Nitrophenyl acetate hydrolysis activity of purified 3A[lipAR9] crystals. (b) Lipase activity of supernatant after release of lipAR9 protein. 3A[lipAR9] crystals were incubated with increasing concentrations of NaCl for 1 h and then removed from the mixture. The increase in lipase activity in the supernatant indicates that lipA was released as a function of NaCl concentration.

DEFINITIONS

The term “Cry protein,” as used herein, refers to any one protein among a class of crystalline proteins produced by strains of *Bacillus thuringiensis* (Bt). Some examples of “Cry proteins” include, but are not limited to, Cry1Aa, Cry1Ab, Cry2Aa, Cry3Aa, Cry4Aa, Cry4Ba, Cry11Aa, Cry11Ba, and Cry19Aa. Their amino acid sequences and polynucleotide coding sequences are known and can be found in publications such as U.S. Patent Application published as US2010/0322977. Their GenBank Accession Nos are:

Cry1Aa AY197341.1
Cry1Ab AY847289.1
Cry2Aa AF273218.1
Cry3Aa AJ237900.1
Cry4Aa AB513706.1
Cry4Ba AB161456.1
Cry11Aa AL731825.1
Cry11Ba LC153032.1
Cry19Aa Y07603.1

In addition to the wild-type Cry proteins, the term “Cry protein” also encompasses functional variants, which (1) share an amino acid sequence identity of at least 80%, 81%, 82%, 83%, 84%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 99.5% to the polypeptide sequence of any one of the Cry proteins listed in US2010/0322977; and (2) retain the ability to spontaneously form crystals within host cells as can be confirmed by known methods such as electron micrograph (see description in, e.g., Park et al., *Appl Environ Microbiol*, 1998, 64, 3932-3938; Schnepf et al., *Microbiol Mol Biol Rev*, 1998, 62, 775-806; Whiteley and Schnepf, *Annu Rev Microbiol*, 1986, 40, 549-576; and Nair et al., *PLoS One*, 2015, 10, e0127669). For example, a “Cry protein” encompasses any variant that confers increased negative charges to the resultant protein by substitutions of at least 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, or more positively charged or neutral amino acids within the domain II of the wild-type Cry protein (e.g., the 295 to 499 segment of SEQ ID NO:4, as well as its corresponding segment in other Cry proteins as

shown in FIG. 27) with negatively charged amino acids such as aspartate (D) and glutamate (E). One example of such mutant is NegCry3Aa, SEQ ID NO:6, which is a Cry3Aa variant containing the following mutations: K384E, N391D, N395D, S425E, Q430E, TQ436437EE, KR442443EE, T461D, and K467E. Further encompassed by the term “Cry protein” are modified Cry proteins containing amino acid insertions or deletions (for example, insertion of at least 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 amino acids, or a deletion of at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more amino acids) at one or more of the 8 specific regions (for example, at least 1, 2, 3, 4, 5, 6, 7, or all 8 regions) shown in FIG. 22, as well as their corresponding regions in other Cry proteins as shown in FIG. 18 of PCT/CN2020/084939, can effectively reduce or enlarge the protein’s channel size, respectively, and can therefore make the resultant crystal-forming protein variant to more effectively entrap a target protein (such as an enzyme) with increased retention rate and reduced diffusion rate for the target protein. Typically, such a channel-size altering Cry protein mutant contains at least 3, 4, or 5 but no more than 10, 12, or 15 point mutations (deletions or insertions) in the entire protein. One example of a Cry3Aa mutant of this nature is termed 3A2-2 (SEQ ID NO:7). In addition, the term “Cry protein” also encompasses Cry protein mutants that have been modified at one or more residues to improve the stability of the resultant protein or its crystal structure, for example, introducing one or more cysteine residues by insertion or substitution to the wild-type Cry protein amino acid sequence so as to form additional disulfide bond or bonds to stabilize the protein or crystal structure. One such example is a double mutant of Cry3Aa (3AS145C, H161R), having the amino acid sequence of SEQ ID NO:9.

Similarly, a “crystal-forming fragment” of a Cry protein is a fragment of any of the known Cry proteins (i.e., less than full length of the wild-type Cry protein) that still retains the ability of self-crystallization, which is demonstrated both by crystallization by the fragment alone and by causing a fusion protein to self-crystallize when the fragment is present in the fusion protein with another protein of interest (e.g., an enzyme). In addition to being a truncated form of a Cry protein, a “crystal-forming fragment” may further contain one or more modifications to the native amino acid sequence such as insertions, deletions, or substitutions, especially conservative modifications, such that the resultant “crystal-forming fragment” shares an amino acid sequence identity of at least 80%, 81%, 82%, 83%, 84%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 99.5% to the polypeptide sequence of the corresponding fragment of a wild-type Cry protein. Exemplary crystal-forming fragments of a Cry protein have been described in earlier disclosures, e.g., WO2018/028371.

The term “nucleic acid” or “polynucleotide” refers to deoxyribonucleic acids (DNA) or ribonucleic acids (RNA) and polymers thereof in either single- or double-stranded form. Unless specifically limited, the term encompasses nucleic acids containing known analogues of natural nucleotides that have similar binding properties as the reference nucleic acid and are metabolized in a manner similar to naturally occurring nucleotides. Unless otherwise indicated, a particular nucleic acid sequence also implicitly encompasses conservatively modified variants thereof (e.g., degenerate codon substitutions), alleles, orthologs, SNPs, and complementary sequences as well as the sequence explicitly indicated. Specifically, degenerate codon substitutions may be achieved by generating sequences in which the third position of one or more selected (or all) codons is substituted

with mixed-base and/or deoxyinosine residues (Batzner et al., *Nucleic Acid Res.* 19:5081 (1991); Ohtsuka et al., *J. Biol. Chem.* 260:2605-2608 (1985); and Rossolini et al., *Mol. Cell. Probes* 8:91-98 (1994)). The term nucleic acid is used interchangeably with gene, cDNA, and mRNA encoded by a gene.

The term “gene” means the segment of DNA involved in producing a polypeptide chain. It may include regions preceding and following the coding region (leader and trailer) as well as intervening sequences (introns) between individual coding segments (exons).

The term “amino acid” refers to naturally occurring and synthetic amino acids, as well as amino acid analogs and amino acid mimetics that function in a manner similar to the naturally occurring amino acids. Naturally occurring amino acids are those encoded by the genetic code, as well as those amino acids that are later modified, e.g., hydroxyproline, γ -carboxyglutamate, and O-phosphoserine. Amino acid analogs refers to compounds that have the same basic chemical structure as a naturally occurring amino acid, i.e., an α carbon that is bound to a hydrogen, a carboxyl group, an amino group, and an R group, e.g., homoserine, norleucine, methionine sulfoxide, methionine methyl sulfonium. Such analogs have modified R groups (e.g., norleucine) or modified peptide backbones, but retain the same basic chemical structure as a naturally occurring amino acid. “Amino acid mimetics” refers to chemical compounds having a structure that is different from the general chemical structure of an amino acid, but that functions in a manner similar to a naturally occurring amino acid.

There are various known methods in the art that permit the incorporation of an unnatural amino acid derivative or analog into a polypeptide chain in a site-specific manner, see, e.g., WO 02/086075.

Amino acids may be referred to herein by either the commonly known three letter symbols or by the one-letter symbols recommended by the IUPAC-IUB Biochemical Nomenclature Commission. Nucleotides, likewise, may be referred to by their commonly accepted single-letter codes.

“Conservatively modified variants” applies to both amino acid and nucleic acid sequences. With respect to particular nucleic acid sequences, “conservatively modified variants” refers to those nucleic acids that encode identical or essentially identical amino acid sequences, or where the nucleic acid does not encode an amino acid sequence, to essentially identical sequences. Because of the degeneracy of the genetic code, a large number of functionally identical nucleic acids encode any given protein. For instance, the codons GCA, GCC, GCG and GCU all encode the amino acid alanine. Thus, at every position where an alanine is specified by a codon, the codon can be altered to any of the corresponding codons described without altering the encoded polypeptide. Such nucleic acid variations are “silent variations,” which are one species of conservatively modified variations. Every nucleic acid sequence herein that encodes a polypeptide also describes every possible silent variation of the nucleic acid. One of skill will recognize that each codon in a nucleic acid (except AUG, which is ordinarily the only codon for methionine, and TGG, which is ordinarily the only codon for tryptophan) can be modified to yield a functionally identical molecule. Accordingly, each silent variation of a nucleic acid that encodes a polypeptide is implicit in each described sequence.

As to amino acid sequences, one of skill will recognize that individual substitutions, deletions or additions to a nucleic acid, peptide, polypeptide, or protein sequence which alters, adds or deletes a single amino acid or a small

percentage of amino acids in the encoded sequence is a “conservatively modified variant” where the alteration results in the substitution of an amino acid with a chemically similar amino acid. Conservative substitution tables providing functionally similar amino acids are well known in the art. Such conservatively modified variants are in addition to and do not exclude polymorphic variants, interspecies homologs, and alleles of the invention.

The following eight groups each contain amino acids that are conservative substitutions for one another:

- 1) Alanine (A), Glycine (G);
 - 2) Aspartic acid (D), Glutamic acid (E);
 - 3) Asparagine (N), Glutamine (Q);
 - 4) Arginine (R), Lysine (K);
 - 5) Isoleucine (I), Leucine (L), Methionine (M), Valine (V);
 - 6) Phenylalanine (F), Tyrosine (Y), Tryptophan (W);
 - 7) Serine (S), Threonine (T); and
 - 8) Cysteine (C), Methionine (M)
- (see, e.g., Creighton, *Proteins*, W. H. Freeman and Co., N. Y. (1984)).

Amino acids may be referred to herein by either their commonly known three letter symbols or by the one-letter symbols recommended by the IUPAC-IUB Biochemical Nomenclature Commission. Nucleotides, likewise, may be referred to by their commonly accepted single-letter codes.

In the present application, amino acid residues are numbered according to their relative positions from the left most residue, which is numbered 1, in an unmodified wild-type polypeptide sequence.

As used in herein, the terms “identical” or percent “identity,” in the context of describing two or more polynucleotide or amino acid sequences, refer to two or more sequences or subsequences that are the same or have a specified percentage of amino acid residues or nucleotides that are the same (for example, a Cry protein or a crystal-forming fragment of a Cry protein sequence comprised in the fusion protein of this invention has at least 80% identity, preferably 85%, 90%, 91%, 92%, 93, 94%, 95%, 96%, 97%, 98%, 99%, or 100% identity, to a reference sequence, e.g., the amino acid sequence of a corresponding wild-type Cry protein or fragment), when compared and aligned for maximum correspondence over a comparison window, or designated region as measured using one of the following sequence comparison algorithms or by manual alignment and visual inspection. Such sequences are then said to be “substantially identical.” With regard to polynucleotide sequences, this definition also refers to the complement of a test sequence. Preferably, the identity exists over a region that is at least about 50 amino acids or nucleotides in length, or more preferably over a region that is 75-100 amino acids or nucleotides in length.

For sequence comparison, typically one sequence acts as a reference sequence, to which test sequences are compared. When using a sequence comparison algorithm, test and reference sequences are entered into a computer, subsequence coordinates are designated, if necessary, and sequence algorithm program parameters are designated. Default program parameters can be used, or alternative parameters can be designated. The sequence comparison algorithm then calculates the percent sequence identities for the test sequences relative to the reference sequence, based on the program parameters. For sequence comparison of nucleic acids and proteins, the BLAST and BLAST 2.0 algorithms and the default parameters discussed below are used.

A “comparison window”, as used herein, includes reference to a segment of any one of the number of contiguous positions selected from the group consisting of from 20 to 600, usually about 50 to about 200, more usually about 100 to about 150 in which a sequence may be compared to a reference sequence of the same number of contiguous positions after the two sequences are optimally aligned. Methods of alignment of sequences for comparison are well-known in the art. Optimal alignment of sequences for comparison can be conducted, e.g., by the local homology algorithm of Smith & Waterman, *Adv. Appl. Math.* 2:482 (1981), by the homology alignment algorithm of Needleman & Wunsch, *J. Mol. Biol.* 48:443 (1970), by the search for similarity method of Pearson & Lipman, *Proc. Nat'l. Acad. Sci. USA* 85:2444 (1988), by computerized implementations of these algorithms (GAP, BESTFIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group, 575 Science Dr., Madison, WI), or by manual alignment and visual inspection, see, e.g., *Current Protocols in Molecular Biology* (Ausubel et al., eds. 1995 supplement).

Examples of algorithms that are suitable for determining percent sequence identity and sequence similarity are the BLAST and BLAST 2.0 algorithms, which are described in Altschul et al., (1990) *J. Mol. Biol.* 215: 403-410 and Altschul et al. (1977) *Nucleic Acids Res.* 25: 3389-3402, respectively. Software for performing BLAST analyses is publicly available at the National Center for Biotechnology Information website, ncbi.nlm.nih.gov. The algorithm involves first identifying high scoring sequence pairs (HSPs) by identifying short words of length W in the query sequence, which either match or satisfy some positive-valued threshold score T when aligned with a word of the same length in a database sequence. T is referred to as the neighborhood word score threshold (Altschul et al., supra). These initial neighborhood word hits acts as seeds for initiating searches to find longer HSPs containing them. The word hits are then extended in both directions along each sequence for as far as the cumulative alignment score can be increased. Cumulative scores are calculated using, for nucleotide sequences, the parameters M (reward score for a pair of matching residues; always >0) and N (penalty score for mismatching residues; always <0). For amino acid sequences, a scoring matrix is used to calculate the cumulative score. Extension of the word hits in each direction are halted when: the cumulative alignment score falls off by the quantity X from its maximum achieved value; the cumulative score goes to zero or below, due to the accumulation of one or more negative-scoring residue alignments; or the end of either sequence is reached. The BLAST algorithm parameters W, T, and X determine the sensitivity and speed of the alignment. The BLASTN program (for nucleotide sequences) uses as defaults a word size (W) of 28, an expectation (E) of 10, M=1, N=-2, and a comparison of both strands. For amino acid sequences, the BLASTP program uses as defaults a word size (W) of 3, an expectation (E) of 10, and the BLOSUM62 scoring matrix (see Henikoff & Henikoff, *Proc. Natl. Acad. Sci. USA* 89:10915 (1989)).

The BLAST algorithm also performs a statistical analysis of the similarity between two sequences (see, e.g., Karlin & Altschul, *Proc. Nat'l. Acad. Sci. USA* 90:5873-5877 (1993)). One measure of similarity provided by the BLAST algorithm is the smallest sum probability (P(N)), which provides an indication of the probability by which a match between two nucleotide or amino acid sequences would occur by chance. For example, a nucleic acid is considered similar to a reference sequence if the smallest sum probability in a

comparison of the test nucleic acid to the reference nucleic acid is less than about 0.2, more preferably less than about 0.01, and most preferably less than about 0.001.

An indication that two nucleic acid sequences or polypeptides are substantially identical is that the polypeptide encoded by the first nucleic acid is immunologically cross reactive with the antibodies raised against the polypeptide encoded by the second nucleic acid, as described below. Thus, a polypeptide is typically substantially identical to a second polypeptide, for example, where the two peptides differ only by conservative substitutions. Another indication that two nucleic acid sequences are substantially identical is that the two molecules or their complements hybridize to each other under stringent conditions, as described below. Yet another indication that two nucleic acid sequences are substantially identical is that the same primers can be used to amplify the sequence.

"Polypeptide," "peptide," and "protein" are used interchangeably herein to refer to a polymer of amino acid residues. All three terms apply to amino acid polymers in which one or more amino acid residue is an artificial chemical mimetic of a corresponding naturally occurring amino acid, as well as to naturally occurring amino acid polymers and non-naturally occurring amino acid polymers. As used herein, the terms encompass amino acid chains of any length, including full-length proteins, wherein the amino acid residues are linked by covalent peptide bonds.

The term "recombinant" when used with reference, e.g., to a cell, or a nucleic acid, protein, or vector, indicates that the cell, nucleic acid, protein or vector, has been modified by the introduction of a heterologous nucleic acid or protein or the alteration of a native nucleic acid or protein, or that the cell is derived from a cell so modified. Thus, for example, recombinant cells express genes that are not found within the native (non-recombinant) form of the cell or express native genes that are otherwise abnormally expressed, under expressed or not expressed at all.

An "expression cassette" is a nucleic acid construct, generated recombinantly or synthetically, with a series of specified nucleic acid elements that permit transcription of a particular polynucleotide sequence in a host cell. An expression cassette may be part of a plasmid, viral vector derived from a viral genome, or nucleic acid fragment/construct. Typically, an expression cassette includes a polynucleotide to be transcribed, operably linked to a promoter. Other elements that may be present in an expression cassette include those that enhance transcription (e.g., enhancers) and terminate transcription (e.g., terminators), as well as those that confer certain binding affinity or antigenicity to the recombinant protein produced from the expression cassette.

A "promoter" is defined as an array of nucleic acid control sequences that direct transcription of a polynucleotide sequence. As used herein, a promoter includes necessary polynucleotide sequences near the start site of transcription, such as, in the case of a polymerase II type promoter, a TATA element. A promoter also optionally includes distal enhancer or repressor elements, which can be located as much as several thousand base pairs from the start site of transcription. A "constitutive" promoter is a promoter that is active under most environmental and developmental conditions. An "inducible" promoter is a promoter that is active under environmental or developmental regulation. The term "operably linked" refers to a functional linkage between a polynucleotide expression control sequence (such as a promoter, or array of transcription factor binding sites) and a second polynucleotide sequence, wherein the expression control

sequence directs transcription of the polynucleotide sequence corresponding to the second sequence.

The term "heterologous" as used in the context of describing the relative location of two elements, refers to the two elements such as polynucleotide sequences (e.g., a promoter or a protein/polypeptide-encoding sequence) or polypeptide sequences (e.g., a Cry protein sequence or another polypeptide sequence) that are not naturally found in the same relative positions. Thus, a "heterologous promoter" of a gene refers to a promoter that is not naturally operably linked to that gene. Similarly, a "heterologous polypeptide" or "heterologous polynucleotide" to a Cry protein or its encoding sequence or a fragment thereof is one derived from an origin other than the Cry protein or, in the case of a fragment of a Cry protein/coding sequence, may be derived from another part of the same Cry protein or coding sequence, but not naturally connected to the fragment in the same fashion. The fusion of a fragment of a Cry protein (or its coding sequence) with a heterologous polypeptide (or polynucleotide sequence) does not result in a longer polypeptide or polynucleotide sequence that can be found naturally in the wild-type Cry protein.

By "host cell" is meant a cell that contains an expression vector and supports the replication or expression of one or more coding sequences harbored in the expression vector. Host cells may be prokaryotic cells such as *Bacillus thuringiensis* (Bt), *Bacillus subtilis* (Bs), or *E. coli*, or eukaryotic cells such as yeast, insect, amphibian, or mammalian cells such as CHO, HeLa and the like, e.g., cultured cells, explants, and cells in vivo.

The term "about" as used herein denotes a range of $\pm 10\%$ of a reference value. For examples, "about 10" defines a range of 9 to 11.

DETAILED DESCRIPTION OF THE INVENTION

I. Introduction

There has been growing interest in using enzymes to catalyze industrial reactions due to their high reactivity, excellent regio- and enantiospecificity, and low environmental toxicity. In order to financially compete with chemical catalysis, biocatalysts are optimized so they can be recycled multiple times. Additionally, biocatalysts are generally optimized so they can withstand high concentrations of organic solvents—conditions that can promote substrate solubility and enzyme activity. Enzyme immobilization can facilitate easy removal and subsequent reuse of enzymes during multiple rounds of catalysis, in addition to improving enzyme stability resistant to harsh industrial conditions. Immobilization by entrapment is particularly attractive since it does not involve any modifications to the enzyme structure, increasing the chance for high activity retention and native enzyme conformation. In this disclosure, a novel one-step method is described for producing entrapped recombinant proteins (especially enzymes, including multi-enzymes) in protein crystals.

To date, all entrapment methods involve first producing the enzyme and carrier separately, and then mixing them to generate the immobilized enzyme. This multi-step process requires purifying and concentrating the enzyme, synthesizing the carrier, and then mixing them, usually including the use of a catalyst to initiate the polymerization of the carrier. This procedure is tedious and expensive. A method that generates an entrapped enzyme or multi-enzyme system in one-step process can significantly reduce production costs

and lead to cheaper commercial catalysts. The method of this invention leads to the entrapped enzyme in one step, avoiding the need for purification of the free enzyme or polymerization of the carrier. Minimizing time, processes, and materials for producing enzyme catalysts allows for greener and more cost-effective products.

II. Production of Recombinant Proteins

A. General Recombinant Technology

Basic texts disclosing general methods and techniques in the field of recombinant genetics include Sambrook and Russell, *Molecular Cloning, A Laboratory Manual* (3rd ed. 2001); Kriegler, *Gene Transfer and Expression: A Laboratory Manual* (1990); and Ausubel et al., eds., *Current Protocols in Molecular Biology* (1994).

For nucleic acids, sizes are given in either kilobases (kb) or base pairs (bp). These are estimates derived from agarose or acrylamide gel electrophoresis, from sequenced nucleic acids, or from published DNA sequences. For proteins, sizes are given in kilodaltons (kDa) or amino acid residue numbers. Proteins sizes are estimated from gel electrophoresis, from sequenced proteins, from derived amino acid sequences, or from published protein sequences.

Oligonucleotides that are not commercially available can be chemically synthesized, e.g., according to the solid phase phosphoramidite triester method first described by Beaucage & Caruthers, *Tetrahedron Lett.* 22: 1859-1862 (1981), using an automated synthesizer, as described in Van Devanter et. al., *Nucleic Acids Res.* 12: 6159-6168 (1984). Purification of oligonucleotides is performed using any art-recognized strategy, e.g., native acrylamide gel electrophoresis or anion-exchange HPLC as described in Pearson & Reanier, *J Chrom.* 255: 137-149 (1983).

The sequence of a gene of interest, such as the polynucleotide sequence encoding an enzyme like lipase or hydrolase, a polynucleotide encoding a Cry protein or a crystal-forming fragment thereof, and synthetic oligonucleotides can be verified after cloning or subcloning using, e.g., the chain termination method for sequencing double-stranded templates of Wallace et al., *Gene* 16: 21-26 (1981).

B. Coding Sequences

Polynucleotide sequences encoding Cry proteins, fragments, or fusion proteins for use in this invention can be readily constructed by using the corresponding coding sequences for the Cry proteins, fragments, or combining the coding sequences for the fusion partners, such as a Cry3Aa protein and *Bacillus subtilis* lipase A (lipA). The sequences for Cry proteins and enzymes are generally known and may be obtained from a commercial supplier.

In addition to the use of full length wild-type Cry proteins for producing crystal-forming proteins for use in this invention, fragments of Cry proteins and/or variants of Cry proteins may also be useful. A DNA sequence encoding a Cry protein can be modified to generate fragments or variants of the Cry protein. So long as the fragments and variants retain the ability to spontaneously form crystals when expressed in a host cell, especially a *Bacillus* bacterial cell, they can be used for producing the protein crystals, either by themselves or by way of fusion proteins capable of undergoing spontaneous crystallization, and therefore producing protein crystals containing one or more recombinant proteins (e.g., one or more enzymes) embedded within. Typically, the variants bear a high percentage of sequence identity (e.g., at least 80, 85, 90, 95, 97, 98, 99% or higher) to the wild-type Cry protein sequence, whereas the fragments may be substantially shorter than the full length Cry

protein, such as having some amino acids (e.g., 10-300 or 20-200 or 50-100 amino acids) removed from the N- or C-terminus of the full length Cry protein. For example, a useful Cry3Aa fragment may be as short as the first 290 amino acids from the N-terminus, encompassing Domain I of the protein. Other examples of such fragments include a Cry protein fragment having its first 57 amino acids from N-terminus removed and a Cry protein fragment having its C-terminal 18 amino acids removed. The ability of a recombinantly produced Cry protein, a fragment thereof, or a fusion protein comprising a Cry protein or fragment to undergo spontaneous crystallization can be verified by electron micrograph, whereas the enzymatic activity of a recombinantly produced enzyme, including in the form of a fusion protein with a Cry protein or a fragment thereof, can be confirmed by established assays for each specific enzyme. Exemplary Cry protein fragments capable of self-crystallizing can be found in the inventors' earlier publications, e.g., WO2018/028371.

In the case of a fusion protein, a peptide linker or spacer is used between the coding sequences for a Cry protein/fragment and one or more heterologous polypeptides. One purpose is to ensure the proper reading frame for the fusion protein such that the coding sequences for both Cry protein/fragment and the heterologous polypeptide(s) are in frame. Another purpose is to provide appropriate spatial relationship between the Cry protein/fragment and the heterologous polypeptide(s), such that each component of the fusion protein may retain its original functionality: the Cry protein/fragment is able to cause self-crystallization of the fusion protein, and the heterologous polypeptide such as an enzyme remains active in its catalytic capacity. Also, one or more linkers may be placed at the very beginning and/or the very end of the open reading frame, so as to facilitate proper start and termination of the coding sequence translation. Such linkage amino acid sequences are usually short and typically no longer than 100 or 50 amino acids, such as between 1 to 100, 1 or 2 to 50, 2 or 3 to 25, 3 or 4 to 10 amino acids.

C. Sequence Modification for Preferred Codon Usage in a Host Organism

The polynucleotide sequence encoding a recombinant protein to be expressed according to the method of this invention can be further altered to coincide with the preferred codon usage of a particular host. For example, the preferred codon usage of one strain of bacterial cells can be used to derive a polynucleotide that encodes a recombinant polypeptide of the invention and includes the codons favored by this strain. The frequency of preferred codon usage exhibited by a host cell can be calculated by averaging frequency of preferred codon usage in a large number of genes expressed by the host cell (e.g., calculation service is available from web site of the Kazusa DNA Research Institute, Japan). This analysis is preferably limited to genes that are highly expressed by the host cell.

At the completion of modification, the coding sequences are verified by sequencing and are then subcloned into an appropriate expression vector for recombinant production of a protein (e.g., an enzyme such as a lipase) along with a Cry protein, a crystal-forming fragment of the Cry protein, or a fusion protein comprising the Cry protein or crystal-forming fragment, such that the protein (e.g., enzyme) is produced and embedded in the protein crystals formed by the Cry protein, fragment, or fusion protein.

III. Expression and Isolation of Proteins Embedded in Protein Crystals

Following verification of the coding sequence, co-expression of a protein of interest (such as an enzyme) and a Cry

protein, a crystal-forming fragment thereof, or a fusion protein comprising the Cry protein or fragment of this invention can be produced using routine techniques in the field of recombinant genetics, relying on the polynucleotide sequences encoding the Cry fusion protein disclosed herein.

A. Expression Systems

To obtain high level expression of a polynucleotide sequence encoding a recombinant protein, one typically subclones a polynucleotide encoding the protein in the correct reading frame into an expression vector that contains a strong promoter to direct transcription, a transcription/translation terminator and a ribosome binding site for translational initiation. Suitable bacterial promoters are well known in the art and described, e.g., in Sambrook and Russell, supra, and Ausubel et al., supra. Bacterial expression systems for expressing the polypeptide are available in, e.g., *E. coli*, *Bacillus* sp., *Salmonella*, and *Caulobacter*. Kits for such expression systems are commercially available.

The promoter used to direct expression of a recombinant protein depends on the particular host cells used for the recombinant protein production. For instance, for effective expression in a *Bacillus* bacterial strain such as *Bacillus thuringiensis* (Bt) or *Bacillus subtilis* (Bs) cells, a promoter known to direct robust protein expression in these particular bacterial cells should be chosen. As shown in the Examples of this disclosure, two separate promoters, *cyt1Aa* and *cry3Aa*, have been successfully used to direct the expression of a recombinant protein as well as a Cry protein. The promoter is optionally positioned about the same distance from an exogenous or recombinant protein transcription start site as it is from the transcription start site in its natural setting. As is known in the art, however, some variation in this distance can be accommodated without loss of promoter function. In some cases, a constitutive promoter is used, whereas in other cases an inducible promoter rather than a constitutive promoter is preferred. Further, the placement of a common promoter or multiple separate promoters (which may be the same or different kind of promoters) and/or the placement of one or more separate ribosome binding sites separating different segments of coding sequences (each encoding a recombinant protein or a crystal-forming protein) in a common expression cassette (e.g., an expression vector such as a plasmid) can allow different expression ratios of the recombinant protein(s) to the crystal-forming protein so as to maximize the percentage of the recombinant protein(s) being entrapped or immobilized within the protein crystals formed by the crystal-forming protein such as a Cry protein, a crystal-forming fragment thereof, or a fusion protein comprising the Cry protein or the fragment.

In addition to the promoter, the expression vector typically includes a transcription unit or expression cassette containing all the additional elements that are required for the expression of the recombinant protein(s) and the crystal-forming protein in host cells. A typical expression cassette thus contains a promoter operably linked to the polynucleotide sequence encoding the recombinant protein and/or crystal-forming protein and signals required for efficient polyadenylation of the transcript, ribosome binding sites, and translation termination. The polynucleotide coding sequence may be linked to a cleavable signal peptide sequence to promote secretion of the polypeptide by the transformed cell. Such signal peptides include, among others, the signal peptides from tissue plasminogen activator, insulin, and neuron growth factor, and juvenile hormone esterase of *Heliothis virescens*. Additional elements of the

cassette may include enhancers and, if genomic DNA is used as the structural gene, introns with functional splice donor and acceptor sites.

In addition to a promoter sequence, the expression cassette should also contain a transcription termination region downstream of the coding sequence to provide for efficient termination. The termination region may be obtained from the same gene as the promoter sequence or may be obtained from different genes. The placement of a commonly shared termination site in combination with the placement of multiple promoters directing transcription of multiple coding sequences can also be used to adjust the expression ratios of a recombinant protein to a crystal-forming protein, see, e.g., illustration in FIG. 4.

The particular expression vector used to transport the genetic information into the cell is not particularly critical. Any of the conventional vectors used for expression in eukaryotic or prokaryotic cells may be used, especially those suitable for expression in cells of *Bacillus* sp. such as Bt and Bs. Standard bacterial expression vectors include plasmids such as pBR322 based plasmids, pSKF, pET23D, and fusion expression systems such as GST and LacZ.

The elements that are typically included in expression vectors also include a replicon that functions in bacteria such as *Bacillus* sp. and *E. coli*, a gene encoding antibiotic resistance to permit selection of bacteria that harbor recombinant plasmids, and unique restriction sites in nonessential regions of the plasmid to allow insertion of coding sequences. The particular antibiotic resistance gene chosen is not critical, any of the many resistance genes known in the art are suitable. Similar to antibiotic resistance selection markers, metabolic selection markers based on known metabolic pathways may also be used as a means for selecting transformed host cells.

B. Transfection Methods

Standard transfection methods are used to produce bacterial, mammalian, yeast, insect, or plant cell lines that express large quantities of a recombinant fusion protein of this invention, which are then purified using standard techniques (see, e.g., Colley et al., *J. Biol. Chem.* 264: 17619-17622 (1989); *Guide to Protein Purification*, in *Methods in Enzymology*, vol. 182 (Deutscher, ed., 1990)). Transformation of eukaryotic and prokaryotic cells are performed according to standard techniques (see, e.g., Morrison, *J. Bact.* 132: 349-351 (1977); Clark-Curtiss & Curtiss, *Methods in Enzymology* 101: 347-362 (Wu et al., eds, 1983)).

Any of the well-known procedures for introducing foreign polynucleotide sequences into host cells may be used. These include the use of calcium phosphate transfection, polybrene, protoplast fusion, electroporation, liposomes, microinjection, plasma vectors, viral vectors and any of the other well-known methods for introducing cloned genomic DNA, cDNA, synthetic DNA, or other foreign genetic material into a host cell (see, e.g., Sambrook and Russell, supra). It is only necessary that the particular genetic engineering procedure used be capable of successfully introducing at least one gene into the host cell capable of expressing the recombinant protein(s) along with a crystal-forming protein (e.g., a Cry protein, a crystal-forming fragment thereof, or a fusion protein comprising a Cry protein or a crystal-forming fragment thereof) according to the method of this invention. As described above and herein, the recombinant protein or proteins (e.g., enzyme or enzymes) and the crystal-forming protein may be contained within one single expression cassette, e.g., in the same expression vector, where all coding sequences may be under the control of one single promoter, or each coding sequence may be under the control

of a separate promoter (which optionally may differ from one another). In the alternative, each one of the coding sequences may be present in a separate expression cassette (e.g., expression vector). In either alternative, different ratios of the recombinant protein(s) to the crystal-forming protein may be achieved by using single copy or multiple copies of any one coding sequence or by placing separate or commonly shared ribosome binding site(s) and/or termination site(s) in the expression cassette, see, e.g., illustration in FIG. 4.

C. Isolation of Crystal-Entrapped Recombinant Proteins

Once the expression of the recombinant protein(s) along with a crystal-forming protein (e.g., a Cry protein, a crystal-forming fragment thereof, or a fusion protein comprising a Cry protein or a crystal-forming fragment thereof) in transfected host cells is confirmed, e.g., via electron micrograph for detecting protein crystals or an immunoassay such as Western blotting analysis, the host cells are then cultured in an appropriate scale for the purpose of purifying or isolating the recombinant protein entrapped within the protein crystals formed by the Cry protein, a crystal-forming fragment thereof, or a fusion protein comprising a Cry protein or a crystal-forming fragment thereof.

When the recombinant protein(s) and the crystal-forming protein are produced recombinantly by transformed bacteria in large amounts, for example after promoter induction, the recombinant protein(s) become entrapped within the crystals formed by the crystal-forming protein. In other words, the recombinantly produced proteins are present in crystalline form or insoluble aggregates within the host cells. Thus, one can readily isolate the crystals from the cell lysate based on their distinct density by utilizing techniques such as centrifugation and density gradient separation followed by one or more rinsing steps to further remove contaminants from the protein crystals.

There are several protocols that are suitable for purification of protein inclusion bodies. For example, purification of aggregate proteins (hereinafter referred to as inclusion bodies) typically involves the extraction, separation and/or purification of inclusion bodies by disruption of bacterial cells, e.g., by incubation in a buffer of about 100-150 µg/ml lysozyme and 0.10% Nonidet P40, a non-ionic detergent. The cell suspension can be ground using a Polytron grinder (Brinkman Instruments, Westbury, NY). Alternatively, the cells can be sonicated on ice. Additional methods of lysing bacteria are described in Ausubel et al. and Sambrook and Russell, both supra, and will be apparent to those of skill in the art.

The cell suspension is generally centrifuged and the pellet containing the inclusion bodies resuspended in buffer which does not dissolve but washes the inclusion bodies, e.g., 20 mM Tris-HCl (pH 7.2), 1 mM EDTA, 150 mM NaCl and 2% Triton-X 100, a non-ionic detergent. It may be necessary to repeat the wash step to remove as much cellular debris as possible. The remaining pellet of inclusion bodies may be resuspended in an appropriate buffer (e.g., 20 mM sodium phosphate, pH 6.8, 150 mM NaCl). Other appropriate buffers will be apparent to those of skill in the art.

Upon isolation, the recombinant protein(s) recovered from host cells in the form of protein crystals, the protein or proteins may be directly used according to their inherent biological activity: for example, a lipase entrapped in Cry protein crystals may be used in a reaction to hydrolyze triglycerides. By virtue of being in an insoluble crystal form, the lipase has a heightened level of resistance to harsh envi-

ronmental conditions such as high temperature, extreme pHs, organic solvents, etc., thus allowing repeated cycles of cleaning and reuse.

In the alternative, following the washing step, the inclusion bodies are solubilized to release the entrapped recombinant protein(s) by the addition of a solvent that is both a strong hydrogen acceptor and a strong hydrogen donor (or a combination of solvents each having one of these properties). The protein(s) from the inclusion bodies may then be renatured by dilution or dialysis with a compatible buffer. Suitable solvents include, but are not limited to, urea (from about 4 M to about 8 M), formamide (at least about 80%, volume/volume basis), and guanidine hydrochloride (from about 4 M to about 8 M). Some solvents that are capable of solubilizing aggregate-forming proteins, such as SDS (sodium dodecyl sulfate) and 70% formic acid, may be inappropriate for use in this procedure due to the possibility of irreversible denaturation of the proteins, accompanied by a lack of immunogenicity and/or activity. Although guanidine hydrochloride and similar agents are denaturants, this denaturation is not irreversible and renaturation may occur upon removal (by dialysis, for example) or dilution of the denaturant, allowing re-formation of the immunologically and/or biologically active protein of interest. After solubilization, the protein(s) can be separated from other bacterial proteins by standard separation techniques. For further description of purifying recombinant polypeptides from bacterial inclusion body, see, e.g., Patra et al., *Protein Expression and Purification* 18: 182-190 (2000).

While the protein crystals tend to remain insoluble at lower or neutral pHs, placing them in alkaline solutions with pH at or greater than 10 or 11 can often effectively dissolve the protein. Once dissolved, the protein can then be analyzed by gel separation (e.g., on an SDS gel) and immunoassays to confirm its identity based on the appropriate molecular weight and immunoreactivity.

IV. Applications of Immobilized Proteins

Another aspect of the present invention relates to the use of a recombinant protein, especially an enzyme, entrapped and immobilized in protein crystals produced according to the methods described herein to exert the protein's inherent biological activity, for example, to perform reactions typically catalyzed by the enzyme present in the protein crystals, such as hydrolysis, esterification, ligation, proteolysis, and the like. As organic solvents are often able to facilitate such reactions and the immobilized recombinant protein produced by the method of this invention is highly tolerant to the presence of organic solvents, a reaction performed using the immobilized protein according to this invention often not only a water-based solvent but also one or more organic solvents, e.g., ethanol, methanol, acetonitrile, and dimethylformamide.

As the inventors discovered that immobilization of recombinant protein(s) within the crystalline Cry protein or fusion proteins leads the protein(s) to have a higher level of resistance to organic solvents and a higher level of thermostability, potentially can retain enzymatic activity for use in more cycles of reactions. In some cases, this reaction process includes a cleaning step, performed after the completion of one round of the reaction and removal of the reaction product(s) as well as any remaining substrate, during which the protein crystals containing recombinant

protein(s) are rinsed or washed in preparation of being used again with fresh substrate in a subsequent round of reaction.

EXAMPLES

The following examples are provided by way of illustration only and not by way of limitation. Those of skill in the art will readily recognize a variety of non-critical parameters that could be changed or modified to yield essentially the same or similar results.

Example 1

Introduction

Immobilization of Enzymes

The major bottleneck to the practical use of enzymes as industrial catalysts is the cost of production and useable lifetime. While biomolecular engineering methods can be used to enhance enzyme stability, these techniques do not aid in rapid production and isolation, particularly when high purities are necessary. A cost-effective biocatalyst should be able to be recycled over many reaction cycles, with minimal loss of activity. One method to improve enzyme stability and make them reusable is immobilization, either by covalent attachment to beads, or by adsorption into porous materials. Immobilization of enzymes allows them to be easily filtered and reused over successive reaction cycles, and in some cases can lead to increased temperature stability and/or organic solvent tolerance due to the reduced conformational flexibility of the enzyme in the matrix¹. This rigidity is also beneficial to mechanical stability by helping to protect against the constant agitation that occurs in large-scale reactors.²

A major limitation of both bead attachment and porous material adsorption approaches, however, is that ~90-99% of the biocatalyst composite is inactive, greatly reducing the catalytic productivity per weight.³ Furthermore, these approaches require multiple steps: (1) production and purification of the enzyme catalyst, (2) production of the support, and (3) anchoring of the enzyme catalyst on the support. These steps can add significantly to the cost and impact catalytic activity. If a strategy can be developed to generate and isolate an immobilized lipase in a single step, it can dramatically lower the cost of the catalyst.

Genetically-Encoded Immobilized Lipases

Genetically-encoded, or in-vivo, immobilization approaches that involve the direct production of immobilized enzymes in bacterial cells, represents a promising direction for more efficient and economical biocatalyst production. Consolidating expression and immobilization into a single step removes the need of columns for enzyme purification, as well as reagents to introduce reactive chemistries, greatly reducing time and production costs.

One of the earliest reports of producing active enzyme particles in cells came from the work of Worrall et al., who showed that they could produce catalytically active inclusion bodies (CatIBs) of β -galactosidase in *E. coli*.⁴ This discovery upended the paradigm that inclusion bodies (IBs) were inactive waste species in cells and prompted the idea that IBs could be used as immobilized catalysts. Since most enzymes are naturally soluble, producing CatIBs can be a challenge. Thus most approaches to produce in vivo immobilized enzymes have involved the incorporation of a fusion-tag to promote aggregation.

Several fusion tags have been exploited to generate immobilized enzymes in vivo including self-aggregating

peptides and protein domains. Peptide tags such as ELK16 and L₆KD have been successfully fused to target enzymes to drive in vivo aggregation. While the use of tags resulted in the formation of highly active enzyme aggregates, none of these aggregates was shown to be reusable.^{5,6} Similarly, several protein domains have been used to promote in vivo enzyme immobilization in cells.⁷⁻⁹ The most successful example has been the work of Diener et al. who used the short coiled-coil domain TDoT of the cell-surface protein tetrabrachion from the hyperthermophilic archaeon *Staphylothermus marinus* to immobilize several enzymes.^{10,11} While enzymes immobilized by the TDoT tag displayed good recyclability, the aggregates produced generally had low activity due to diffusion limitations. Identifying a platform capable of promoting the production of in vivo immobilized enzymes with both high activity and good reusability had until recently, remained elusive.

Cry-Enzyme Fusion Proteins as Genetically-Encoded Immobilized Catalysts

Recently a new strategy was developed to produce genetically-encoded immobilized enzymes based on crystal proteins (Cry), such as Cry3Aa. Cry proteins are insecticidal proteins that innately form crystals inside the bacterium *Bacillus thuringiensis* (Bt) (FIG. 3).¹²⁻¹⁴ It was previously demonstrated that expression of Cry3Aa fused to GFP or mCherry yielded fluorescent protein crystals in Bt—indicating that the fusion partners were folded and functional.¹⁵ Electron microscopy carried out on Cry3Aa crystals and Cry3Aa-GFP crystals revealed them to be similar in size, indicating that the fusion partner does not change the crystal packing (FIG. 3b,c).¹⁵

The use of this Cry crystal platform to produce genetically-immobilized lipases as potential biodiesel catalysts was explored, see, e.g., WO2018/028371. The first-generation biodiesel catalyst was generated by genetically fusing *Bacillus subtilis* lipase A (lipA) to the C-terminus of a truncated Cry3Aa (Cry3Aa*) and then producing the fusion protein in Bt. The resultant Cry3Aa*-lipA fusion crystals exhibited higher catalytic activities compared to free lipA, as well as improved solvent and temperature stability.¹⁶ Significantly, the Cry3Aa*-lipA fusion crystals can be used to produce biodiesel from coconut oil with high conversion efficiency over 10 reaction cycles (FIG. 4). These properties made the Cry3Aa*-lipA catalyst the first example of a genetically-encoded immobilized enzyme that retained both high activity and recyclability.

Cry3Aa as a Carrier for Enzyme Entrapment

A potential limitation of using Cry3Aa-fusion crystals, however, is that the enzyme of interest is chemically modified at its N-terminus, which can hamper enzyme activity. Moreover, fusion to a bulky Cry3Aa protein may cause problems to the enzyme during folding. Indeed, Cry3Aa-fusion enzymes of a *Proteus mirabilis* lipase have been generated, which only retained 5% activity of retention compared to wild-type. This motivated further efforts to explore novel approaches to produced immobilized enzymes using Cry3Aa.

Cry3Aa proteins crystals contain large porous 50 Å×50 Å channels. It is envisaged that these channels can act as cages to trap the enzyme of interest inside (FIG. 3). Since Cry3Aa forms a crystal the target enzyme should be spatially confined and ordered such that each monomer does not interfere with the other. Moreover, since the unit cell dimensions of Cry3Aa are known, one can quantify the amount of enzyme molecules loaded into the Cry3Aa crystals, a characteristic can be measured using the Cry3Aa-fusion crystals based on the structural information. Most importantly, this strategy

retains the feature of a one-step immobilization platform, and the entrapped enzyme is much more likely to be in its native state since there are no covalent modifications.

Results and Discussions

Entrapment of PML Inside Cry3Aa Crystals

Proteus mirabilis lipase (PML) is a bacterial lipase capable conversion of vegetable oils including waste cooking oil to biodiesel. PML is highly soluble and expresses well in *E. coli*. The potential of PML fused to Cry3Aa was recently explored as a catalyst for biodiesel production from canola oil and waste cooking oil, and an optimized mutant containing two-point mutations (I118V, E130G, PML^{VG}) was produced. PML is a good target for Cry3Aa entrapment since the fusion of PML to Cry3Aa drastically reduces the activity. Additionally, the size of PML (55 Å×41 Å×26 Å) is a good fit for the Cry3Aa channel (FIG. 1). For clarity, square brackets ([and]) are used to indicate the entrapment of the PML molecules (e.g., Cry3Aa[PML]), whereas a hyphen (-) indicates a genetic fusion (e.g., Cry3Aa-PML).

Several plasmid variations were generated to improve the amount of PML loaded into the crystals. A list of these constructs is displayed in FIG. 4. The initial Cry3Aa[PML] coexpression cassette was generated by cloning the PML gene containing a separate ribosome binding site downstream of Cry3Aa. This plasmid was transformed, expressed and the resulting crystals were purified as described previously.¹⁵ Solubilization of the crystals and analysis by SDS-PAGE indicated that the Cry3Aa crystals contained PML (FIG. 5a).

Several methods were tried to improve the PML loading into the Cry3Aa crystals. First, Cry3Aa was put under the control of two *cyt1Aa* promoters¹⁷ and PML under one *cry3Aa* promoter (3PPML^{VG}[Cyt13A]). It was anticipated that the temporal difference between the promoters (*cyt1Aa* is activated during late-stage sporulation while *cry3Aa* is activated during vegetative and early-stage sporulation) would cause PML to accumulate in the cell first and increase the relative ratio of PML to Cry3Aa during crystallization. Although PML entrapment was achieved, the loading was much lower than Cry3Aa[PML^{VG}](FIG. 5a).

Another approach was based on increasing the ratio of PML to Cry3Aa in the transcript. This was accomplished implementing two strategies. First, by inserting another copy of PML downstream of Cry3Aa[PML^{VG}] making Cry3Aa[PML^{VG}]². Since the proteins are under one promoter, each transcript will have double the amount of PML per Cry3Aa. Second, by putting Cry3Aa and PML under two separate *cry3Aa* promoters but only one terminator (3P3A[3PPML^{VG}]). In the latter case, for each transcript containing Cry3Aa there will be two transcripts containing PML. Each of these strategies should result in a theoretical 2:1 PML:Cry3Aa ratio. Both constructs resulted in higher loading of PML into the Cry3Aa crystals, albeit 3P3A[3PPML^{VG}] demonstrated the highest loading efficiency at 64-75% depending on the batch (FIG. 5b). This corresponds to around 4×10⁵ to 6×10⁵ molecules of PML^{VG} per crystal, which is unprecedented in terms of enzyme loading. Notably, these crystals retained near identical morphology and size to native Cry3Aa crystals (FIG. 6).

The activities of all the resulting PML entrapped Cry3Aa crystals were determined by measuring the hydrolysis of p-nitrophenyl palmitate. As demonstrated in FIG. 7, higher PML loading into the crystals resulted in higher activity per pg crystal, with the 3P3A3P[PML^{VG}] construct displaying the highest activity. Moreover, entrapment of PML^{VG} inside

Cry3Aa crystals significantly stabilized it against thermal denaturation, displaying a 200-fold increase in residual activity after incubation for 1 h incubation at 58° C. (FIG. 8B).

With the optimized immobilized lipase at hand, the practicality of the catalyst for industrial reactions was tested. Two common applications of lipases are biodiesel production (Scheme 1 in FIG. 16A) and ester synthesis in organic solvents such as the cosmetic compound benzyl laurate (Scheme 2 in FIG. 16A).

The optimized Cry3Aa[PML^{VG}] crystals were capable of converting waste cooking oil into biodiesel in 10 h and retained maximal conversion for 7 reaction cycles (FIG. 8A). These data support the high catalytic activity and stability of the immobilized catalyst for biodiesel production.

For benzyl laurate synthesis, the optimized Cry3Aa[PML^{VG}] crystals were lyophilized in water for 24 h. The dried crystals were then reacted with 10 mM benzyl alcohol and 10 mM lauric acid in neat isooctane for 24 h. Gas chromatography-mass spectrometry (GCMS) analysis of the reaction mixture revealed that benzyl laurate was produced (FIG. 9). While this reaction has not been optimized to maximize yield, it demonstrates that the Cry3Aa entrapped crystals can be used in neat organic solvents for synthesis reactions.

As further demonstration that PML can be entrapped within Cry3Aa crystals, in vitro co-crystallization of carboxy-rhodamine-labelled PML and Cry3Aa was performed by vapor diffusion (FIG. 10). Since only Cry3Aa exhibits fluorescence upon UV (280 nm) irradiation, and PML is dye-labelled, formation of Cry3Aa-entrapped PML crystals could be confirmed based on their fluorescent properties (FIG. 10). Notably, these data demonstrate that Cry3Aa-mediated entrapment can also be performed in vitro.

Coentrapment of Multiple Proteins Inside Cry3Aa Crystals

One major advantage of using enzyme over chemical catalysis is that enzymes are specific, allowing for multiple reactions to occur in one-pot without unwanted side reactions or side products. It was hypothesized using the expression platform one could entrap multiple enzymes simultaneously in a single crystal. Notably, this would offer a significant advantage over other multi-enzyme systems since there is no need to purify the enzymes separately. As a proof of concept, mCherry, green fluorescent protein (GFP) and Cry3Aa were incorporated on the same plasmid and expressed in Bt. Cry3Aa was expressed under one Cry3Aa promoter, while GFP and mCherry were expressed on a separate Cry3Aa promoter separated by a ribosome binding site. Crystals produced contained both red and green fluorescence indicating co-entrapment of both proteins occurred in a single crystal (FIG. 11).

Co-immobilizing two or more enzymes on a single platform can be beneficial to either impart the catalyst with a broader substrate specificity or generate a coupled system that promotes the synthesis of higher value products. In the latter case, active site proximity can lead to much faster turnover rates since intermediates are released in close proximity to the subsequent active site, so that they can be quickly turned over before diffusing away.

Glycerol—a byproduct of FAME biodiesel production is usually extracted from the aqueous layer and purified prior to its separate commercialization. A more efficient and value-added approach would be to simultaneously convert glycerol to a cosmetic product during the transesterification process. Not only would this save time by doing both reactions in one pot, but also removal of glycerol might

increase the efficiency and yield of FAME biodiesel by pulling the equilibrium of transesterification reaction towards the products via Le Chatelier's principle.

Among glycerol-derived cosmetics, glycosylated glycerol such as 2-O-(α -D-glucopyranosyl)-sn-glycerol (uGG) has gained a lot of attention as a powerful moisturizing agent.³⁰ Interestingly, Goedl et al. demonstrated that they could produce this compound enzymatically from sucrose and glycerol using a sucrose phosphorylase enzyme from *Leuconostoc mesenteroides* (LmSP).³⁰ Therefore, it would be plausible to simultaneously convert oil to biodiesel and α GG using a one-pot combination of PML and LmSP (Scheme 3 in FIG. 16B).

Having already entrapped PML inside Cry3Aa crystals, the inventors' next step was to first verify that LmSP can be entrapped inside Cry3Aa crystals as well. A plasmid was constructed that coexpresses Cry3Aa and LmSP on separate cry3Aa promoters. This plasmid was transformed and expressed in Bt. SDS-PAGE analysis of the crystals indicated that LmSP was entrapped inside Cry3Aa crystals with high efficiency—33% of the protein molecules in the crystals are LmSP (FIG. 12a). For the LmSP:PML dual catalyst construct, a copy of PML^{VG} was inserted downstream of LmSP with its own ribosome binding site. After expression of this new construct 3P3A3P[LmSP][PML^{VG}] in Bt and purification, SDS-PAGE analysis revealed that both LmSP and PML^{VG} were entrapped in Cry3Aa crystals with high efficiency (FIG. 12a). These 3P3A3P[LmSP][PML^{VG}] crystals displayed both high sucrose phosphorylase activity (FIG. 12b) and pNPP hydrolysis activity (FIG. 12c).

Entrapment of PML Inside Cry3Aa* Crystals

A potential limitation of using native Cry3Aa crystals is that they are soluble at alkaline pH. Thus, reactions performed at alkaline pH will cause Cry3Aa to solubilize and release the entrapped protein or enzyme. It was previously demonstrated that truncation of 18 amino acids from the C-terminus of Cry3Aa and expression in Bt resulted in the production of particles that were much more resistant to solubilization. It was hypothesized that if these C-terminally truncated crystals (Cry3Aa*) still retained pores they could also entrap proteins or enzymes in vivo. To test this, Cry3Aa* and two copies of PML^{VG} were put under a single cry3Aa promoter with each protein having its own ribosome binding site. As expected, SDS-PAGE analysis demonstrated that PML^{VG} was entrapped in Cry3Aa* crystals with high efficiency and the resulting Cry3Aa*[PML^{VG}]₂ crystals displayed high activity (FIG. 13).

Entrapment of PML Inside Cry3Aa-Fusion Crystals

One can envisage that if Cry3Aa-fusion crystals formed pore-like structures they should also be capable of enzyme entrapment. Notably, this would be a facile strategy to produce multi-protein or multi-enzyme immobilized systems. Some advantages to this system over co-entrapment in native Cry3Aa crystals are that the fusion partner will be expressed in a 1:1 ratio as Cry3Aa, so proteins or enzymes that exhibit poor expression levels can be fused to Cry3Aa or Cry3Aa*. The feasibility of this approach by coexpressing Cry3Aa* fused to SpyCatcher (Cry3Aa*-SpyCat) and PML^{VG} to produce Cry3Aa*-SpyCat[PML^{VG}] crystals (FIG. 13a) has already been demonstrated. While SpyCatcher (10 kDa) is not an enzyme, it can act as a reaction handle to conjugate candidate enzymes fused with a Spytag to produce immobilized dual enzyme crystals.

It was recently discovered that CotA-laccase can convert linoleic acid, an abundant fatty acid in waste oils, into azelaic acid, a valuable chemical used in the cosmetic, pharmaceutical and polymer industries.¹⁶ Thus instead of

using methanol to produce biodiesel, lipase mediated hydrolysis of the triacylglycerols in water will generate free linoleic acid, which can be subsequently converted to azelaic acid by CotA-laccase (Scheme 4 in FIG. 16B). Since CotA expression is low, it is a good candidate for conjugation to the Cry3Aa*-SpyCat[PML^{VG}] crystals to produce the lipase:laccase catalyst. Preliminary results described herein indicate successful conjugation of CotA-Spytag Cry3Aa*-SpyCat[PML^{VG}] crystals, with full retention of laccase activity (FIG. 13).

Another example of entrapment of proteins inside Cry3Aa-fusion crystals is the entrapment of PML^{VG} inside Cry3Aa*-lipA crystals. Waste cooking oils are economical feedstocks for biodiesel production, but they have various compositions from source to source. Therefore, production of a single catalyst containing lipases with different chain specificities would be a versatile catalyst for waste cooking oil conversion to biodiesel. The plan was to coexpress Cry3Aa, PML mentioned previously, and lipase A from *Bacillus subtilis* (lipA) in the same Bt cell. PML prefers long chain fatty acids C₁₄-C₂₀ while lipA prefers medium chain fatty acids C₆-C₁₂. Experimental data showed that PML^{VG} could be entrapped inside Cry3Aa*-lipA crystals and the resulting dual lipase construct was an effective catalyst for waste cooking oil conversion to biodiesel (FIG. 14).

Example 2

Crystals Formed from Cry3Aa Mutants with Multiple Negatively Charged Amino Acid Substitutions (NegCry3Aa) Enhance the Binding of Certain Proteins

To investigate whether Cry3Aa protein crystal can be engineered to enhance its affinity to the encapsulated protein cargo, a negatively-charged Cry3Aa mutant (negCry3Aa) was generated by substituting specific residues within Domain II that are exposed to the crystal solvent channel with negatively-charged amino acids, namely, aspartate (D) and glutamate (E). For initial studies, the following mutations: K384E, N391D, N395D, S425E, Q430E, TQ436437EE, KR442443EE, T461D, and K467E (bold in the protein sequence alignment of negCry3Aa (SEQ ID NO:6) to Cry3Aa (SEQ ID NO:4) shown in FIG. 17) were engineered into the wild-type cry3aa gene, which was then used to make the final expression construct comprised of the negCry3Aa, a ribosome binding site and the supercharged GFP (scGFP), see Lawrence M S, Phillips K J, Liu DR. Supercharging proteins can impart unusual resilience. *J Am Chem Soc.* 2007; 129(33):10110-10112. This expression plasmid was then transformed into Bt cells and grown for 48 h or 72 h. Cells were harvested and the resultant crystals (NegCry3Aa-RBS-scGFP) were purified by sucrose density gradient centrifugation, followed by extensive washing in ice-cold ddH₂O. As shown in FIG. 18A, the NegCry3Aa-RBS-scGFP crystal pellets exhibited bright green fluorescence when imaged at the GFP excitation wavelength while no such fluorescence was observed for the control pellet. To further verify the encapsulation of scGFP in negCry3Aa, the negCry3Aa-RBS-scGFP crystal pellets were solubilized in standard SDS sample buffer and boiled for 5 min, followed by SDS-PAGE. As indicated in the SDS-PAGE gel in FIG. 18B, two major bands corresponding to negCry3Aa (theoretical MW=73.1 kDa) and scGFP (theoretical MW=28.5 kDa), respectively, can be observed.

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Entrapment of lipAR9 in negCry3Aa Crystals

The negCry3Aa mutant also can be used to improve the loading of a lipase. When lipase A fused to polyarginine tail (lipAR9) was co-expressed with negCry3Aa the loading improved (FIG. 19).

Example 3

Cry3Aa Fusion Protein Capable of Protein-Entrapping

The present inventors have demonstrated in their previous studies that the fusion of 1-3 repeats of a 56-residue methallothionein protein having the following amino acid sequence (SEQ ID NO:8): MTSTILVKCA-CEPCLCNVDPSKAIDRNGLYYCSEACADGHTGG-SKGC GHTGCNCHG to the C-terminus of Cry3Aa via genetic fusion can still form fusion crystals (Cry3Aa-[SmtA]₁₋₃) in Bt (Sun, Q., Cheng, S W., Cheung, K., Lee, M M., Chan, M K., "Cry protein crystal-immobilized metallothioneins for bioremediation of heavy metals from water", Crystals, 2019, 9(6): 287).

The Cry3Aa-[SmtA]₁ construct (a fusion of SEQ ID NO:4 and SEQ ID NO:8) was then tested to reveal whether it can entrap mCherry in vivo when both coding sequences for Cry3Aa-[SmtA]₁ and mCherry are co-expressed in Bt. A pHT315 plasmid harboring the gene encoding sequences of the Cry3Aa-[SmtA]₁, a ribosome binding site, and mCherry was transformed into Bt and grown for 48-72 h. As indicated on the SDS-PAGE gel shown in FIG. 20, the extracts from 40/60% sucrose layer (lane 4) contained the Cry3Aa-[SmtA]₁ (MW~78 kDa) and mCherry (MW~26 kDa), thus indicating the entrapment of mCherry in Cry3Aa-[SmtA]₁.

Example 4

Changing the Size of Cry3Aa Channel

The size of the Cry3Aa solvent channel will influence the size of the protein (e.g., an enzyme) that can be entrapped inside the crystal. For instance, a larger protein may require more space in the channel to be accommodated without steric clash, while a smaller protein might necessitate a smaller channel to prevent its diffusion. In order to increase the size of the channel, amino acids exposed to the solvent channel (FIG. 21) would need to be removed from the Cry3Aa sequence (SEQ ID NO:4). These regions include segments 110-128, 181-197, 211-226, 250-264, 423-436, 457-471, 593-598, and 627-632 of Cry3Aa (SEQ ID NO:4). The specific amino acids that can be deleted are indicated in bold in the Cry3Aa sequence (FIG. 22) and are listed in FIG. 23.

In order to illustrate this for a potential target enzyme, COOT modeling was used to fit a sucrose phosphorylase (LmSP) homolog *Thermoanaerobacter thermosaccharolyticum* 6F-phosphate phosphorylase (PDB ID: 6S9V) in the Cry3Aa crystal channel. While the majority of LmSP can be accommodated in the crystal, several loops (colored in red) clash with the enzyme model (FIG. 24a). Notably, several of these regions are the same as listed in FIG. 22. One specific area that interferes with LmSP accommodation is the region from 423-436. Therefore, amino acids Gln 426, Tyr 427, Glu 433 and Ala 434 in this region were deleted from Cry3Aa to expand the channel size. When LmSP was co-expressed with this new mutant of Cry3Aa (3A2-2, SEQ ID NO:7), significantly improved entrapment of LmSP in the crystal was observed (FIG. 24b).

Thus, the present inventors have illustrated for the first time that modifying a Cry protein by amino acid insertions or deletions (for example, insertion of at least 1, 2, 3, 4, 5,

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6, or 7 amino acid, or a deletion of at least 1, 2, 3, 4, 5, 6, or 7 amino acids) at one or more of the 8 specific regions (for example, at least 1, 2, 3, 4, 5, 6, or 7 regions) shown in FIG. 22 can effectively reduce or enlarge the protein's channel size and can therefore make the resultant crystal-forming protein to entrap a target protein (e.g., an enzyme) with higher level of efficacy, i.e., improved entrapment/retention rate and limited diffusion rate for the target protein.

Example 5

Incorporating Mutations in Cry3Aa to Stabilize the Crystal

A very important characteristic of a catalyst is its reusability. Cry3Aa crystals are inherently reusable since they are solid materials. If the crystals solubilize, however, they will release the entrapped enzyme into the bulk media. Chemical cross-linkers can be used to reduce the solubility, but this adds an additional step to catalyst production and poses environmental concerns. Taking advantage of the genetically encoded nature of the Cry3Aa crystal, a more elegant approach is to introduce stabilizing mutations at the intermolecular interfaces to generate an intrinsically insoluble crystal.

One position where is Ser 145, whose Ca atom is only 3.7 Å away from the Ca of Ser 145 of an adjacent Cry3Aa monomer, making them suitable for disulfide bond formation (FIG. 25a). When Ser 145 was mutated to cysteine (3AS145C) the crystal became much more resistant to solubilization at alkaline pH (FIG. 25b). Solubilization of the crystals in DTT confirmed disulfide bond formation. Another potential region to stabilize is His 161, which interacts with Glu 168 of a neighboring Cry3Aa monomer (FIG. 25c). His 161 was mutated to Arg (3AH161R) to increase the pKa of the charged residue thereby maintaining an electrostatic interaction with Glu 168 across a broader range of pH values. This mutation also reduced the solubility of the Cry3Aa crystal (FIG. 25b). More importantly, when a modified Cry3Aa protein containing both the S145C and H161R mutations was co-expressed with three copies of PML^{VG} (3ADM[PML^{VG}]₃), the reusability of the catalyst was enhanced during the production of biodiesel from waste cooking oil (FIG. 25d), increasing the total turnover number (TTN) by 50% after 25 reaction cycles.

Example 6

Cry3Aa Crystal Entrapment for Protein Purification

It was previously demonstrated that lysozyme can bind to Cry3Aa crystals in high occupancy, but can be easily released at high salt concentration. It was later demonstrated that the highly positive charge nature of lysozyme was responsible for its binding to Cry3Aa, which has a negatively charged patch of amino acids within its crystal channels. Based on this data, it was speculated that a positively charged patch of amino acids (poly-arginine) can be fused to a protein, coexpress the fusion protein with Cry3Aa in *Bacillus thuringiensis*, capture the fusion protein in vivo by the Cry3Aa crystal, and release the fusion protein in vitro by adding NaCl. If successful, this can offer anew approach to express and purify soluble protein without the need for columns or expensive reagents. To test this hypothesis, the coding sequence for a poly-arginine tail with 9 consecutive arginine residues (R9) was fused to the coding sequence for lipase A from *Bacillus subtilis* (lipA, 19 kDa) and inserted

downstream of Cry3Aa coding sequence to generate the construct 3A[lipAR9]. As expected, the purified Cry3Aa crystals contained lipAR9 as demonstrated by the high lipase activity of the resulting crystals (FIG. 26a). Notably, lipAR9 can be easily released from the crystal by addition of NaCl (FIG. 26b). These data show that not only can R9 be used as a fusion tag to encapsulate proteins and enzymes

within Cry3Aa crystals, but they also confirm that the Cry3Aa platform could be implemented as a general strategy for protein purification.

All patents, patent applications, and other publications, including GenBank Accession Numbers, cited in this application are incorporated by reference in the entirety for all purposes.

SEQUENCE LISTING

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Ser Asn Glu Val Arg Gly Lys Gln Leu Trp Gln Phe Val Gln Thr Leu
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Leu Gln Glu Thr Gln Ala Lys Lys Val Asn Phe Ile Gly His Ser Gln
65 70 75 80

Gly Pro Leu Ala Cys Arg Tyr Val Ala Ala Asn Tyr Pro Asp Ser Val
85 90 95

Ala Ser Val Thr Ser Ile Asn Gly Val Asn His Gly Ser Glu Ile Ala
100 105 110

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Gly His Arg Gly Asp Pro Gln Asp Ala Ile Ala Ala Leu Glu Ser Leu
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Pro Lys Thr Pro Gly Gly Glu Gly Asp Glu Ile Val Asn Gly Val His
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Tyr Tyr Cys Phe Gly Ser Tyr Ile Gln Gly Leu Ile Ala Gly Glu Lys
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Phe Phe Thr Glu Lys Gln Asn Asp Gly Leu Val Gly Arg Ser Ser Met
225 230 235 240

Arg Leu Gly Lys Leu Ile Lys Asp Asp Tyr Ala Gln Asp His Ile Asp
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Ser Asn Glu Val Arg Gly Lys Gln Leu Trp Gln Phe Val Gln Thr Leu
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Gly Pro Leu Ala Cys Arg Tyr Val Ala Ala Asn Tyr Pro Asp Ser Val
 85 90 95

Ala Ser Val Thr Ser Ile Asn Gly Val Asn His Gly Ser Glu Ile Ala
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Gly His Arg Gly Asp Pro Gln Asp Ala Ile Ala Ala Leu Glu Ser Leu
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 165 170 175

Pro Lys Thr Pro Gly Gly Glu Gly Asp Glu Ile Val Asn Gly Val His
 180 185 190

Tyr Tyr Cys Phe Gly Ser Tyr Ile Gln Gly Leu Ile Ala Gly Glu Lys
 195 200 205

Gly Asn Leu Leu Asp Pro Thr His Ala Ala Met Arg Val Leu Asn Thr
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Phe Phe Thr Glu Lys Gln Asn Asp Gly Leu Val Gly Arg Ser Ser Met
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Arg Leu Gly Lys Leu Ile Lys Asp Asp Tyr Ala Gln Asp His Ile Asp
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Asp	His	Cys	Val	Lys 245	Trp	Tyr	Asn	Val	Gly 250	Leu	Asp	Lys	Leu	Arg
Ser	Ser	Tyr	Glu	Ser 260	Trp	Val	Asn	Phe	Asn 265	Arg	Tyr	Arg	Arg 270	Glu
Thr	Leu	Thr	Val	Leu 275	Asp	Leu	Ile	Ala	Leu 280	Phe	Pro	Leu 285	Tyr	Asp
Arg	Leu	Tyr	Pro	Lys 290	Glu	Val	Lys 295	Thr	Glu	Leu	Thr	Arg	Asp	Val
Thr 305	Asp	Pro	Ile	Val	Gly 310	Val	Asn	Asn	Leu	Arg	Gly	Tyr	Gly	Thr
Phe	Ser	Asn	Ile	Glu 325	Asn	Tyr	Ile	Arg	Lys 330	Pro	His	Leu	Phe	Asp
Leu	His	Arg	Ile	Gln 340	Phe	His	Thr	Arg	Phe 345	Gln	Pro	Gly	Tyr	Tyr
Asn	Asp	Ser	Phe	Asn 355	Tyr	Trp	Ser	Gly	Asn 360	Tyr	Val	Ser	Thr	Arg
Ser	Ile	Gly	Ser	Asn 370	Asp	Ile	Ile	Thr	Ser	Pro	Phe	Tyr	Gly	Asn
Ser 385	Ser	Glu	Pro	Val	Gln	Asn	Leu	Glu	Phe	Asn 395	Gly	Glu	Lys	Val
Arg	Ala	Val	Ala	Asn 405	Thr	Asn	Leu	Ala	Val	Trp	Pro	Ser	Ala	Val
Ser	Gly	Val	Thr	Lys 420	Val	Glu	Phe	Ser	Gln	Tyr	Asn	Asp	Gln	Thr
Glu	Ala	Ser	Thr	Gln	Thr	Tyr	Asp	Ser	Lys	Arg	Asn	Val	Gly	Ala

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Ser Trp Asp Ser Ile Asp Gln Leu Pro Pro Glu Thr Thr Asp Glu Pro
 450 455 460
 Leu Glu Lys Gly Tyr Ser His Gln Leu Asn Tyr Val Met Cys Phe Leu
 465 470 475 480
 Met Gln Gly Ser Arg Gly Thr Ile Pro Val Leu Thr Trp Thr His Lys
 485 490 495
 Ser Val Asp Phe Phe Asn Met Ile Asp Ser Lys Lys Ile Thr Gln Leu
 500 505 510
 Pro Leu Val Lys Ala Tyr Lys Leu Gln Ser Gly Ala Ser Val Val Ala
 515 520 525
 Gly Pro Arg Phe Thr Gly Gly Asp Ile Ile Gln Cys Thr Glu Asn Gly
 530 535 540
 Ser Ala Ala Thr Ile Tyr Val Thr Pro Asp Val Ser Tyr Ser Gln Lys
 545 550 555 560
 Tyr Arg Ala Arg Ile His Tyr Ala Ser Thr Ser Gln Ile Thr Phe Thr
 565 570 575
 Leu Ser Leu Asp Gly Ala Pro Phe Asn Gln Tyr Tyr Phe Asp Lys Thr
 580 585 590
 Ile Asn Lys Gly Asp Thr Leu Thr Tyr Asn Ser Phe Asn Leu Ala Ser
 595 600 605
 Phe Ser Thr Pro Phe Glu Leu Ser Gly Asn Asn Leu Gln Ile Gly Val
 610 615 620
 Thr
 625

<210> SEQ ID NO 6
 <211> LENGTH: 644
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of sequence: amino acid sequence of
 negCry3Aa

<400> SEQUENCE: 6

Met Asn Pro Asn Asn Arg Ser Glu His Asp Thr Ile Lys Thr Thr Glu
 1 5 10 15
 Asn Asn Glu Val Pro Thr Asn His Val Gln Tyr Pro Leu Ala Glu Thr
 20 25 30
 Pro Asn Pro Thr Leu Glu Asp Leu Asn Tyr Lys Glu Phe Leu Arg Met
 35 40 45
 Thr Ala Asp Asn Asn Thr Glu Ala Leu Asp Ser Ser Thr Thr Lys Asp
 50 55 60
 Val Ile Gln Lys Gly Ile Ser Val Val Gly Asp Leu Leu Gly Val Val
 65 70 75 80
 Gly Phe Pro Phe Gly Gly Ala Leu Val Ser Phe Tyr Thr Asn Phe Leu
 85 90 95
 Asn Thr Ile Trp Pro Ser Glu Asp Pro Trp Lys Ala Phe Met Glu Gln
 100 105 110
 Val Glu Ala Leu Met Asp Gln Lys Ile Ala Asp Tyr Ala Lys Asn Lys
 115 120 125
 Ala Leu Ala Glu Leu Gln Gly Leu Gln Asn Asn Val Glu Asp Tyr Val
 130 135 140
 Ser Ala Leu Ser Ser Trp Gln Lys Asn Pro Val Ser Ser Arg Asn Pro
 145 150 155 160
 His Ser Gln Gly Arg Ile Arg Glu Leu Phe Ser Gln Ala Glu Ser His
 165 170 175

Phe	Arg	Asn	Ser	Met	Pro	Ser	Phe	Ala	Ile	Ser	Gly	Tyr	Glu	Val	Leu
		180						185					190		
Phe	Leu	Thr	Thr	Tyr	Ala	Gln	Ala	Ala	Asn	Thr	His	Leu	Phe	Leu	Leu
		195					200					205			
Lys	Asp	Ala	Gln	Ile	Tyr	Gly	Glu	Glu	Trp	Gly	Tyr	Glu	Lys	Glu	Asp
		210				215					220				
Ile	Ala	Glu	Phe	Tyr	Lys	Arg	Gln	Leu	Lys	Leu	Thr	Gln	Glu	Tyr	Thr
225					230					235					240
Asp	His	Cys	Val	Lys	Trp	Tyr	Asn	Val	Gly	Leu	Asp	Lys	Leu	Arg	Gly
				245					250					255	
Ser	Ser	Tyr	Glu	Ser	Trp	Val	Asn	Phe	Asn	Arg	Tyr	Arg	Arg	Glu	Met
			260					265					270		
Thr	Leu	Thr	Val	Leu	Asp	Leu	Ile	Ala	Leu	Phe	Pro	Leu	Tyr	Asp	Val
		275					280					285			
Arg	Leu	Tyr	Pro	Lys	Glu	Val	Lys	Thr	Glu	Leu	Thr	Arg	Asp	Val	Leu
		290					295					300			
Thr	Asp	Pro	Ile	Val	Gly	Val	Asn	Asn	Leu	Arg	Gly	Tyr	Gly	Thr	Thr
305					310					315					320
Phe	Ser	Asn	Ile	Glu	Asn	Tyr	Ile	Arg	Lys	Pro	His	Leu	Phe	Asp	Tyr
			325						330					335	
Leu	His	Arg	Ile	Gln	Phe	His	Thr	Arg	Phe	Gln	Pro	Gly	Tyr	Tyr	Gly
			340					345				350			
Asn	Asp	Ser	Phe	Asn	Tyr	Trp	Ser	Gly	Asn	Tyr	Val	Ser	Thr	Arg	Pro
		355					360					365			
Ser	Ile	Gly	Ser	Asn	Asp	Ile	Ile	Thr	Ser	Pro	Phe	Tyr	Gly	Asn	Glu
		370				375					380				
Ser	Ser	Glu	Pro	Val	Gln	Asp	Leu	Glu	Phe	Asp	Gly	Glu	Lys	Val	Tyr
385					390					395					400
Arg	Ala	Val	Ala	Asn	Thr	Asn	Leu	Ala	Val	Trp	Pro	Ser	Ala	Val	Tyr
				405					410					415	
Ser	Gly	Val	Thr	Lys	Val	Glu	Phe	Glu	Gln	Tyr	Asn	Asp	Glu	Thr	Asp
			420					425					430		
Glu	Ala	Ser	Glu	Glu	Thr	Tyr	Asp	Ser	Glu	Glu	Asn	Val	Gly	Ala	Val
		435					440					445			
Ser	Trp	Asp	Ser	Ile	Asp	Gln	Leu	Pro	Pro	Glu	Thr	Asp	Asp	Glu	Pro
		450				455						460			
Leu	Glu	Glu	Gly	Tyr	Ser	His	Gln	Leu	Asn	Tyr	Val	Met	Cys	Phe	Leu
465					470					475					480
Met	Gln	Gly	Ser	Arg	Gly	Thr	Ile	Pro	Val	Leu	Thr	Trp	Thr	His	Lys
				485					490					495	
Ser	Val	Asp	Phe	Phe	Asn	Met	Ile	Asp	Ser	Lys	Lys	Ile	Thr	Gln	Leu
			500					505					510		
Pro	Leu	Val	Lys	Ala	Tyr	Lys	Leu	Gln	Ser	Gly	Ala	Ser	Val	Val	Ala
		515					520					525			
Gly	Pro	Arg	Phe												

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Ile Asn Lys Gly Asp Thr Leu Thr Tyr Asn Ser Phe Asn Leu Ala Ser
   595                                600                                605

Phe Ser Thr Pro Phe Glu Leu Ser Gly Asn Asn Leu Gln Ile Gly Val
   610                                615                                620

Thr Gly Leu Ser Ala Gly Asp Lys Val Tyr Ile Asp Lys Ile Glu Phe
   625                                630                                635                                640

Ile Pro Val Asn

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<210> SEQ ID NO 7
<211> LENGTH: 640
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: amino acid sequence of
      3A2-2

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<400> SEQUENCE: 7

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```

Met Asn Pro Asn Asn Arg Ser Glu His Asp Thr Ile Lys Thr Thr Glu
  1                                5                                10                                15

Asn Asn Glu Val Pro Thr Asn His Val Gln Tyr Pro Leu Ala Glu Thr
   20                                25                                30

Pro Asn Pro Thr Leu Glu Asp Leu Asn Tyr Lys Glu Phe Leu Arg Met
   35                                40                                45

Thr Ala Asp Asn Asn Thr Glu Ala Leu Asp Ser Ser Thr Thr Lys Asp
   50                                55                                60

Val Ile Gln Lys Gly Ile Ser Val Val Gly Asp Leu Leu Gly Val Val
   65                                70                                75                                80

Gly Phe Pro Phe Gly Gly Ala Leu Val Ser Phe Tyr Thr Asn Phe Leu
   85                                90                                95

Asn Thr Ile Trp Pro Ser Glu Asp Pro Trp Lys Ala Phe Met Glu Gln
  100                                105                                110

Val Glu Ala Leu Met Asp Gln Lys Ile Ala Asp Tyr Ala Lys Asn Lys
  115                                120                                125

Ala Leu Ala Glu Leu Gln Gly Leu Gln Asn Asn Val Glu Asp Tyr Val
  130                                135                                140

Ser Ala Leu Ser Ser Trp Gln Lys Asn Pro Val Ser Ser Arg Asn Pro
  145                                150                                155                                160

His Ser Gln Gly Arg Ile Arg Glu Leu Phe Ser Gln Ala Glu Ser His
  165                                170                                175

Phe Arg Asn Ser Met Pro Ser Phe Ala Ile Ser Gly Tyr Glu Val Leu
  180                                185                                190

Phe Leu Thr Thr Tyr Ala Gln Ala Ala Asn Thr His Leu Phe Leu Leu
  195                                200                                205

Lys Asp Ala Gln Ile Tyr Gly Glu Glu Trp Gly Tyr Glu Lys Glu Asp
  210                                215                                220

Ile Ala Glu Phe Tyr Lys Arg Gln Leu Lys Leu Thr Gln Glu Tyr Thr
  225                                230                                235                                240

Asp His Cys Val Lys Trp Tyr Asn Val Gly Leu Asp Lys Leu Arg Gly
  245                                250                                255

Ser Ser Tyr Glu Ser Trp Val Asn Phe Asn Arg Tyr Arg Arg Glu Met
  260                                265                                270

Thr Leu Thr Val Leu Asp Leu Ile Ala Leu Phe Pro Leu Tyr Asp Val
  275                                280                                285

Arg Leu Tyr Pro Lys Glu Val Lys Thr Glu Leu Thr Arg Asp Val Leu
  290                                295                                300

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Thr Asp Pro Ile Val Gly Val Asn Asn Leu Arg Gly Tyr Gly Thr Thr
 305 310 315 320
 Phe Ser Asn Ile Glu Asn Tyr Ile Arg Lys Pro His Leu Phe Asp Tyr
 325 330 335
 Leu His Arg Ile Gln Phe His Thr Arg Phe Gln Pro Gly Tyr Tyr Gly
 340 345 350
 Asn Asp Ser Phe Asn Tyr Trp Ser Gly Asn Tyr Val Ser Thr Arg Pro
 355 360 365
 Ser Ile Gly Ser Asn Asp Ile Ile Thr Ser Pro Phe Tyr Gly Asn Lys
 370 375 380
 Ser Ser Glu Pro Val Gln Asn Leu Glu Phe Asn Gly Glu Lys Val Tyr
 385 390 395 400
 Arg Ala Val Ala Asn Thr Asn Leu Ala Val Trp Pro Ser Ala Val Tyr
 405 410 415
 Ser Gly Val Thr Lys Val Glu Phe Ser Asn Asp Gln Thr Asp Ser Thr
 420 425 430
 Gln Thr Tyr Asp Ser Lys Arg Asn Val Gly Ala Val Ser Trp Asp Ser
 435 440 445
 Ile Asp Gln Leu Pro Pro Glu Thr Thr Asp Glu Pro Leu Glu Lys Gly
 450 455 460
 Tyr Ser His Gln Leu Asn Tyr Val Met Cys Phe Leu Met Gln Gly Ser
 465 470 475 480
 Arg Gly Thr Ile Pro Val Leu Thr Trp Thr His Lys Ser Val Asp Phe
 485 490 495
 Phe Asn Met Ile Asp Ser Lys Lys Ile Thr Gln Leu Pro Leu Val Lys
 500 505 510
 Ala Tyr Lys Leu Gln Ser Gly Ala Ser Val Val Ala Gly Pro Arg Phe
 515 520 525
 Thr Gly Gly Asp Ile Ile Gln Cys Thr Glu Asn Gly Ser Ala Ala Thr
 530 535 540
 Ile Tyr Val Thr Pro Asp Val Ser Tyr Ser Gln Lys Tyr Arg Ala Arg
 545 550 555 560
 Ile His Tyr Ala Ser Thr Ser Gln Ile Thr Phe Thr Leu Ser Leu Asp
 565 570 575
 Gly Ala Pro Phe Asn Gln Tyr Tyr Phe Asp Lys Thr Ile Asn Lys Gly
 580 585 590
 Asp Thr Leu Thr Tyr Asn Ser Phe Asn Leu Ala Ser Phe Ser Thr Pro
 595 600 605
 Phe Glu Leu Ser Gly Asn Asn Leu Gln Ile Gly Val Thr Gly Leu Ser
 610 615 620
 Ala Gly Asp Lys Val Tyr Ile Asp Lys Ile Glu Phe Ile Pro Val Asn
 625 630 635 640

<210> SEQ ID NO 8

<211> LENGTH: 56

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

 <223> OTHER INFORMATION: Description of sequence: amino acid sequence of
 methallothionein protein

<400> SEQUENCE: 8

Met Thr Ser Thr Thr Leu Val Lys Cys Ala Cys Glu Pro Cys Leu Cys
 1 5 10 15
 Asn Val Asp Pro Ser Lys Ala Ile Asp Arg Asn Gly Leu Tyr Tyr Cys
 20 25 30

-continued

Ser Glu Ala Cys Ala Asp Gly His Thr Gly Gly Ser Lys Gly Cys Gly
 35 40 45

His Thr Gly Cys Asn Cys His Gly
 50 55

<210> SEQ ID NO 9

<211> LENGTH: 644

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of sequence: amino acid sequence of
 Cry3Aa double mutant (S145C, H161R)

<400> SEQUENCE: 9

Met Asn Pro Asn Asn Arg Ser Glu His Asp Thr Ile Lys Thr Thr Glu
 1 5 10 15

Asn Asn Glu Val Pro Thr Asn His Val Gln Tyr Pro Leu Ala Glu Thr
 20 25 30

Pro Asn Pro Thr Leu Glu Asp Leu Asn Tyr Lys Glu Phe Leu Arg Met
 35 40 45

Thr Ala Asp Asn Asn Thr Glu Ala Leu Asp Ser Ser Thr Thr Lys Asp
 50 55 60

Val Ile Gln Lys Gly Ile Ser Val Val Gly Asp Leu Leu Gly Val Val
 65 70 75 80

Gly Phe Pro Phe Gly Gly Ala Leu Val Ser Phe Tyr Thr Asn Phe Leu
 85 90 95

Asn Thr Ile Trp Pro Ser Glu Asp Pro Trp Lys Ala Phe Met Glu Gln
 100 105 110

Val Glu Ala Leu Met Asp Gln Lys Ile Ala Asp Tyr Ala Lys Asn Lys
 115 120 125

Ala Leu Ala Glu Leu Gln Gly Leu Gln Asn Asn Val Glu Asp Tyr Val
 130 135 140

Cys Ala Leu Ser Ser Trp Gln Lys Asn Pro Val Ser Ser Arg Asn Pro
 145 150 155 160

Arg Ser Gln Gly Arg Ile Arg Glu Leu Phe Ser Gln Ala Glu Ser His
 165 170 175

Phe Arg Asn Ser Met Pro Ser Phe Ala Ile Ser Gly Tyr Glu Val Leu
 180 185 190

Phe Leu Thr Thr Tyr Ala Gln Ala Ala Asn Thr His Leu Phe Leu Leu
 195 200 205

Lys Asp Ala Gln Ile Tyr Gly Glu Glu Trp Gly Tyr Glu Lys Glu Asp
 210 215 220

Ile Ala Glu Phe Tyr Lys Arg Gln Leu Lys Leu Thr Gln Glu Tyr Thr
 225 230 235 240

Asp His Cys Val Lys Trp Tyr Asn Val Gly Leu Asp Lys Leu Arg Gly
 245 250 255

Ser Ser Tyr Glu Ser Trp Val Asn Phe Asn Arg Tyr Arg Arg Glu Met
 260 265 270

Thr Leu Thr Val Leu Asp Leu Ile Ala Leu Phe Pro Leu Tyr Asp Val
 275 280 285

Arg Leu Tyr Pro Lys Glu Val Lys Thr Glu Leu Thr Arg Asp Val Leu
 290 295 300

Thr Asp Pro Ile Val Gly Val Asn Asn Leu Arg Gly Tyr Gly Thr Thr
 305 310 315 320

Phe Ser Asn Ile Glu Asn Tyr Ile Arg Lys Pro His Leu Phe Asp Tyr

-continued

35	40	45
Asn Tyr Asn Asn Gly Pro Val Leu Ser Arg Phe Val Gln Lys Val Leu		
50	55	60
Asp Glu Thr Gly Ala Lys Lys Val Asp Ile Val Ala His Ser Met Gly		
65	70	75 80
Gly Ala Asn Thr Leu Tyr Tyr Ile Lys Asn Leu Asp Gly Gly Asn Lys		
85	90	95
Val Ala Asn Val Val Thr Val Gly Gly Ala Asn Arg Leu Thr Thr Gly		
100	105	110
Lys Ala Leu Pro Gly Thr Asp Pro Asn Gln Lys Ile Leu Tyr Thr Ser		
115	120	125
Ile Tyr Ser Ser Ala Asp Met Ile Val Met Asn Tyr Leu Ser Arg Leu		
130	135	140
Asp Gly Ala Arg Asn Val Gln Ile His Gly Val Gly His Ile Gly Leu		
145	150	155 160
Leu Tyr Ser Ser Gln Val Asn Ser Leu Ile Lys Glu Gly Leu Asn Gly		
165	170	175
Gly Gly Gln Asn Thr Asn		
180		

<210> SEQ ID NO 11

<211> LENGTH: 195

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Description of sequence: amino acid sequence of lipAR9

<400> SEQUENCE: 11

Met Ala Glu His Asn Pro Val Val Met Val His Gly Ile Gly Gly Ala		
1	5	10 15
Ser Phe Asn Phe Ala Gly Ile Lys Ser Tyr Leu Val Ser Gln Gly Trp		
20	25	30
Ser Arg Asp Lys Leu Tyr Ala Val Asp Phe Trp Asp Lys Thr Gly Thr		
35	40	45
Asn Tyr Asn Asn Gly Pro Val Leu Ser Arg Phe Val Gln Lys Val Leu		
50	55	60
Asp Glu Thr Gly Ala Lys Lys Val Asp Ile Val Ala His Ser Met Gly		
65	70	75 80
Gly Ala Asn Thr Leu Tyr Tyr Ile Lys Asn Leu Asp Gly Gly Asn Lys		
85	90	95
Val Ala Asn Val Val Thr Val Gly Gly Ala Asn Arg Leu Thr Thr Gly		
100	105	110
Lys Ala Leu Pro Gly Thr Asp Pro Asn Gln Lys Ile Leu Tyr Thr Ser		
115	120	125
Ile Tyr Ser Ser Ala Asp Met Ile Val Met Asn Tyr Leu Ser Arg Leu		
130	135	140
Asp Gly Ala Arg Asn Val Gln Ile His Gly Val Gly His Ile Gly Leu		
145	150	155 160
Leu Tyr Ser Ser Gln Val Asn Ser Leu Ile Lys Glu Gly Leu Asn Gly		
165	170	175
Gly Gly Gln Asn Thr Asn Gly Ser Ser Gly Arg Arg Arg Arg Arg Arg		
180	185	190
Arg Arg Arg		
195		

-continued

<210> SEQ ID NO 12
 <211> LENGTH: 633
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of sequence: consensus sequence
 from Fig. 17

<400> SEQUENCE: 12

```

Met Asn Pro Asn Asn Arg Ser Glu His Asp Thr Ile Lys Thr Thr Glu
1          5          10          15

Asn Asn Glu Val Pro Thr Asn His Val Gln Tyr Pro Leu Ala Glu Thr
20          25          30

Pro Asn Pro Thr Leu Glu Asp Leu Asn Tyr Lys Glu Phe Leu Arg Met
35          40          45

Thr Ala Asp Asn Asn Thr Glu Ala Leu Asp Ser Ser Thr Thr Lys Asp
50          55          60

Val Ile Gln Lys Gly Ile Ser Val Val Gly Asp Leu Leu Gly Val Val
65          70          75          80

Gly Phe Pro Phe Gly Gly Ala Leu Val Ser Phe Tyr Thr Asn Phe Leu
85          90          95

Asn Thr Ile Trp Pro Ser Glu Asp Pro Trp Lys Ala Phe Met Glu Gln
100         105         110

Val Glu Ala Leu Met Asp Gln Lys Ile Ala Asp Tyr Ala Lys Asn Lys
115         120         125

Ala Leu Ala Glu Leu Gln Gly Leu Gln Asn Asn Val Glu Asp Tyr Val
130         135         140

Ser Ala Leu Ser Ser Trp Gln Lys Asn Pro Val Ser Ser Arg Asn Pro
145         150         155         160

His Ser Gln Gly Arg Ile Arg Glu Leu Phe Ser Gln Ala Glu Ser His
165         170         175

Phe Arg Asn Ser Met Pro Ser Phe Ala Ile Ser Gly Tyr Glu Val Leu
180         185         190

Phe Leu Thr Thr Tyr Ala Gln Ala Ala Asn Thr His Leu Phe Leu Leu
195         200         205

Lys Asp Ala Gln Ile Tyr Gly Glu Glu Trp Gly Tyr Glu Lys Glu Asp
210         215         220

Ile Ala Glu Phe Tyr Lys Arg Gln Leu Lys Leu Thr Gln Glu Tyr Thr
225         230         235         240

Asp His Cys Val Lys Trp Tyr Asn Val Gly Leu Asp Lys Leu Arg Gly
245         250         255

Ser Ser Tyr Glu Ser Trp Val Asn Phe Asn Arg Tyr Arg Arg Glu Met
260         265         270

Thr Leu Thr Val Leu Asp Leu Ile Ala Leu Phe Pro Leu Tyr Asp Val
275         280         285

Arg Leu Tyr Pro Lys Glu Val Lys Thr Glu Leu Thr Arg Asp Val Leu
290         295         300

Thr Asp Pro Ile Val Gly Val Asn Asn Leu Arg Gly Tyr Gly Thr Thr
305         310         315         320

Phe Ser Asn Ile Glu Asn Tyr Ile Arg Lys Pro His Leu Phe Asp Tyr
325         330         335

Leu His Arg Ile Gln Phe His Thr Arg Phe Gln Pro Gly Tyr Tyr Gly
340         345         350

Asn Asp Ser Phe Asn Tyr Trp Ser Gly Asn Tyr Val Ser Thr Arg Pro
355         360         365

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Ser Ile Gly Ser Asn Asp Ile Ile Thr Ser Pro Phe Tyr Gly Asn Ser
 370                               375                               380

Ser Glu Pro Val Gln Leu Glu Phe Gly Glu Lys Val Tyr Arg Ala Val
385                               390                               395                               400

Ala Asn Thr Asn Leu Ala Val Trp Pro Ser Ala Val Tyr Ser Gly Val
                               405                               410                               415

Thr Lys Val Glu Phe Gln Tyr Asn Asp Thr Asp Glu Ala Ser Thr Tyr
                               420                               425                               430

Asp Ser Asn Val Gly Ala Val Ser Trp Asp Ser Ile Asp Gln Leu Pro
                               435                               440                               445

Pro Glu Thr Asp Glu Pro Leu Glu Gly Tyr Ser His Gln Leu Asn Tyr
                               450                               455                               460

Val Met Cys Phe Leu Met Gln Gly Ser Arg Gly Thr Ile Pro Val Leu
465                               470                               475                               480

Thr Trp Thr His Lys Ser Val Asp Phe Phe Asn Met Ile Asp Ser Lys
                               485                               490                               495

Lys Ile Thr Gln Leu Pro Leu Val Lys Ala Tyr Lys Leu Gln Ser Gly
                               500                               505                               510

Ala Ser Val Val Ala Gly Pro Arg Phe Thr Gly Gly Asp Ile Ile Gln
                               515                               520                               525

Cys Thr Glu Asn Gly Ser Ala Ala Thr Ile Tyr Val Thr Pro Asp Val
530                               535                               540

Ser Tyr Ser Gln Lys Tyr Arg Ala Arg Ile His Tyr Ala Ser Thr Ser
545                               550                               555                               560

Gln Ile Thr Phe Thr Leu Ser Leu Asp Gly Ala Pro Phe Asn Gln Tyr
                               565                               570                               575

Tyr Phe Asp Lys Thr Ile Asn Lys Gly Asp Thr Leu Thr Tyr Asn Ser
                               580                               585                               590

Phe Asn Leu Ala Ser Phe Ser Thr Pro Phe Glu Leu Ser Gly Asn Asn
595                               600                               605

Leu Gln Ile Gly Val Thr Gly Leu Ser Ala Gly Asp Lys Val Tyr Ile
610                               615                               620

Asp Lys Ile Glu Phe Ile Pro Val Asn
625                               630

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<210> SEQ ID NO 13

<211> LENGTH: 1176

<212> TYPE: PRT

<213> ORGANISM: Bacillus thuringiensis

<220> FEATURE:

<223> OTHER INFORMATION: Description of sequence: protein CryIAa from Fig. 27

<400> SEQUENCE: 13

```

Met Asp Asn Asn Pro Asn Ile Asn Glu Cys Ile Pro Tyr Asn Cys Leu
 1                               5                               10                               15

Ser Asn Pro Glu Val Glu Val Leu Gly Gly Glu Arg Ile Glu Thr Gly
20                               25                               30

Tyr Thr Pro Ile Asp Ile Ser Leu Ser Leu Thr Gln Phe Leu Leu Ser
35                               40                               45

Glu Phe Val Pro Gly Ala Gly Phe Val Leu Gly Leu Val Asp Ile Ile
50                               55                               60

Trp Gly Ile Phe Gly Pro Ser Gln Trp Asp Ala Phe Pro Val Gln Ile
65                               70                               75                               80

Glu Gln Leu Ile Asn Gln Arg Ile Glu Glu Phe Ala Arg Asn Gln Ala

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85								90				95			
Ile	Ser	Arg	Leu	Glu	Gly	Leu	Ser	Asn	Leu	Tyr	Gln	Ile	Tyr	Ala	Glu
			100					105					110		
Ser	Phe	Arg	Glu	Trp	Glu	Ala	Asp	Pro	Thr	Asn	Pro	Ala	Leu	Arg	Glu
		115					120					125			
Glu	Met	Arg	Ile	Gln	Phe	Asn	Asp	Met	Asn	Ser	Ala	Leu	Thr	Thr	Ala
	130					135					140				
Ile	Pro	Leu	Leu	Ala	Val	Gln	Asn	Tyr	Gln	Val	Pro	Leu	Leu	Ser	Val
145					150					155					160
Tyr	Val	Gln	Ala	Ala	Asn	Leu	His	Leu	Ser	Val	Leu	Arg	Asp	Val	Ser
			165					170						175	
Val	Phe	Gly	Gln	Arg	Trp	Gly	Phe	Asp	Ala	Ala	Thr	Ile	Asn	Ser	Arg
		180						185					190		
Tyr	Asn	Asp	Leu	Thr	Arg	Leu	Ile	Gly	Asn	Tyr	Thr	Asp	Tyr	Ala	Val
	195					200						205			
Arg	Trp	Tyr	Asn	Thr	Gly	Leu	Glu	Arg	Val	Trp	Gly	Pro	Asp	Ser	Arg
	210				215						220				
Asp	Trp	Val	Arg	Tyr	Asn	Gln	Phe	Arg	Arg	Glu	Leu	Thr	Leu	Thr	Val
225					230					235					240
Leu	Asp	Ile	Val	Ala	Leu	Phe	Ser	Asn	Tyr	Asp	Ser	Arg	Arg	Tyr	Pro
			245					250						255	
Ile	Arg	Thr	Val	Ser	Gln	Leu	Thr	Arg	Glu	Ile	Tyr	Thr	Asn	Pro	Val
		260						265					270		
Leu	Glu	Asn	Phe	Asp	Gly	Ser	Phe	Arg	Gly	Met	Ala	Gln	Arg	Ile	Glu
		275					280					285			
Gln	Asn	Ile	Arg	Gln	Pro	His	Leu	Met	Asp	Ile	Leu	Asn	Ser	Ile	Thr
	290					295					300				
Ile	Tyr	Thr	Asp	Val	His	Arg	Gly	Phe	Asn	Tyr	Trp	Ser	Gly	His	Gln
305					310					315					320
Ile	Thr	Ala	Ser	Pro	Val	Gly	Phe	Ser	Gly	Pro	Glu	Phe	Ala	Phe	Pro
			325						330					335	
Leu	Phe	Gly	Asn	Ala	Gly	Asn	Ala	Ala	Pro	Pro	Val	Leu	Val	Ser	Leu
		340						345					350		
Thr	Gly	Leu	Gly	Ile	Phe	Arg	Thr	Leu	Ser	Ser	Pro	Leu	Tyr	Arg	Arg
		355					360						365		
Ile	Ile	Leu	Gly	Ser	Gly	Pro	Asn	Asn	Gln	Glu	Leu	Phe	Val	Leu	Asp
	370				375						380				
Gly	Thr	Glu	Phe	Ser	Phe	Ala	Ser	Leu	Thr	Thr	Asn	Leu	Pro	Ser	Thr
385					390					395					400
Ile	Tyr	Arg	Gln	Arg	Gly	Thr	Val	Asp	Ser	Leu	Asp	Val	Ile	Pro	Pro
			405						410					415	
Gln	Asp	Asn	Ser	Val	Pro	Pro	Arg	Ala	Gly	Phe	Ser	His	Arg	Leu	Ser
		420						425					430		
His	Val	Thr	Met	Leu	Ser	Gln	Ala	Ala	Gly	Ala	Val	Tyr	Thr	Leu	Arg
		435					440					445			
Ala	Pro	Thr	Phe	Ser	Trp	Gln	His	Arg	Ser	Ala	Glu	Phe	Asn	Asn	Ile
	450					455					460				
Ile	Pro	Ser	Ser	Gln	Ile	Thr	Gln	Ile	Pro	Leu	Thr	Lys	Ser	Thr	Asn
465					470					475					480
Leu	Gly	Ser	Gly	Thr	Ser	Val	Val	Lys	Gly	Pro	Gly	Phe	Thr	Gly	Gly
			485						490					495	
Asp	Ile	Leu	Arg	Arg	Thr	Ser	Pro	Gly	Gln	Ile	Ser	Thr	Leu	Arg	Val
		500						505					510		

Asn	Ile	Thr	Ala	Pro	Leu	Ser	Gln	Arg	Tyr	Arg	Val	Arg	Ile	Arg	Tyr
515						520						525			
Ala	Ser	Thr	Thr	Asn	Leu	Gln	Phe	His	Thr	Ser	Ile	Asp	Gly	Arg	Pro
530						535						540			
Ile	Asn	Gln	Gly	Asn	Phe	Ser	Ala	Thr	Met	Ser	Ser	Gly	Ser	Asn	Leu
545				550						555				560	
Gln	Ser	Gly	Ser	Phe	Arg	Thr	Val	Gly	Phe	Thr	Thr	Pro	Phe	Asn	Phe
				565				570						575	
Ser	Asn	Gly	Ser	Ser	Val	Phe	Thr	Leu	Ser	Ala	His	Val	Phe	Asn	Ser
				580				585						590	
Gly	Asn	Glu	Val	Tyr	Ile	Asp	Arg	Ile	Glu	Phe	Val	Pro	Ala	Glu	Val
		595				600						605			
Thr	Phe	Glu	Ala	Glu	Tyr	Asp	Leu	Glu	Arg	Ala	Gln	Lys	Ala	Val	Asn
610						615						620			
Glu	Leu	Phe	Thr	Ser	Ser	Asn	Gln	Ile	Gly	Leu	Lys	Thr	Asp	Val	Thr
625				630						635				640	
Asp	Tyr	His	Ile	Asp	Gln	Val	Ser	Asn	Leu	Val	Glu	Cys	Leu	Ser	Asp
				645				650						655	
Glu	Phe	Cys	Leu	Asp	Glu	Lys	Gln	Glu	Leu	Ser	Glu	Lys	Val	Lys	His
		660						665						670	
Ala	Lys	Arg	Leu	Ser	Asp	Glu	Arg	Asn	Leu	Leu	Gln	Asp	Pro	Asn	Phe
675						680						685			
Arg	Gly	Ile	Asn	Arg	Gln	Leu	Asp	Arg	Gly	Trp	Arg	Gly	Ser	Thr	Asp
690						695				700					
Ile	Thr	Ile	Gln	Gly	Gly	Asp	Asp	Val	Phe	Lys	Glu	Asn	Tyr	Val	Thr
705				710						715				720	
Leu	Leu	Gly	Thr	Phe	Asp	Glu	Cys	Tyr	Pro	Thr	Tyr	Leu	Tyr	Gln	Lys
				725				730						735	
Ile	Asp	Glu	Ser	Lys	Leu	Lys	Ala	Tyr	Thr	Arg	Tyr	Gln	Leu	Arg	Gly
		740						745						750	
Tyr	Ile	Glu	Asp	Ser	Gln	Asp	Leu	Glu	Ile	Tyr	Leu	Ile	Arg	Tyr	Asn
755						760						765			
Ala	Lys	His	Glu	Thr	Val	Asn	Val	Pro	Gly	Thr	Gly	Ser	Leu	Trp	Pro
770						775						780			
Leu	Ser	Ala	Gln	Ser	Pro	Ile	Gly	Lys	Cys	Gly	Glu	Pro	Asn	Arg	Cys
785				790						795				800	
Ala	Pro	His	Leu	Glu	Trp	Asn	Pro	Asp	Leu	Asp	Cys	Ser	Cys	Arg	Asp
		805						810						815	
Gly	Glu	Lys	Cys	Ala	His	His	Ser	His	His	Phe	Ser	Leu	Asp	Ile	Asp
		820						825						830	
Val	Gly	Cys	Thr	Asp	Leu	Asn	Glu	Asp	Leu	Gly	Val	Trp	Val	Ile	Phe
835						840						845			
Lys	Ile	Lys	Thr	Gln	Asp	Gly	His	Ala	Arg	Leu	Gly	Asn	Leu	Glu	Phe
850						855						860			
Leu	Glu	Glu	Lys	Pro	Leu	Val	Gly	Glu	Ala	Leu	Ala	Arg	Val	Lys	Arg
865				870						875				880	
Ala	Glu														

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His Ala Ala Asp Lys Arg Val His Ser Ile Arg Glu Ala Tyr Leu Pro
 930 935 940
 Glu Leu Ser Val Ile Pro Gly Val Asn Ala Ala Ile Phe Glu Glu Leu
 945 950 955 960
 Glu Gly Arg Ile Phe Thr Ala Phe Ser Leu Tyr Asp Ala Arg Asn Val
 965 970 975
 Ile Lys Asn Gly Asp Phe Asn Asn Gly Leu Ser Cys Trp Asn Val Lys
 980 985 990
 Gly His Val Asp Val Glu Glu Gln Asn Asn Gln Arg Ser Val Leu Val
 995 1000 1005
 Val Pro Glu Trp Glu Ala Glu Val Ser Gln Glu Val Arg Val Cys
 1010 1015 1020
 Pro Gly Arg Gly Tyr Ile Leu Arg Val Thr Ala Tyr Lys Glu Gly
 1025 1030 1035
 Tyr Gly Glu Gly Cys Val Thr Ile His Glu Ile Glu Asn Asn Thr
 1040 1045 1050
 Asp Glu Leu Lys Phe Ser Asn Cys Val Glu Glu Glu Ile Tyr Pro
 1055 1060 1065
 Asn Asn Thr Val Thr Cys Asn Asp Tyr Thr Val Asn Gln Glu Glu
 1070 1075 1080
 Tyr Gly Gly Ala Tyr Thr Ser Arg Asn Arg Gly Tyr Asn Glu Ala
 1085 1090 1095
 Pro Ser Val Pro Ala Asp Tyr Ala Ser Val Tyr Glu Glu Lys Ser
 1100 1105 1110
 Tyr Thr Asp Gly Arg Arg Glu Asn Pro Cys Glu Phe Asn Arg Gly
 1115 1120 1125
 Tyr Arg Asp Tyr Thr Pro Leu Pro Val Gly Tyr Val Thr Lys Glu
 1130 1135 1140
 Leu Glu Tyr Phe Pro Glu Thr Asp Lys Val Trp Ile Glu Ile Gly
 1145 1150 1155
 Glu Thr Glu Gly Thr Phe Ile Val Asp Ser Val Glu Leu Leu Leu
 1160 1165 1170
 Met Glu Glu
 1175

<210> SEQ ID NO 14

<211> LENGTH: 633

<212> TYPE: PRT

<213> ORGANISM: Bacillus thuringiensis

<220> FEATURE:

<223> OTHER INFORMATION: Description of sequence: protein Cry2Aa from Fig. 27

<400> SEQUENCE: 14

Met Asn Asn Val Leu Asn Ser Gly Arg Thr Thr Ile Cys Asp Ala Tyr
 1 5 10 15
 Asn Val Val Ala His Asp Pro Phe Ser Phe Glu His Lys Ser Leu Asp
 20 25 30
 Thr Ile Gln Lys Glu Trp Met Glu Trp Lys Arg Thr Asp His Ser Leu
 35 40 45
 Tyr Val Ala Pro Val Val Gly Thr Val Ser Ser Phe Leu Leu Lys Lys
 50 55 60
 Val Gly Ser Leu Ile Gly Lys Arg Ile Leu Ser Glu Leu Trp Gly Ile
 65 70 75 80
 Ile Phe Pro Ser Gly Ser Thr Asn Leu Met Gln Asp Ile Leu Arg Glu
 85 90 95

Thr	Glu		Gln	Phe	Leu	Asn	Gln	Arg	Leu	Asn	Thr	Asp	Thr	Leu	Ala	Arg
				100					105					110		
Val	Asn	Ala	Glu	Leu	Ile	Gly	Leu	Gln	Ala	Asn	Ile	Arg	Glu	Phe	Asn	
		115					120					125				
Gln	Gln	Val	Asp	Asn	Phe	Leu	Asn	Pro	Thr	Gln	Asn	Pro	Val	Pro	Leu	
		130				135					140					
Ser	Ile	Thr	Ser	Ser	Val	Asn	Thr	Met	Gln	Gln	Leu	Phe	Leu	Asn	Arg	
145					150					155					160	
Leu	Pro	Gln	Phe	Gln	Ile	Gln	Gly	Tyr	Gln	Leu	Leu	Leu	Leu	Pro	Leu	
				165				170						175		
Phe	Ala	Gln	Ala	Ala	Asn	Met	His	Leu	Ser	Phe	Ile	Arg	Asp	Val	Ile	
			180					185					190			
Leu	Asn	Ala	Asp	Glu	Trp	Gly	Ile	Ser	Ala	Ala	Thr	Leu	Arg	Thr	Tyr	
		195				200						205				
Arg	Asp	Tyr	Leu	Arg	Asn	Tyr	Thr	Arg	Asp	Tyr	Ser	Asn	Tyr	Cys	Ile	
		210				215					220					
Asn	Thr	Tyr	Gln	Thr	Ala	Phe	Arg	Gly	Leu	Asn	Thr	Arg	Leu	His	Asp	
225					230					235					240	
Met	Leu	Glu	Phe	Arg	Thr	Tyr	Met	Phe	Leu	Asn	Val	Phe	Glu	Tyr	Val	
				245					250					255		
Ser	Ile	Trp	Ser	Leu	Phe	Lys	Tyr	Gln	Ser	Leu	Met	Val	Ser	Ser	Gly	
			260					265					270			
Ala	Asn	Leu	Tyr	Ala	Ser	Gly	Ser	Gly	Pro	Gln	Gln	Thr	Gln	Ser	Phe	
		275					280					285				
Thr	Ala	Gln	Asn	Trp	Pro	Phe	Leu	Tyr	Ser	Leu	Phe	Gln	Val	Asn	Ser	
		290				295						300				
Asn	Tyr	Ile	Leu	Ser	Gly	Ile	Ser	Gly	Thr	Arg	Leu	Ser	Ile	Thr	Phe	
305					310					315					320	
Pro	Asn	Ile	Gly	Gly	Leu	Pro	Gly	Ser	Thr	Thr	Thr	His	Ser	Leu	Asn	
				325					330					335		
Ser	Ala	Arg	Val	Asn	Tyr	Ser	Gly	Gly	Val	Ser	Ser	Gly	Leu	Ile	Gly	
			340					345					350			
Ala	Thr	Asn	Leu	Asn	His	Asn	Phe	Asn	Cys	Ser	Thr	Val	Leu	Pro	Pro	
		355					360					365				
Leu	Ser	Thr	Pro	Phe	Val	Arg	Ser	Trp	Leu	Asp	Ser	Gly	Thr	Asp	Arg	
		370				375					380					
Glu	Gly	Val	Ala	Thr	Ser	Thr	Asn	Trp	Gln	Thr	Glu	Ser	Phe	Gln	Thr	
385					390					395					400	
Thr	Leu	Ser	Leu	Arg	Cys	Gly	Ala	Phe	Ser	Ala	Arg	Gly	Asn	Ser	Asn	
				405					410					415		
Tyr	Phe	Pro	Asp	Tyr	Phe	Ile	Arg	Asn	Ile	Ser	Gly	Val	Pro	Leu	Val	
			420					425					430			
Ile	Arg	Asn	Glu</													

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Ser Glu Lys Phe Gly Asn Gln Gly Asp Ser Leu Arg Phe Glu Gln Ser
515 520 525

Asn Thr Thr Ala Arg Tyr Thr Leu Arg Gly Asn Gly Asn Ser Tyr Asn
530 535 540

Leu Tyr Leu Arg Val Ser Ser Ile Gly Asn Ser Thr Ile Arg Val Thr
545 550 555 560

Ile Asn Gly Arg Val Tyr Thr Val Ser Asn Val Asn Thr Thr Thr Asn
565 570 575

Asn Asp Gly Val Asn Asp Asn Gly Ala Arg Phe Ser Asp Ile Asn Ile
580 585 590

Gly Asn Ile Val Ala Ser Asp Asn Thr Asn Val Thr Leu Asp Ile Asn
595 600 605

Val Thr Leu Asn Ser Gly Thr Pro Phe Asp Leu Met Asn Ile Met Phe
610 615 620

Val Pro Thr Asn Leu Pro Pro Leu Tyr
625 630

<210> SEQ ID NO 15
 <211> LENGTH: 1180
 <212> TYPE: PRT
 <213> ORGANISM: Bacillus thuringiensis
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of sequence: protein Cry4Aa from
 Fig. 27

<400> SEQUENCE: 15

Met Asn Pro Tyr Gln Asn Lys Asn Glu Tyr Glu Thr Leu Asn Ala Ser
1 5 10 15

Gln Lys Lys Leu Asn Ile Ser Asn Asn Tyr Thr Arg Tyr Pro Ile Glu
20 25 30

Asn Ser Pro Lys Gln Leu Leu Gln Ser Thr Asn Tyr Lys Asp Trp Leu
35 40 45

Asn Met Cys Gln Gln Asn Gln Gln Tyr Gly Gly Asp Phe Glu Thr Phe
50 55 60

Ile Asp Ser Gly Glu Leu Ser Ala Tyr Thr Ile Val Val Gly Thr Val
65 70 75 80

Leu Thr Gly Phe Gly Phe Thr Thr Pro Leu Gly Leu Ala Leu Ile Gly
85 90 95

Phe Gly Thr Leu Ile Pro Val Leu Phe Pro Ala Gln Asp Gln Ser Asn
100 105 110

Thr Trp Ser Asp Phe Ile Thr Gln Thr Lys Asn Ile Ile Lys Lys Glu
115 120 125

Ile Ala Ser Thr Tyr Ile Ser Asn Ala Asn Lys Ile Leu Asn Arg Ser
130 135 140

Phe Asn Val Ile Ser Thr Tyr His Asn His Leu Lys Thr Trp Glu Asn
145 150 155 160

Asn Pro Asn Pro Gln Asn Thr Gln Asp Val Arg Thr Gln Ile Gln Leu
165 170 175

Val His Tyr His Phe Gln Asn Val Ile Pro Glu Leu Val Asn Ser Cys
180 185 190

Pro Pro Asn Pro Ser Asp Cys Asp Tyr Tyr Asn Ile Leu Val Leu Ser
195 200 205

Ser Tyr Ala Gln Ala Ala Asn Leu His Leu Thr Val Leu Asn Gln Ala
210 215 220

Val Lys Phe Glu Ala Tyr Leu Lys Asn Asn Arg Gln Phe Asp Tyr Leu
225 230 235 240

Glu	Pro	Leu	Pro	Thr	Ala	Ile	Asp	Tyr	Tyr	Pro	Val	Leu	Thr	Lys	Ala
				245					250						
Ile	Glu	Asp	Tyr	Thr	Asn	Tyr	Cys	Val	Thr	Thr	Tyr	Lys	Lys	Gly	Leu
				260					265						
Asn	Leu	Ile	Lys	Thr	Thr	Pro	Asp	Ser	Asn	Leu	Asp	Gly	Asn	Ile	Asn
				275					280						
Trp	Asn	Thr	Tyr	Asn	Thr	Tyr	Arg	Thr	Lys	Met	Thr	Thr	Ala	Val	Leu
				290					295						
Asp	Leu	Val	Ala	Leu	Phe	Pro	Asn	Tyr	Asp	Val	Gly	Lys	Tyr	Pro	Ile
				305					310						
Gly	Val	Gln	Ser	Glu	Leu	Thr	Arg	Glu	Ile	Tyr	Gln	Val	Leu	Asn	Phe
				325					330						
Glu	Glu	Ser	Pro	Tyr	Lys	Tyr	Tyr	Asp	Phe	Gln	Tyr	Gln	Glu	Asp	Ser
				340					345						
Leu	Thr	Arg	Arg	Pro	His	Leu	Phe	Thr	Trp	Leu	Asp	Ser	Leu	Asn	Phe
				355					360						
Tyr	Glu	Lys	Ala	Gln	Thr	Thr	Pro	Asn	Asn	Phe	Phe	Thr	Ser	His	Tyr
				370					375						
Asn	Met	Phe	His	Tyr	Thr	Leu	Asp	Asn	Ile	Ser	Gln	Lys	Ser	Ser	Val
				385					390						
Phe	Gly	Asn	His	Asn	Val	Thr	Asp	Lys	Leu	Lys	Ser	Leu	Gly	Leu	Ala
				405					410						
Thr	Asn	Ile	Tyr	Ile	Phe	Leu	Leu	Asn	Val	Ile	Ser	Leu	Asp	Asn	Lys
				420					425						
Tyr	Leu	Asn	Asp	Tyr	Asn	Asn	Ile	Ser	Lys	Met	Asp	Phe	Phe	Ile	Thr
				435					440						
Asn	Gly	Thr	Arg	Leu	Leu	Glu	Lys	Glu	Leu	Thr	Ala	Gly	Ser	Gly	Gln
				450					455						
Ile	Thr	Tyr	Asp	Val	Asn	Lys	Asn	Ile	Phe	Gly	Leu	Pro	Ile	Leu	Lys
				465					470						
Arg	Arg	Glu	Asn	Gln	Gly	Asn	Pro	Thr	Leu	Phe	Pro	Thr	Tyr	Asp	Asn
				485					490						
Tyr	Ser	His	Ile	Leu	Ser	Phe	Ile	Lys	Ser	Leu	Ser	Ile	Pro	Ala	Thr
				500					505						
Tyr	Lys	Thr	Gln	Val	Tyr	Thr	Phe	Ala	Trp	Thr	His	Ser	Ser	Val	Asp
				515					520						
Pro	Lys	Asn	Thr	Ile	Tyr	Thr	His	Leu	Thr	Thr	Gln	Ile	Pro	Ala	Val
				530					535						
Lys	Ala	Asn	Ser	Leu	Gly	Thr	Ala	Ser	Lys	Val	Val	Gln	Gly	Pro	Gly
				545					550						
His	Thr	Gly	Gly	Asp	Leu	Ile	Asp	Phe	Lys	Asp	His	Phe	Lys	Ile	Thr
				565					570						
Cys	Gln	His	Ser	Asn	Phe	Gln	Gln	Ser	Tyr	Phe	Ile	Arg	Ile	Arg	Tyr
				580					585						
Ala	Ser	Asn	Gly	Ser	Ala	Asn	Thr	Arg	Ala	Val	Ile	Asn	Leu	Ser	Ile
				595					600						
Pro	Gly	Val	Ala	Glu	Leu	Gly	Met	Ala	Leu	Asn	Pro	Thr	Phe	Ser	Gly
				610					615						
Thr	Asp	Tyr	Thr	Asn	Leu	Lys	Tyr	Lys	Asp	Phe	Gln	Tyr	Leu	Glu	Phe
				625					630						
Ser	Asn	Glu	Val	Lys	Phe	Ala	Pro	Asn	Gln	Asn	Ile	Ser	Leu	Val	Phe
				645					650						

Asn	Arg	Ser	Asp	Val	Tyr	Thr	Asn	Thr	Val	Leu	Ile	Asp	Lys	Ile	
			660						665				670		
Glu	Phe	Leu	Pro	Ile	Thr	Arg	Ser	Ile	Arg	Glu	Asp	Arg	Glu	Lys	Gln
		675					680					685			
Lys	Leu	Glu	Thr	Val	Gln	Gln	Ile	Ile	Asn	Thr	Phe	Tyr	Ala	Asn	Pro
		690					695					700			
Ile	Lys	Asn	Thr	Leu	Gln	Ser	Glu	Leu	Thr	Asp	Tyr	Asp	Ile	Asp	Gln
705					710					715					720
Ala	Ala	Asn	Leu	Val	Glu	Cys	Ile	Ser	Glu	Glu	Leu	Tyr	Pro	Lys	Glu
				725					730					735	
Lys	Met	Leu	Leu	Leu	Asp	Glu	Val	Lys	Asn	Ala	Lys	Gln	Leu	Ser	Gln
			740					745					750		
Ser	Arg	Asn	Val	Leu	Gln	Asn	Gly	Asp	Phe	Glu	Ser	Ala	Thr	Leu	Gly
		755					760					765			
Trp	Thr	Thr	Ser	Asp	Asn	Ile	Thr	Ile	Gln	Glu	Asp	Asp	Pro	Ile	Phe
	770					775						780			
Lys	Gly	His	Tyr	Leu	His	Met	Ser	Gly	Ala	Arg	Asp	Ile	Asp	Gly	Thr
785					790					795					800
Ile	Phe	Pro	Thr	Tyr	Ile	Phe	Gln	Lys	Ile	Asp	Glu	Ser	Lys	Leu	Lys
				805					810					815	
Pro	Tyr	Thr	Arg	Tyr	Leu	Val	Arg	Gly	Phe	Val	Gly	Ser	Ser	Lys	Asp
			820					825					830		
Val	Glu	Leu	Val	Val	Ser	Arg	Tyr	Gly	Glu	Glu	Ile	Asp	Ala	Ile	Met
		835					840					845			
Asn	Val	Pro	Ala	Asp	Leu	Asn	Tyr	Leu	Tyr	Pro	Ser	Thr	Phe	Asp	Cys
		850				855					860				
Glu	Gly	Ser	Asn	Arg	Cys	Glu	Thr	Ser	Ala	Val	Pro	Ala	Asn	Ile	Gly
865					870					875					880
Asn	Thr	Ser	Asp	Met	Leu	Tyr	Ser	Cys	Gln	Tyr	Asp	Thr	Gly	Lys	Lys
				885					890					895	
His	Val	Val	Cys	Gln	Asp	Ser	His	Gln	Phe	Ser	Phe	Thr	Ile	Asp	Thr
			900					905					910		
Gly	Ala	Leu	Asp	Thr	Asn	Glu	Asn	Ile	Gly	Val	Trp	Val	Met	Phe	Lys
		915					920					925			
Ile	Ser	Ser	Pro	Asp	Gly	Tyr	Ala	Ser	Leu	Asp	Asn	Leu	Glu	Val	Ile
						935					940				
Glu	Glu	Gly	Pro	Ile	Asp	Gly	Glu	Ala	Leu	Ser	Arg	Val	Lys	His	Met
945					950					955					960
Glu	Lys	Lys	Trp	Asn	Asp	Gln	Met	Glu	Ala	Lys	Arg	Ser	Glu	Thr	Gln
				965					970					975	
Gln	Ala	Tyr	Asp	Val	Ala	Lys	Gln	Ala	Ile	Asp	Ala	Leu	Phe	Thr	Asn
			980					985					990		
Val	Gln	Asp	Glu	Ala	Leu	Gln	Phe	Asp	Thr	Thr	Leu	Ala	Gln	Ile	Gln
		995					1000					1005			
Tyr	Ala	Glu													

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1070	1075	1080
Val Leu Val Leu Ser Asn Trp Ser Ala Gly Val Ser Gln Asn Val		
1085	1090	1095
His Leu Gln His Asn His Gly Tyr Val Leu Arg Val Ile Ala Lys		
1100	1105	1110
Lys Glu Gly Pro Gly Asn Gly Tyr Val Thr Leu Met Asp Cys Glu		
1115	1120	1125
Glu Asn Gln Glu Lys Leu Thr Phe Thr Ser Cys Glu Glu Gly Tyr		
1130	1135	1140
Ile Thr Lys Thr Val Asp Val Phe Pro Asp Thr Asp Arg Val Arg		
1145	1150	1155
Ile Glu Ile Gly Glu Thr Glu Gly Ser Phe Tyr Ile Glu Ser Ile		
1160	1165	1170
Glu Leu Ile Cys Met Asn Glu		
1175	1180	

<210> SEQ ID NO 16
 <211> LENGTH: 1245
 <212> TYPE: PRT
 <213> ORGANISM: *Bacillus thuringiensis*
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of sequence: protein Cry5B from
 Fig. 27

<400> SEQUENCE: 16

Met Ala Thr Ile Asn Glu Leu Tyr Pro Val Pro Tyr Asn Val Leu Ala			
1	5	10	15
His Pro Ile Lys Glu Val Asp Asp Pro Tyr Ser Trp Ser Asn Leu Leu			
	20	25	30
Lys Gly Ile Gln Glu Gly Trp Glu Glu Trp Gly Lys Thr Gly Gln Lys			
	35	40	45
Lys Leu Phe Glu Asp His Leu Thr Ile Ala Trp Asn Leu Tyr Lys Thr			
	50	55	60
Gly Lys Leu Asp Tyr Phe Ala Leu Thr Lys Ala Ser Ile Ser Leu Ile			
	65	70	75
Gly Phe Ile Pro Gly Ala Glu Ala Ala Val Pro Phe Ile Asn Met Phe			
	85	90	95
Val Asp Phe Val Trp Pro Lys Leu Phe Gly Ala Asn Thr Glu Gly Lys			
	100	105	110
Asp Gln Gln Leu Phe Asn Ala Ile Met Asp Ala Val Asn Lys Met Val			
	115	120	125
Asp Asn Lys Phe Leu Arg Tyr Asn Leu Ser Thr Leu Asn Lys Thr Ile			
	130	135	140
Glu Gly Leu Gln Gly Asn Leu Gly Leu Phe Gln Asn Pro Ile Gln Val			
	145	150	155
Ala Ile Cys Gln Gly Ser Thr Pro Glu Arg Val Ile Phe Asp Gln Asn			
	165	170	175
Cys Thr Pro Cys Asn Pro Asn Gln Pro Cys Lys Asp Asp Leu Asp Arg			
	180	185	190
Val Ala Ser Arg Phe Asp Thr Ala Asn Ser Gln Phe Thr Gln His Leu			
	195	200	205
Pro Glu Phe Lys Asn Pro Trp Ser Asp Glu Asn Ser Thr Gln Glu Phe			
	210	215	220
Lys Arg Thr Ser Val Glu Leu Thr Leu Pro Leu Tyr Thr Thr Val Ala			
	225	230	235
			240

Thr	Leu	His	Leu	Leu	Leu	Tyr	Lys	Gly	Tyr	Ile	Glu	Phe	Met	Thr	Lys	
				245					250					255		
Trp	Asn	Phe	His	Asn	Ala	Gln	Tyr	Leu	Asn	Asn	Leu	Lys	Val	Glu	Leu	
				260					265					270		
Gln	Gln	Leu	Ile	His	Ser	Tyr	Ser	Glu	Thr	Val	Arg	Thr	Ser	Phe	Leu	
				275					280					285		
Gln	Phe	Leu	Pro	Thr	Leu	Asn	Asn	Arg	Ser	Lys	Ser	Ser	Val	Asn	Ala	
				290					295					300		
Tyr	Asn	Gly	Tyr	Val	Arg	Asn	Met	Thr	Val	Asn	Cys	Leu	Asp	Ile	Ala	
				305					310					315		
Ala	Thr	Trp	Pro	Thr	Phe	Asp	Thr	His	Asn	Tyr	His	Gln	Gly	Arg	Lys	
				325					330					335		
Leu	Asp	Leu	Thr	Arg	Ile	Ile	Leu	Ser	Asp	Thr	Ala	Gly	Pro	Ile	Lys	
				340					345					350		
Glu	Tyr	Thr	Thr	Gly	Asp	Lys	Thr	Ser	Gly	Pro	Gln	His	Ser	Asn	Ile	
				355					360					365		
Thr	Pro	Asn	Asn	Ile	Leu	Asp	Thr	Pro	Ser	Pro	Thr	Tyr	Gln	His	Ser	
				370					375					380		
Phe	Val	Ser	Val	Asp	Ser	Ile	Val	Tyr	Ser	Arg	Lys	Glu	Leu	Gln	Gln	
				385					390					395		
Leu	Asp	Ile	Ala	Thr	Tyr	Ser	Thr	Asn	Asn	Ser	Asn	Asn	Cys	His	Pro	
				405					410					415		
Tyr	Gly	Leu	Arg	Leu	Ser	Tyr	Thr	Asp	Gly	Ser	Arg	Tyr	Asp	Tyr	Gly	
				420					425					430		
Asp	Asn	Gln	Pro	Asp	Phe	Thr	Thr	Ser	Asn	Asn	Asn	Tyr	Cys	His	Asn	
				435					440					445		
Ser	Tyr	Thr	Ala	Pro	Ile	Thr	Leu	Val	Asn	Ala	Arg	His	Leu	Tyr	Asn	
				450					455					460		
Ala	Lys	Gly	Ser	Leu	Gln	Asn	Val	Glu	Ser	Leu	Val	Val	Ser	Thr	Val	
				465					470					475		
Asn	Gly	Gly	Ser	Gly	Ser	Cys	Ile	Cys	Asp	Ala	Trp	Ile	Asn	Tyr	Leu	
				485					490					495		
Arg	Pro	Pro	Gln	Thr	Ser	Lys	Asn	Glu	Ser	Arg	Pro	Asp	Gln	Lys	Ile	
				500					505					510		
Asn	Val	Leu	Tyr	Pro	Ile	Thr	Glu	Thr	Val	Asn	Lys	Gly	Thr	Gly	Gly	
				515					520					525		
Asn	Leu	Gly	Val	Ile	Ser	Ala	Tyr	Val	Pro	Met	Glu	Leu	Val	Pro	Glu	
				530					535					540		
Asn	Val	Ile	Gly	Asp	Val	Asn	Ala	Asp	Thr	Lys	Leu	Pro	Leu	Thr	Gln	
				545					550					555		
Leu	Lys	Gly	Phe	Pro	Phe	Glu	Lys	Tyr	Gly	Ser	Glu	Tyr	Asn	Asn	Arg	
				565					570					575		
Gly	Ile	Ser	Leu	Val	Arg	Glu	Trp	Ile	Asn	Gly	Asn	Asn	Ala	Val	Lys	
				580					585					590		
Leu	Ser	Asn	Ser	Gln	Ser	Val	Gly	Ile	Gln	Ile	Thr	Asn	Gln	Thr	Lys	
				595					600					605		
Gln	Lys	Tyr	Glu	Ile	Arg	Cys	Arg	Tyr	Ala	Ser	Lys	Gly	Asp	Asn	Asn	
				610					615					620		
Val	Tyr	Phe	Asn	Val	Asp	Leu	Ser	Glu	Asn	Pro	Phe	Arg	Asn	Ser	Ile	
				625					630					635		
Ser	Phe	Gly	Ser	Thr	Glu	Ser	Ser	Val	Val	Gly	Val	Gln</				

660										665										670									
Ser	Phe	Tyr	Val	His	Ile	Thr	Asn	Gln	Gly	Ser	Ser	Asp	Leu	Phe	Leu														
675										680										685									
Asp	Arg	Ile	Glu	Phe	Val	Pro	Lys	Ile	Gln	Phe	Gln	Phe	Cys	Asp	Asn														
690										695										700									
Asn	Asn	Leu	His	Cys	Asp	Cys	Asn	Asn	Pro	Val	Asp	Thr	Asp	Cys	Thr														
705										710										715									
Phe	Cys	Cys	Val	Cys	Thr	Ser	Leu	Thr	Asp	Cys	Asp	Cys	Asn	Asn	Pro														
720										725										730									
Arg	Gly	Leu	Asp	Cys	Thr	Leu	Cys	Cys	Gln	Val	Glu	Asn	Gln	Leu	Pro														
735										740										745									
Ser	Phe	Val	Thr	Leu	Thr	Asp	Leu	Gln	Asn	Ile	Thr	Thr	Gln	Val	Asn														
750										755										760									
Ala	Leu	Val	Ala	Ser	Ser	Glu	His	Asp	Thr	Leu	Ala	Thr	Asp	Val	Ser														
765										770										775									
Asp	Tyr	Glu	Ile	Glu	Glu	Val	Val	Leu	Lys	Val	Asp	Ala	Leu	Ser	Gly														
780										785										790									
Glu	Val	Phe	Gly	Lys	Glu	Lys	Lys	Ala	Leu	Arg	Lys	Leu	Val	Asn	His														
795										800										805									
Thr	Lys	Arg	Leu	Ser	Lys	Ala	Arg	Asn	Leu	Leu	Ile	Gly	Gly	Asn	Phe														
810										815										820									
Asp	Asn	Leu	Asp	Ala	Trp	Tyr	Arg	Gly	Arg	Asn	Val	Val	Asn	Val	Ser														
825										830										835									
Asp	His	Glu	Leu	Phe	Lys	Ser	Asp	His	Val	Leu	Leu	Pro	Pro	Pro	Thr														
840										845										850									
Leu	Tyr	Ser	Ser	Tyr	Met	Phe	Gln	Lys	Val	Glu	Glu	Ser	Lys	Leu	Lys														
855										860										865									
Ala	Asn	Thr	Arg	Tyr	Thr	Val	Ser	Gly	Phe	Ile	Ala	His	Ala	Glu	Asp														
870										875										880									
Leu	Glu	Ile	Val	Val	Ser	Arg	Tyr	Gly	Gln	Glu	Val	Lys	Lys	Val	Val														
885										890										895									
Gln	Val	Pro	Tyr	Gly	Glu	Ala	Phe	Pro	Leu	Thr	Ser	Arg	Gly	Ala	Ile														
900										905										910									
Cys	Cys	Pro	Pro	Arg	Ser	Thr	Cys	Asn	Gly	Lys	Pro	Ala	Asp	Pro	His														
915										920										925									
Phe	Phe	Ser	Tyr	Ser	Ile	Asp	Val	Gly	Thr	Leu	Asp	Val	Glu	Ala	Asn														
930										935										940									
Pro	Gly	Ile	Glu	Leu	Gly	Leu	Arg	Ile	Val	Glu	Arg	Thr	Gly	Met	Ala														
945										950										955									
Arg	Val	Ser	Asn	Leu	Glu	Ile	Arg	Glu	Asp	Arg	Pro	Leu	Lys	Lys	Asn														
960										965										970									
Glu	Leu	Arg	Asn	Val	Gln	Arg	Ala	Ala	Arg	Asn	Trp	Arg	Thr	Ala	Tyr														
975										980										985									
Asp	Gln	Glu	Arg	Ala	Glu	Val	Thr	Ala	Leu	Ile	Gln	Pro	Val	Leu															
990										995										1000									
Asn	Gln	Ile	Asn	Ala	Leu	Tyr	Glu	Asn	Glu	Asp	Trp	Asn	Arg	Ala															
1005										1010										1015									
Ile	Arg	Ser	Gly	Val	Ser	Tyr	His	Asp	Leu	Glu	Ala	Ile	Val	Leu															
1020										1025										1030									
Pro	Thr	Leu	Pro	Lys	Leu	Asn	His	Trp	Phe	Met	Ser	Asp	Met	Leu															
1035										1040										1045									
Gly	Glu	Gln	Gly	Ser																									

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Arg Ala Tyr Thr Gln Leu Glu Glu Ser Thr Ile Leu His Asn Gly
 1085 1090 1095

His Phe Thr Thr Asp Ala Ala Asn Trp Thr Ile Glu Gly Asp Ala
 1100 1105 1110

His His Ala Ile Leu Glu Asp Gly Arg Arg Val Leu Arg Leu Pro
 1115 1120 1125

Asp Trp Ser Ser Ser Val Ser Gln Thr Ile Glu Ile Glu Asn Phe
 1130 1135 1140

Asp Pro Asp Lys Glu Tyr Gln Leu Val Phe His Ala Gln Gly Glu
 1145 1150 1155

Gly Thr Val Ser Leu Gln His Gly Glu Glu Gly Glu Tyr Val Glu
 1160 1165 1170

Thr His Pro His Lys Ser Ala Asn Phe Thr Thr Ser His Arg Gln
 1175 1180 1185

Gly Val Thr Phe Glu Thr Asn Lys Val Thr Val Glu Ile Thr Ser
 1190 1195 1200

Glu Asp Gly Glu Phe Leu Val Asp His Ile Ala Leu Val Glu Ala
 1205 1210 1215

Pro Leu Pro Thr Asp Asp Gln Ser Ser Asp Gly Asn Thr Thr Ser
 1220 1225 1230

Asn Thr Asn Ser Asn Thr Ser Met Asn Asn Asn Gln
 1235 1240 1245

<210> SEQ ID NO 17

<211> LENGTH: 609

<212> TYPE: PRT

<213> ORGANISM: Bacillus thuringiensis

<220> FEATURE:

<223> OTHER INFORMATION: Description of sequence: protein Cry7Ca1 from Fig. 27

<400> SEQUENCE: 17

Met Arg Gly Ser His His His His His Gly Ser Met Ile Phe Ser
 1 5 10 15

Ser Ile Ser Ile Ile Arg Thr Phe Met Gly Phe Ala Gly His Gly Thr
 20 25 30

Ala Gly Gly Ile Ile Gly Leu Phe Thr Glu Val Leu Arg Leu Leu Trp
 35 40 45

Pro Asn Lys Gln Asn Asp Leu Trp Glu Ser Phe Met Asn Glu Val Glu
 50 55 60

Ala Leu Ile Asn Gln Glu Ile Thr Glu Ala Val Val Ser Lys Ala Leu
 65 70 75 80

Ser Glu Leu Glu Gly Leu Arg Asn Ala Leu Glu Gly Tyr Thr Ser Ala
 85 90 95

Leu Glu Ala Trp Gln Asn Asn Arg Ser Asp Lys Leu Lys Gln Leu Leu
 100 105 110

Val Tyr Glu Arg Phe Val Ser Thr Glu Asn Leu Phe Lys Phe Ala Met
 115 120 125

Pro Ser Phe Arg Ser Val Gly Phe Glu Gly Pro Leu Leu Thr Val Tyr
 130 135 140

Ala Gln Ala Ala Asn Leu His Leu Phe Leu Leu Lys Asn Ala Glu Leu
 145 150 155 160

Phe Gly Ala Glu Trp Gly Met Gln Gln Tyr Glu Ile Asp Leu Phe Tyr
 165 170 175

Asn Glu Gln Lys Gly Tyr Val Glu Glu Tyr Thr Asp His Cys Val Lys

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180							185					190				
Trp	Tyr	Lys	Glu	Gly	Leu	Asn	Lys	Leu	Lys	Asn	Ala	Ser	Gly	Val	Lys	
		195					200					205				
Gly	Lys	Val	Trp	Glu	Asn	Tyr	Asn	Arg	Phe	Arg	Arg	Glu	Met	Thr	Ile	
	210					215					220					
Met	Val	Leu	Asp	Leu	Leu	Pro	Leu	Phe	Pro	Ile	Tyr	Asp	Ala	Arg	Thr	
225					230					235					240	
Tyr	Pro	Met	Glu	Thr	Val	Thr	Glu	Leu	Thr	Arg	Gln	Ile	Phe	Thr	Asp	
				245					250					255		
Pro	Ile	Gly	Leu	Thr	Gly	Ile	Asn	Glu	Thr	Lys	Tyr	Pro	Asp	Trp	Tyr	
		260						265					270			
Gly	Ala	Ala	Ser	Ser	Glu	Phe	Val	Leu	Ile	Glu	Asn	Arg	Ala	Ile	Pro	
		275						280				285				
Lys	Pro	Gly	Leu	Phe	Gln	Trp	Leu	Thr	Lys	Ile	Asn	Val	Arg	Ala	Arg	
	290					295					300					
Val	Val	Glu	Pro	Asn	Asp	Arg	Phe	Ala	Ile	Trp	Thr	Gly	His	Ser	Val	
305					310					315					320	
Val	Thr	Gln	Tyr	Thr	Lys	Ser	Thr	Thr	Glu	Asn	Thr	Phe	Asn	Tyr	Gly	
				325					330					335		
Thr	Ser	Ser	Gly	Ser	Thr	Leu	Ser	His	Thr	Phe	Asp	Leu	Leu	Ser	Lys	
			340					345					350			
Asp	Ile	Tyr	Gln	Thr	Tyr	Ser	Ile	Ala	Ala	Ala	Asn	Lys	Ser	Ala	Thr	
	355						360					365				
Trp	Tyr	Gln	Ala	Val	Pro	Leu	Leu	Arg	Leu	Tyr	Gly	Ile	Asn	Ser	Ser	
	370					375					380					
Asn	Val	Leu	Ser	Glu	Asp	Ala	Phe	Ser	Phe	Ser	Asn	Asn	Ile	Pro	Ser	
385					390					395					400	
Ser	Lys	Cys	Lys	Ser	Thr	Tyr	Ser	Ser	Asp	Gln	Leu	Pro	Ile	Glu	Leu	
			405						410					415		
Leu	Asp	Glu	Pro	Ile	Tyr	Gly	Asp	Leu	Glu	Glu	Tyr	Gly	His	Arg	Leu	
		420					425						430			
Ser	Tyr	Val	Ser	Glu	Ile	Phe	Lys	Glu	Thr	Gly	Ser	Gly	Thr	Ile	Pro	
	435						440					445				
Val	Leu	Gly	Trp	Thr	His	Val	Ser	Val	Arg	Pro	Asp	Asn	Lys	Leu	Tyr	
	450					455					460					
Pro	Asp	Lys	Ile	Thr	Gln	Ile	Pro	Ala	Val	Lys	Ala	Phe	Glu	Thr	Asn	
465					470					475					480	
Thr	Ala	Gly	Val	Glu	Ile	Ile	Asp	Ser	Ala	Ser	Thr	Gly	Gly	Pro	Ile	
			485						490					495		
Leu	Lys	Ile	Val	Asn	Asn	Asn	Leu	Pro	Ser	Asn	Gln	Val	Phe	Arg	Met	
		500						505					510			
Arg	Leu	Ser	Phe	Ser	Glu	Pro	Gln	Lys	Ile	Lys	Val	Arg	Val	Arg	Tyr	
	515						520					525				
Ala	Ala	Thr	Gly	Asp	Gly	Val	Met	Ser	Phe	Ser	Gly	Ile	Ala	His	Asp	
	530					535					540					
Glu	Tyr	Phe	Thr	Ala	Thr	Met	Lys	Glu	Gly	Glu	Ala	Leu	Lys	Tyr	Ser	
545					550					555					560	
Tyr	Leu	Thr	Met	Gly	Asn	Asp	Tyr	Ala	Gly	Thr	Ala	Ala	Glu	Leu	Ser	
			565						570					575		
Met	Leu	Tyr	Ile	Ile	Lys	Ala	Asn	Thr	Ser	Asn	Cys	Thr	Ile	Tyr	Ile	
		580						585					590			
Asp	Lys	Ile	Glu	Phe	Ile	Pro	Val	Val	Asp	Leu	Gln	Pro	Ser	Leu	Ile	
		595					600					605				

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Ser

<210> SEQ ID NO 18
 <211> LENGTH: 1164
 <212> TYPE: PRT
 <213> ORGANISM: Bacillus thuringiensis
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of sequence: protein Cry8Ea1 from
 Fig. 27

<400> SEQUENCE: 18

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Met Ser Pro Asn Asn Gln Asn Glu Tyr Glu Ile Ile Asp Met Ala Pro
1          5          10          15

Ser Thr Ser Val Ser Asn Asp Ser Asn Arg Tyr Pro Phe Ala Ser Asp
20          25          30

Pro Thr Asn Ala Leu Gln Asn Met Asn Tyr Lys Glu Tyr Leu Arg Met
35          40          45

Ser Glu Gly Tyr Asp Ser Glu Tyr Ser Gly Ser Pro Glu Val Leu Ile
50          55          60

Ser Glu Arg Asp Ala Val Lys Thr Ala Ile Ser Leu Val Gly Thr Ile
65          70          75          80

Leu Gly Lys Leu Gly Val Pro Leu Val Gly Pro Ile Val Ser Leu Tyr
85          90          95

Ser Thr Leu Ile Asp Val Leu Trp Pro Gly Gly Lys Ser Gln Trp Glu
100         105         110

Ile Phe Met Glu Gln Val Glu Ala Leu Ile Asn Gln Lys Ile Ala Glu
115         120         125

Tyr Ala Arg Ala Lys Ala Leu Ala Glu Leu Glu Gly Leu Gly Asn Asn
130         135         140

Tyr Gln Leu Tyr Leu Thr Ala Leu Glu Glu Trp Gln Glu Asn Pro Ser
145         150         155         160

Ser Thr Arg Val Leu Arg Asp Val Arg Asn Arg Phe Glu Ile Leu Asp
165         170         175

Ser Leu Phe Thr Gln Tyr Met Pro Ser Phe Arg Val Thr Gly Tyr Glu
180         185         190

Val Pro Leu Leu Ser Val Tyr Ala Gln Ala Ala Asn Leu His Leu Leu
195         200         205

Leu Leu Lys Asp Ala Ser Ile Phe Gly Glu Glu Trp Gly Phe Ser Thr
210         215         220

Thr Ala Ile Asn Asn Tyr Tyr Asn Arg Gln Met Ser Leu Ile Ala Gln
225         230         235         240

Tyr Ser Asp His Cys Val Gln Trp Tyr Arg Thr Gly Leu Asp Arg Leu
245         250         255

Lys Gly Ser Asn Ala Lys Gln Trp Val Glu Tyr Asn Arg Phe Arg Arg
260         265         270

Glu Met Thr Leu Ser Val Leu Asp Ile Met Thr Leu Phe Pro Met Tyr
275         280         285

Asp Met Arg Thr Tyr Pro Met Glu Thr Lys Ala Gln Leu Thr Arg Glu
290         295         300

Val Tyr Thr Asp Pro Ile Gly Ala Ile Gly Ala Gln Gly Ser Trp Tyr
305         310         315         320

Asp Ser Ala Pro Ser Phe Asn Thr Leu Glu Ser Thr Phe Ile Arg Gly
325         330         335

Lys His Leu Phe Asp Phe Ile Thr Arg Leu Ser Ile Tyr Thr Gly Arg
340         345         350
  
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Ser	Ser	Phe	Ser	Ala	Ser	Asn	Tyr	Leu	Lys	Lys	Trp	Ile	Gly	His	Gln	
		355					360					365				
Ile	Ser	Ser	Gln	Pro	Ile	Gly	Gly	Ser	Ile	Gln	Thr	Gln	Thr	Tyr	Gly	
	370					375					380					
Thr	Thr	Ser	Gly	Ser	Ser	Val	Ile	Ala	Thr	Gln	Gln	Ile	Gly	Phe	Thr	
385					390					395					400	
Gly	Phe	Asp	Val	Tyr	Lys	Thr	Leu	Ser	Thr	Ala	Gly	Val	Leu	Phe	Ala	
			405					410						415		
Tyr	Thr	Ser	Lys	Tyr	Tyr	Gly	Val	Ser	Lys	Val	Val	Phe	Asp	Ala	Ile	
			420					425					430			
Tyr	Pro	Asp	Asn	Lys	Tyr	Lys	Thr	Thr	Phe	Thr	Tyr	Asn	Pro	Gly	Ser	
		435					440					445				
Glu	Gly	Ile	Gly	Ala	Gln	Glu	Lys	Asp	Ser	Glu	Val	Glu	Leu	Pro	Pro	
	450					455					460					
Glu	Thr	Leu	Asp	Gln	Pro	Asn	Tyr	Glu	Ala	Tyr	Ser	His	Arg	Leu	Asn	
465				470						475					480	
Tyr	Val	Thr	Phe	Ile	Arg	Asn	Pro	Asp	Val	Pro	Val	Phe	Ser	Trp	Thr	
			485						490					495		
His	Arg	Ser	Ala	Asp	Arg	Thr	Asn	Thr	Val	Tyr	Ser	Asp	Lys	Ile	Thr	
			500				505						510			
Gln	Ile	Pro	Val	Val	Lys	Ala	Ser	Asp	Gly	Pro	Lys	Pro	Ser	Ala	Asn	
		515					520					525				
Glu	Val	Gly	His	Tyr	Leu	Gly	Gly	Asp	Pro	Ile	Ser	Phe	Asn	Ser	Ser	
	530					535					540					
Gly	Ser	Thr	Gly	Val	Ile	Arg	Leu	Asn	Ile	Asn	Ser	Pro	Leu	Ser	Gln	
545				550						555					560	
Lys	Tyr	Arg	Val	Arg	Ile	Arg	Tyr	Cys	Ser	Ser	Val	Asp	Phe	Asp	Leu	
			565					570						575		
Asp	Val	Val	Arg	Gly	Gly	Thr	Thr	Val	Asn	Asn	Gly	Arg	Phe	Asn	Lys	
		580					585						590			
Ser	Ala	Pro	Asn	Val	Gly	Trp	Gln	Ser	Leu	Lys	Tyr	Glu	Asn	Phe	Lys	
		595					600					605				
Phe	Ala	Ser	Phe	Ser	Thr	Pro	Phe	Thr	Phe	Asn	Gln	Ala	Gln	Asp	Thr	
	610					615					620					
Leu	Lys	Ile	Ser	Val	Arg	Asn	Phe	Ser	Ser	Ile	Val	Gly	Gly	Ser	Val	
625				630						635					640	
Val	Tyr	Ile	Asp	Arg	Ile	Glu	Leu	Ile	Pro	Val	Asn	Ala	Thr	Tyr	Glu	
			645					650						655		
Ala	Glu	Gln	Asp	Leu	Asp	Ser	Ala	Lys	Lys	Ala	Val	Asn	Thr	Leu	Phe	
		660					665						670			
Thr	Asn	Thr	Lys	Asp	Gly	Leu	Arg	Pro	Gly	Val	Thr	Asp	Tyr	Glu	Val	
	675					680						685				
Asn	Gln	Ala	Ala	Asn	Leu	Val	Glu	Cys	Leu	Ser	Asp	Asp	Leu	Tyr	Pro	
	690				695						700					
Asn	Glu	Lys	Arg	Leu	Leu	Phe	Asp	Ala	Val	Lys	Glu	Ala	Lys	Arg	Leu	
705				710						715					720	
Ser	Glu	Ala	Arg	Asn	Leu	Leu	Gln	Asp	Pro	Asp	Phe	Gln	Glu	Ile	Asn	
			725					730						735		
Gly	Glu	Asn	Gly	Trp	Thr	Ala	Ser	Thr	Gly	Ile	Glu	Val	Val	Glu	Gly	
		740						745					750			
Asp	Ala	Leu	Phe	Lys	Gly	Arg	Tyr	Leu	Arg	Leu	Pro	Gly	Ala	Arg	Glu	
	755						760						765			

[illegible]

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<210> SEQ ID NO 19
<211> LENGTH: 675
<212> TYPE: PRT
<213> ORGANISM: Bacillus thuringiensis
<220> FEATURE:
<223> OTHER INFORMATION: Description of sequence: protein Cry10Aa from
        Fig. 27

<400> SEQUENCE: 19

Met Asn Pro Tyr Gln Asn Lys Asn Glu Tyr Glu Ile Phe Asn Ala Pro
 1             5             10            15

Ser Asn Gly Phe Ser Lys Ser Asn Asn Tyr Ser Arg Tyr Pro Leu Ala
 20            25            30

Asn Lys Pro Asn Gln Pro Leu Lys Asn Thr Asn Tyr Lys Asp Trp Leu
 35            40            45

Asn Val Cys Gln Asp Asn Gln Gln Tyr Gly Asn Asn Ala Gly Asn Phe
 50            55            60

Ala Ser Ser Glu Thr Ile Val Gly Val Ser Ala Gly Ile Ile Val Val
 65            70            75            80

Gly Thr Met Leu Gly Ala Phe Ala Ala Pro Val Leu Ala Ala Gly Ile
 85            90            95

Ile Ser Phe Gly Thr Leu Leu Pro Ile Phe Trp Gln Gly Ser Asp Pro
100           105           110

Ala Asn Val Trp Gln Asp Leu Leu Asn Ile Gly Gly Arg Pro Ile Gln
115           120           125

Glu Ile Asp Lys Asn Ile Ile Asn Val Leu Thr Ser Ile Val Thr Pro
130           135           140

Ile Lys Asn Gln Leu Asp Lys Tyr Gln Glu Phe Phe Asp Lys Trp Glu
145           150           155           160

Pro Ala Arg Thr His Ala Asn Ala Lys Ala Val His Asp Leu Phe Thr
165           170           175

Thr Leu Glu Pro Ile Ile Asp Lys Asp Leu Asp Met Leu Lys Asn Asn
180           185           190

Ala Ser Tyr Arg Ile Pro Thr Leu Pro Ala Tyr Ala Gln Ile Ala Thr
195           200           205

Trp His Leu Asn Leu Leu Lys His Ala Ala Thr Tyr Tyr Asn Ile Trp
210           215           220

Leu Gln Asn Gln Gly Ile Asn Pro Ser Thr Phe Asn Ser Ser Asn Tyr
225           230           235           240

Tyr Gln Gly Tyr Leu Lys Arg Lys Ile Gln Glu Tyr Thr Asp Tyr Cys
245           250           255

Ile Gln Thr Tyr Asn Ala Gly Leu Thr Met Ile Arg Thr Asn Thr Asn
260           265           270

Ala Thr Trp Asn Met Tyr Asn Thr Tyr Arg Leu Glu Met Thr Leu Thr
275           280           285

Val Leu Asp Leu Ile Ala Ile Phe Pro Asn Tyr Asp Pro Glu Lys Tyr
290           295           300

Pro Ile Gly Val Lys Ser Glu Leu Ile Arg Glu Val Tyr Thr Asn Val
305           310           315           320

Asn Ser Asp Thr Phe Arg Thr Ile Thr Glu Leu Glu Asn Gly Leu Thr
325           330           335

Arg Asn Pro Thr Leu Phe Thr Trp Ile Asn Gln Gly Arg Phe Tyr Thr
340           345           350

Arg Asn Ser Arg Asp Ile Leu Asp Pro Tyr Asp Ile Phe Ser Phe Thr
355           360           365

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Gly Asn Gln Met Ala Phe Thr His Thr Asn Asp Asp Arg Asn Ile Ile
 370 375 380
 Trp Gly Ala Val His Gly Asn Ile Ile Ser Gln Asp Thr Ser Lys Val
 385 390 395 400
 Phe Pro Phe Tyr Arg Asn Lys Pro Ile Asp Lys Val Glu Ile Val Arg
 405 410 415
 His Arg Glu Tyr Ser Asp Ile Ile Tyr Glu Met Ile Phe Phe Ser Asn
 420 425 430
 Ser Ser Glu Val Phe Arg Tyr Ser Ser Asn Ser Thr Ile Glu Asn Asn
 435 440 445
 Tyr Lys Arg Thr Asp Ser Tyr Met Ile Pro Lys Gln Thr Trp Lys Asn
 450 455 460
 Glu Glu Tyr Gly His Thr Leu Ser Tyr Ile Lys Thr Asp Asn Tyr Ile
 465 470 475 480
 Phe Ser Val Val Arg Glu Arg Arg Arg Val Ala Phe Ser Trp Thr His
 485 490 495
 Thr Ser Val Asp Phe Gln Asn Thr Ile Asp Leu Asp Asn Ile Thr Gln
 500 505 510
 Ile His Ala Leu Lys Ala Leu Lys Val Ser Ser Asp Ser Lys Ile Val
 515 520 525
 Lys Gly Pro Gly His Thr Gly Gly Asp Leu Val Ile Leu Lys Asp Ser
 530 535 540
 Met Asp Phe Arg Val Arg Phe Leu Lys Asn Val Ser Arg Gln Tyr Gln
 545 550 555 560
 Val Arg Ile Arg Tyr Ala Thr Asn Ala Pro Lys Thr Thr Val Phe Leu
 565 570 575
 Thr Gly Ile Asp Thr Ile Ser Val Glu Leu Pro Ser Thr Thr Ser Arg
 580 585 590
 Gln Asn Pro Asn Ala Thr Asp Leu Thr Tyr Ala Asp Phe Gly Tyr Val
 595 600 605
 Thr Phe Pro Arg Thr Val Pro Asn Lys Thr Phe Glu Gly Glu Asp Thr
 610 615 620
 Leu Leu Met Thr Leu Tyr Gly Thr Pro Asn His Ser Tyr Asn Ile Tyr
 625 630 635 640
 Ile Asp Lys Ile Glu Phe Ile Pro Ile Thr Gln Ser Val Leu Asp Tyr
 645 650 655
 Thr Glu Lys Gln Asn Ile Glu Lys Thr Gln Lys Ile Val Asn Asp Leu
 660 665 670
 Phe Val Asn
 675

<210> SEQ ID NO 20
 <211> LENGTH: 643
 <212> TYPE: PRT
 <213> ORGANISM: *Bacillus thuringiensis*
 <220> FEATURE:
 <223> OTHER INFORMATION: Description of sequence: protein Cry11Aa from
 Fig. 27

<400> SEQUENCE: 20

Met Glu Asp Ser Ser Leu Asp Thr Leu Ser Ile Val Asn Glu Thr Asp
 1 5 10 15
 Phe Pro Leu Tyr Asn Asn Tyr Thr Glu Pro Thr Ile Ala Pro Ala Leu
 20 25 30
 Ile Ala Val Ala Pro Ile Ala Gln Tyr Leu Ala Thr Ala Ile Gly Lys
 35 40 45

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Trp	Ala	Ala	Lys	Ala	Ala	Phe	Ser	Lys	Val	Leu	Ser	Leu	Ile	Phe	Pro
50						55					60				
Gly	Ser	Gln	Pro	Ala	Thr	Met	Glu	Lys	Val	Arg	Thr	Glu	Val	Glu	Thr
65					70					75					80
Leu	Ile	Asn	Gln	Lys	Leu	Ser	Gln	Asp	Arg	Val	Asn	Ile	Leu	Asn	Ala
				85					90					95	
Glu	Tyr	Arg	Gly	Ile	Ile	Glu	Val	Ser	Asp	Val	Phe	Asp	Ala	Tyr	Ile
			100					105					110		
Lys	Gln	Pro	Gly	Phe	Thr	Pro	Ala	Thr	Ala	Lys	Gly	Tyr	Phe	Leu	Asn
		115					120					125			
Leu	Ser	Gly	Ala	Ile	Ile	Gln	Arg	Leu	Pro	Gln	Phe	Glu	Val	Gln	Thr
	130					135					140				
Tyr	Glu	Gly	Val	Ser	Ile	Ala	Leu	Phe	Thr	Gln	Met	Cys	Thr	Leu	His
145					150					155					160
Leu	Thr	Leu	Leu	Lys	Asp	Gly	Ile	Leu	Ala	Gly	Ser	Ala	Trp	Gly	Phe
				165					170					175	
Thr	Gln	Ala	Asp	Val	Asp	Ser	Phe	Ile	Lys	Leu	Phe	Asn	Gln	Lys	Val
			180					185					190		
Leu	Asp	Tyr	Arg	Thr	Arg	Leu	Met	Arg	Met	Tyr	Thr	Glu	Glu	Phe	Gly
	195					200						205			
Arg	Leu	Cys	Lys	Val	Ser	Leu	Lys	Asp	Gly	Leu	Thr	Phe	Arg	Asn	Met
	210					215					220				
Cys	Asn	Leu	Tyr	Val	Phe	Pro	Phe	Ala	Glu	Ala	Trp	Ser	Leu	Met	Arg
225					230					235					240
Tyr	Glu	Gly	Leu	Lys	Leu	Gln	Ser	Ser	Leu	Ser	Leu	Trp	Asp	Tyr	Val
			245						250					255	
Gly	Val	Ser	Ile	Pro	Val	Asn	Tyr	Asn	Glu	Trp	Gly	Gly	Leu	Val	Tyr
			260					265					270		
Lys	Leu	Leu	Met	Gly	Glu	Val	Asn	Gln	Arg	Leu	Thr	Thr	Val	Lys	Phe
	275						280					285			
Asn	Tyr	Ser	Phe	Thr	Asn	Glu	Pro	Ala	Asp	Ile	Pro	Ala	Arg	Glu	Asn
	290				295						300				
Ile	Arg	Gly	Val	His	Pro	Ile	Tyr	Asp	Pro	Ser	Ser	Gly	Leu	Thr	Gly
305					310					315					320
Trp	Ile	Gly	Asn	Gly	Arg	Thr	Asn	Asn	Phe	Asn	Phe	Ala	Asp	Asn	Asn
			325						330					335	
Gly	Asn	Glu	Ile	Met	Glu	Val	Arg	Thr	Gln	Thr	Phe	Tyr	Gln	Asn	Pro
			340					345					350		
Asn	Asn	Glu	Pro	Ile	Ala	Pro	Arg	Asp	Ile	Ile	Asn	Gln	Ile	Leu	Thr
	355					360						365			
Ala	Pro	Ala	Pro	Ala	Asp	Leu	Phe	Phe	Lys	Asn	Ala	Asp	Ile	Asn	Val
	370					375					380				
Lys	Phe	Thr	Gln	Trp	Phe	Gln	Ser	Thr	Leu	Tyr	Gly	Trp	Asn	Ile	Lys
385					390					395					400
Leu	Gly	Thr	Gln	Thr	Val	Leu	Ser	Ser	Arg	Thr	Gly	Thr	Ile	Pro	Pro
			405						410					415	
Asn	Tyr	Leu	Ala	Tyr	Asp	Gly	Tyr	Tyr	Ile	Arg	Ala	Ile	Ser	Ala	Cys
		420						425					430		
Pro	Arg	Gly	Val	Ser	Leu	Ala	Tyr	Asn	His	Asp	Leu	Thr	Thr	Leu	Thr
	435							440				445			
Tyr	Asn	Arg	Ile	Glu	Tyr	Asp	Ser	Pro	Thr	Thr	Glu	Asn	Ile	Ile	Val
450						455					460				

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Gly Phe Ala Pro Asp Asn Thr Lys Asp Phe Tyr Ser Lys Lys Ser His
 465 470 475 480
 Tyr Leu Ser Glu Thr Asn Asp Ser Tyr Val Ile Pro Ala Leu Gln Phe
 485 490 495
 Ala Glu Val Ser Asp Arg Ser Phe Leu Glu Asp Thr Pro Asp Gln Ala
 500 505 510
 Thr Asp Gly Ser Ile Lys Phe Ala Arg Thr Phe Ile Ser Asn Glu Ala
 515 520 525
 Lys Tyr Ser Ile Arg Leu Asn Thr Gly Phe Asn Thr Ala Thr Arg Tyr
 530 535 540
 Lys Leu Ile Ile Arg Val Arg Val Pro Tyr Arg Leu Pro Ala Gly Ile
 545 550 555 560
 Arg Val Gln Ser Gln Asn Ser Gly Asn Asn Arg Met Leu Gly Ser Phe
 565 570 575
 Thr Ala Asn Ala Asn Pro Glu Trp Val Asp Phe Val Thr Asp Ala Phe
 580 585 590
 Thr Phe Asn Asp Leu Gly Ile Thr Thr Ser Ser Thr Asn Ala Leu Phe
 595 600 605
 Ser Ile Ser Ser Asp Ser Leu Asn Ser Gly Glu Glu Trp Tyr Leu Ser
 610 615 620
 Gln Leu Phe Leu Val Lys Glu Ser Ala Phe Thr Thr Gln Ile Asn Pro
 625 630 635 640
 Leu Leu Lys

What is claimed is:

1. A method for recombinantly expressing a protein, comprising:

- (1) obtaining bacterial cells comprising an expression cassette encoding the protein and an expression cassette encoding a modified Cry protein, wherein the bacterial cells are *Bacillus subtilis* (Bs), *Bacillus thuringiensis* (Bt), or *Escherichia coli* (*E. coli*) cells,
- (2) culturing the bacterial cells under conditions permissible for the expression of the protein and the modified Cry protein, thereby expressing separately the protein and the modified Cry protein, and
- (3) permitting the modified Cry protein to form crystal containing the protein in the bacterial cells, wherein the modified Cry protein comprises the amino acid sequence set forth in SEQ ID NO:4, which is modified to comprise:
 - (i) substitutions with aspartate (D) or glutamate (E) at residues K384, N391, N395, S425, Q430, T436, Q437, K442, R443, T461, and K467;
 - (ii) addition of the amino acid sequence set forth in SEQ ID NO:8 in 1-3 repeats at the C-terminus;
 - (iii) one or more deletions in segment 110-128, 181-197, 211-226, 250-264, 423-436, 457-471, 593-598, or 627-632;
 - (iv) one or more substitutions at residue 5145 or H161, or
 - (v) the amino acid sequence set forth in SEQ ID NO:5.

2. The method of claim 1, wherein the protein is an enzyme.

3. The method of claim 2, wherein the enzyme is a lipase, ligase, hydrolase, esterase, protease, or glycosidase.

4. The method of claim 3, wherein the lipase is a *Proteus mirabilis* lipase (PML), an I118V+E130G mutant of PML (PML^{VG}), a *Bacillus subtilis* lipase A (lipA), or a lipase A fused to a polyarginine tail comprising nine consecutive arginine residues (lipAR9).

5. The method of claim 1, wherein the modified cry protein comprises the amino acid sequence set forth in SEQ ID NO:6.

6. The method of claim 1, wherein the expression cassette encoding the protein and the expression cassette encoding the modified Cry protein are one single expression cassette.

7. The method of claim 6, wherein the one single expression cassette comprises (1) one copy of polynucleotide sequence encoding the modified Cry protein and (2) one copy or two or more copies of polynucleotide sequence encoding the protein.

8. The method of claim 7, wherein (1) the polynucleotide sequence encoding the modified Cry protein and (2) the polynucleotide sequence encoding the protein are operably linked to one single promoter.

9. The method of claim 8, wherein the one single promoter is operably linked to (1) one copy of the polynucleotide sequence encoding the modified Cry protein, followed by (2) one copy of the polynucleotide sequence encoding the protein, with one ribosome binding site between (1) and (2).

10. The method of claim 8, wherein the one single promoter is operably linked to (1) one copy of the polynucleotide sequence encoding the modified Cry protein, followed by (2) two or more copies of the polynucleotide sequence encoding the protein, with one ribosome binding site between (1) and (2) and between two copies of polynucleotide sequence encoding the protein.

11. The method of claim 7, wherein (1) the polynucleotide sequence encoding the modified Cry protein and (2) the polynucleotide sequence encoding the protein are operably linked to two separate promoters.

12. The method of claim 11, wherein the two separate promoters are two different promoters.

13. The method of claim 11, wherein (1) the polynucleotide sequence encoding the modified Cry protein and (2)

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the polynucleotide sequence encoding the protein share one single termination codon, resulting in one copy of the modified Cry protein and two copies of the protein being expressed.

14. The method of claim 1, wherein the expression cassette encoding the protein and the expression cassette encoding the modified Cry protein are two separate expression cassettes.

15. The method of claim 1, wherein the modified Cry protein comprises the amino acid sequence set forth in SEQ ID NO:7.

16. The method of claim 1, wherein the modified Cry protein comprises the amino acid sequence set forth in SEQ ID NO:8 fused to the C-terminus of the amino acid sequence set forth in SEQ ID NO:4.

17. The method of claim 1, wherein two or more proteins are expressed with the modified Cry protein and are contained within the crystal formed by the modified Cry protein.

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18. The method of claim 1, wherein the modified Cry protein comprises the amino acid sequence set forth in SEQ ID NO:9.

19. The method of claim 1, further comprising, prior to step (1), introducing into the bacterial cells the expression cassette encoding the protein and the expression cassette encoding the modified Cry protein.

20. The method of claim 1, further comprising, after step (3), isolating the crystal formed by the modified Cry protein containing the protein.

21. The method of claim 20, further comprising, after the isolating step, washing the crystal under conditions permissible for releasing the protein from the crystal, thereby releasing from the crystal and isolating the protein.

22. The method of claim 20, wherein the protein is an enzyme.

23. The method of claim 1, wherein the bacterial cells are *Bacillus thuringiensis* (Bt) cells.

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